

# Toward a Unified Model for Understanding and Generation

One root question —can a single model be both a understander and a generator?

My personal view: **representation can bridge understanding and generation together.**

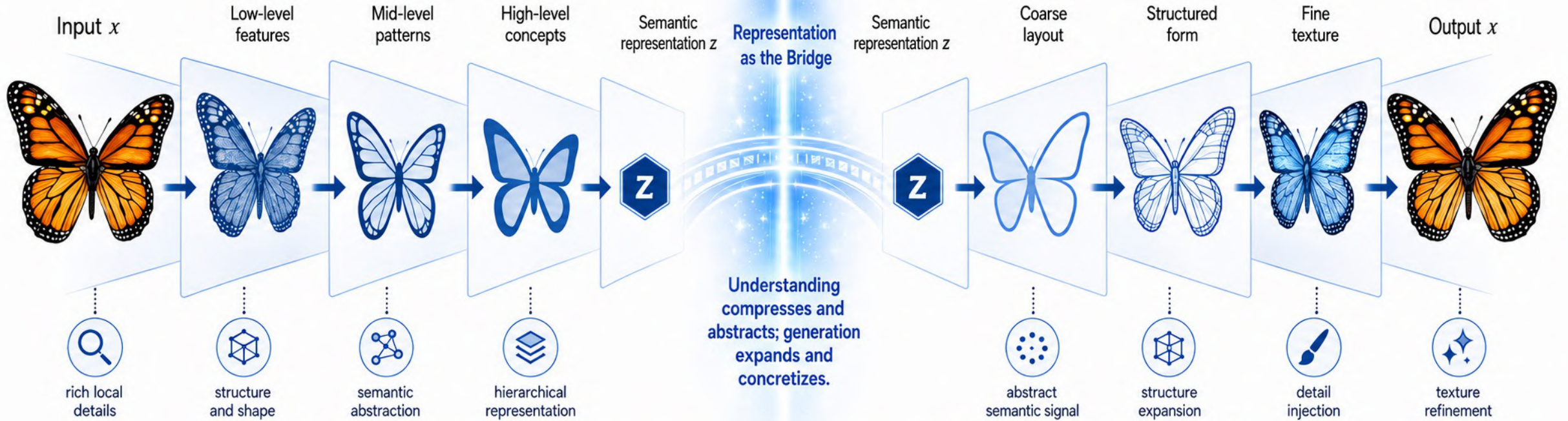
Tao Zhang · 2026-08-14

吴泰霖团队组会报告

课题组主页: <https://ai4s.lab.westlake.edu.cn/>

## Understanding Process

## Generation Process



**Duality:** a good understanding process induces a good reverse generation process.



### Abstraction & Compression

Extract what matters; discard the rest.



### Hierarchical Representation

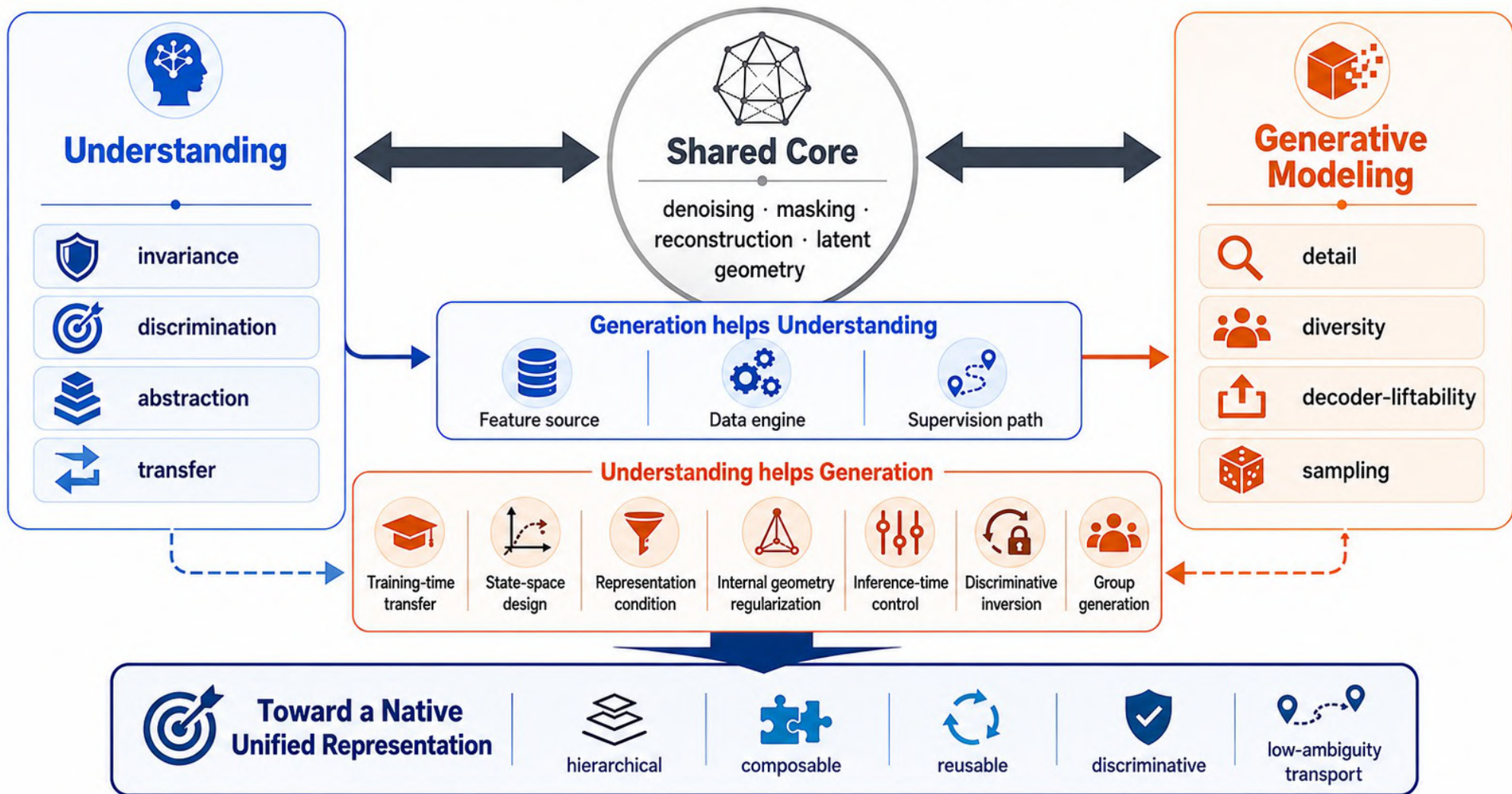
Organize knowledge across multiple levels.



### Detail Synthesis & Reconstruction

Add back what matters to the task.

# Toward a Unified Model for Understanding and Generation



# Roadmap

§1 Outline

§2 Motivation — why a unified model, why now, the several gaps, and my personal views on what a good representation needs.

§3 Intro — understanding (survey taxonomy) + generation (DDPM).

§4 Generation helps Understanding — diffusion's own representations: where, when good, when broken.

§5 Understanding helps Generation — five ways representation is injected into generation.

§6 The Story of Hiddens of Generative Model: A Quick Tour

§7 Building a Unified Model — existing attempts and their limits.

§8 Open Question & Wild Ideas & Summary — answer the root question.

1 Outline

2 Motiv

3 Intro

4 Gen→Rep

5 Rep→Gen

6 Quick Tour

7 Unified

8 Summary

# The root question

**【Problem】** How do we build a single unified model that does both understanding and generation — better than specialized models at each?

**【Why-important】** Understanding is winning, gen paradigms stall, the two are one coin; a native unified model would let each side lift the other instead of being bolted together.

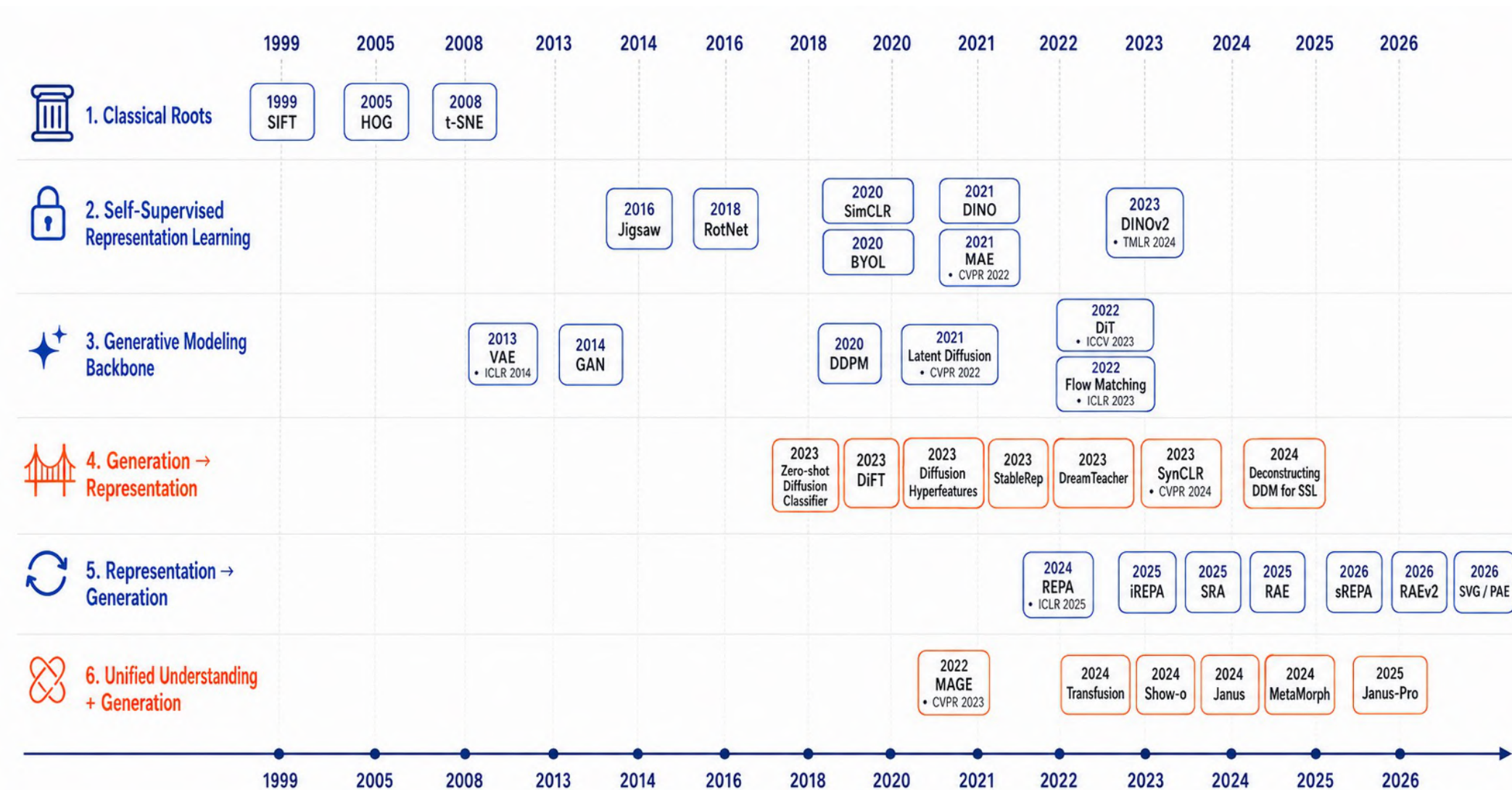
What I cannot create, I do not understand! By Richard Feynman



What I cannot understand, I do not create!

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## Literature timeline

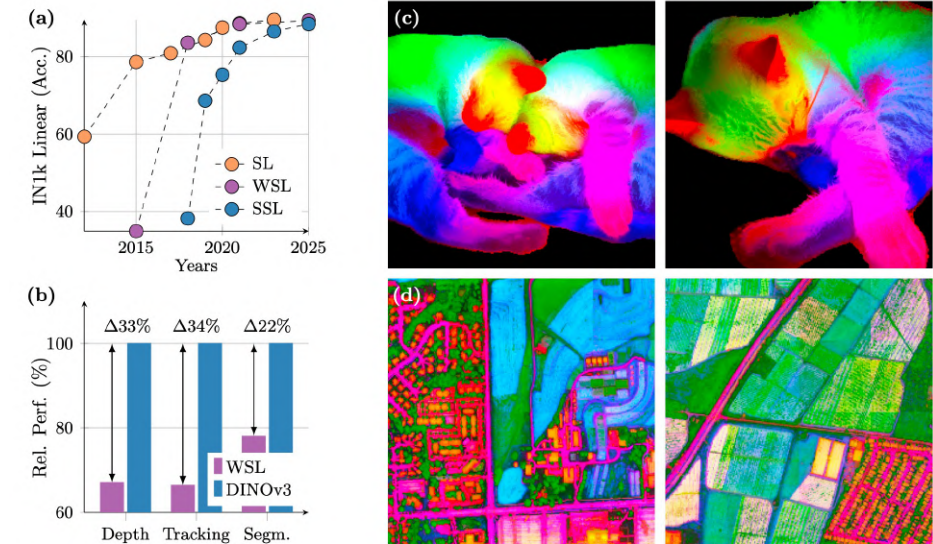


Main year = first public appearance; small text = venue/year when useful.

## §2.1 Background — Understanding is winning

- Understanding (Representation learning) keeps getting stronger: DINOv3-class features excel across nearly every downstream task (classification, segmentation, correspondence, depth) with no labels.
- Pure self-supervised backbones are now the default feature source for the whole vision stack.

| TASK                              | BENCHMARK        | DINO<br>ViT-B/8<br>0.09B | DINOv2<br>ViT-G/14<br>1.1B | DINOv3<br>ViT-7B/16<br>7B | SIGLIP 2<br>ViT-G-OPT/16<br>1.8B | PE<br>ViT-G/14<br>1.9B |
|-----------------------------------|------------------|--------------------------|----------------------------|---------------------------|----------------------------------|------------------------|
| Segmentation                      | ADE-20k          | 31.8                     | 49.5                       | <b>55.9</b>               | 42.7                             | 38.9                   |
| Depth estimation                  | NYU ↓            | 0.537                    | 0.372                      | <b>0.309</b>              | 0.494                            | 0.436                  |
| Video tracking                    | DAVIS            | 68.7                     | 76.6                       | <b>83.3</b>               | 62.9                             | 49.8                   |
| Instance retrieval                | Met              | 17.1                     | 44.6                       | <b>55.4</b>               | 13.9                             | 10.6                   |
| Image classification              | ImageNet ReaL    | 85.9                     | 89.9                       | 90.4                      | <b>90.5</b>                      | 90.4                   |
| Image classification              | ObjectNet        | 39.9                     | 66.4                       | 79.0                      | 78.6                             | <b>80.2</b>            |
| Fine-grained Image classification | iNaturalist 2021 | 68.3                     | 86.1                       | <b>89.8</b>               | 82.7                             | 87.0                   |



**Figure 1:** (a) Evolution of linear probing results on ImageNet1k (IN1k) over the years, comparing fully-supervised (SL), weakly-supervised (WSL) and self-supervised learning (SSL) methods. Despite coming into the picture later, SSL has quickly progressed and now reached the Imagenet accuracy plateau of recent years. On the other hand, we demonstrate that SSL offers the unique promise of high-quality dense features. With DINOv3, we markedly improve over weakly-supervised models on dense tasks, as shown by the relative performance of the best-in-class WSL models to DINOv3 (b). We also produce PCA maps of features obtained from high resolution images with DINOv3 trained on natural (c) and aerial images (d).

## §2.1 Background — generative paradigms are stalling



- By contrast, *paradigm-level* innovation in generation is slowing: since DDPM/LDM/DiT the gains are increasingly incremental (schedule tweaks, guidance tricks, scaling).
- The big open lever is no longer "a new sampler" but "a better representation/geometry to generate in".

### Generative Models: Four Families Converging on One Recipe

2013 → 2026: separate families → latent-space diffusion/flow transformers — and the recipe's weak point is exactly where representations enter

#### Latent / likelihood



→ survives as everyone's latent space

#### Adversarial



→ survives inside tokenizer losses

#### Autoregressive



→ merges with LLMs → unified models

#### Diffusion / score / flow



#### 2026 default recipe

VAE latent space  
+ transformer backbone  
+ flow-matching objective  
+ (adv. loss in tokenizer)  
all four families, one model

#### THE RECIPE'S WEAK POINT (→ §4)

The denoiser must learn semantics from pixels/latents ALONE, slowly and implicitly — and the VAE latent is optimized for reconstruction, not meaning.

This is exactly where representations help.

#### READ THE TREND

Families did not die — they specialized into LAYERS of one stack (latent / critic / sequence prior / transport).  
Generation converged from the model side; §3-§4 ask what the stack still lacks: explicit representations.

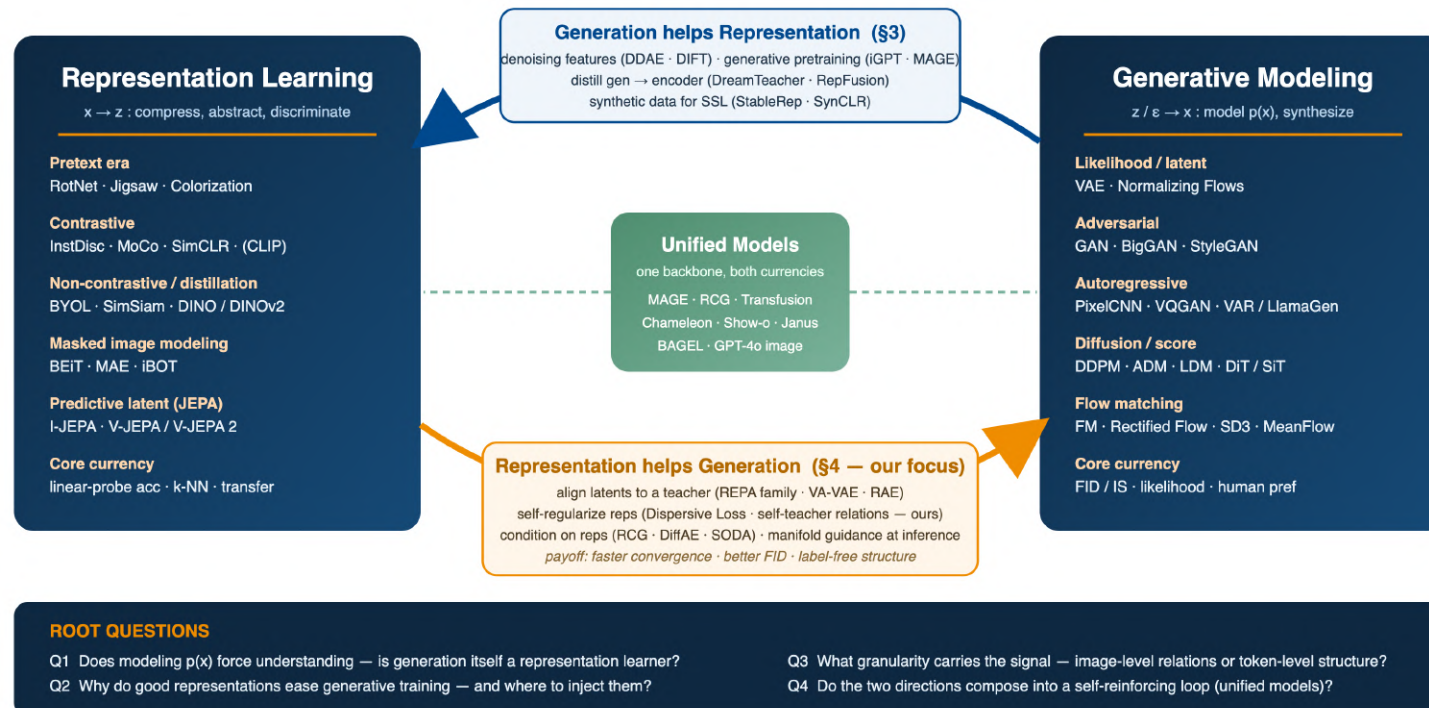
## §2.1 Background — two sides of one coin



- Left: representation methods & products are flowing *into* generation (VA-VAE / REPA / RAE align or borrow SSL features).
- Right: generation models' internal features are used *for* downstream understanding (diffusion features for classification / correspondence).
- The two directions are not separate research areas — they are the same coin.

### Representation Learning × Generative Modeling: Two Fields, One Loop

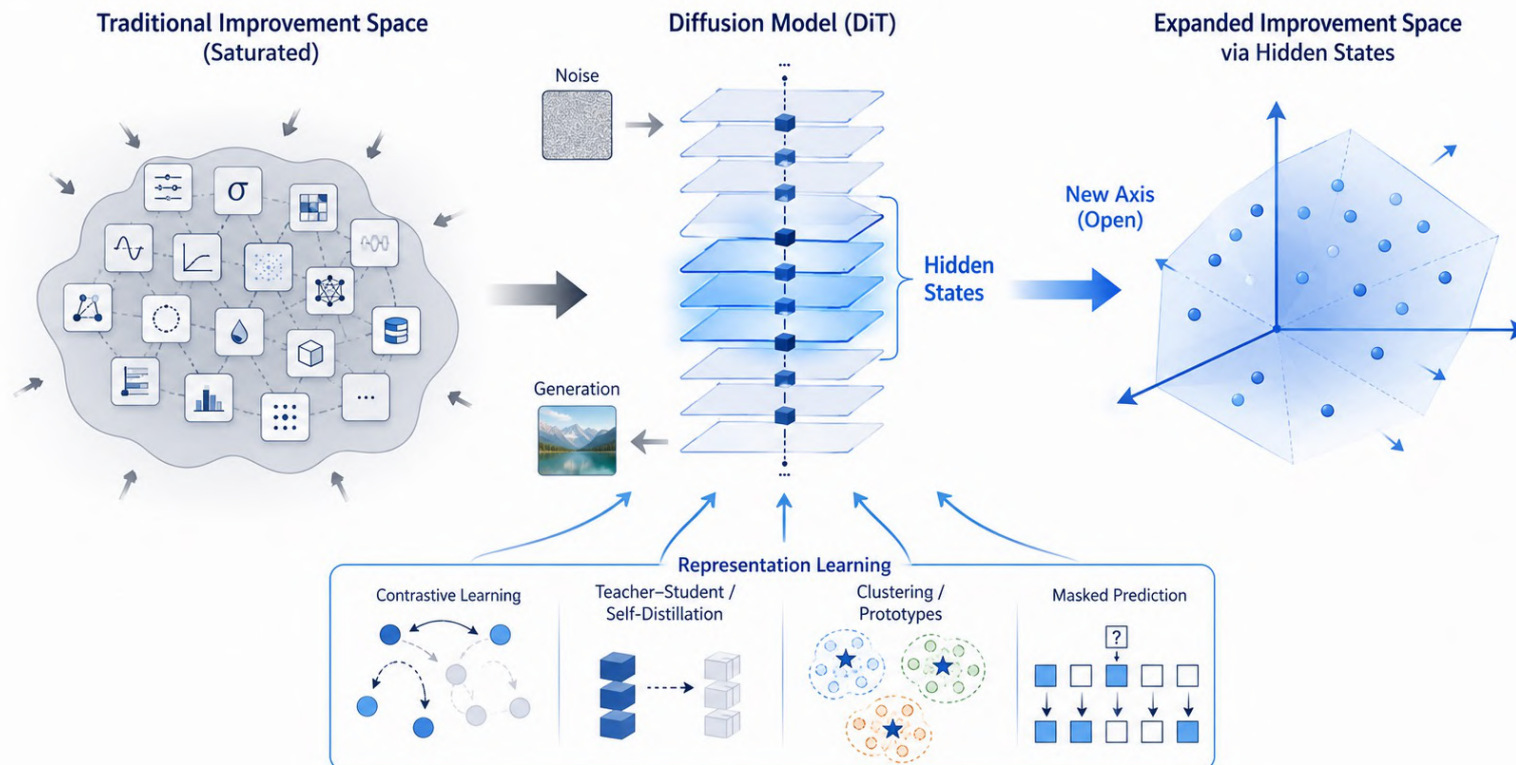
understanding compresses the world · generation reconstructs it — each is the other's supervision signal



## §2.1 Background — Interesting trend



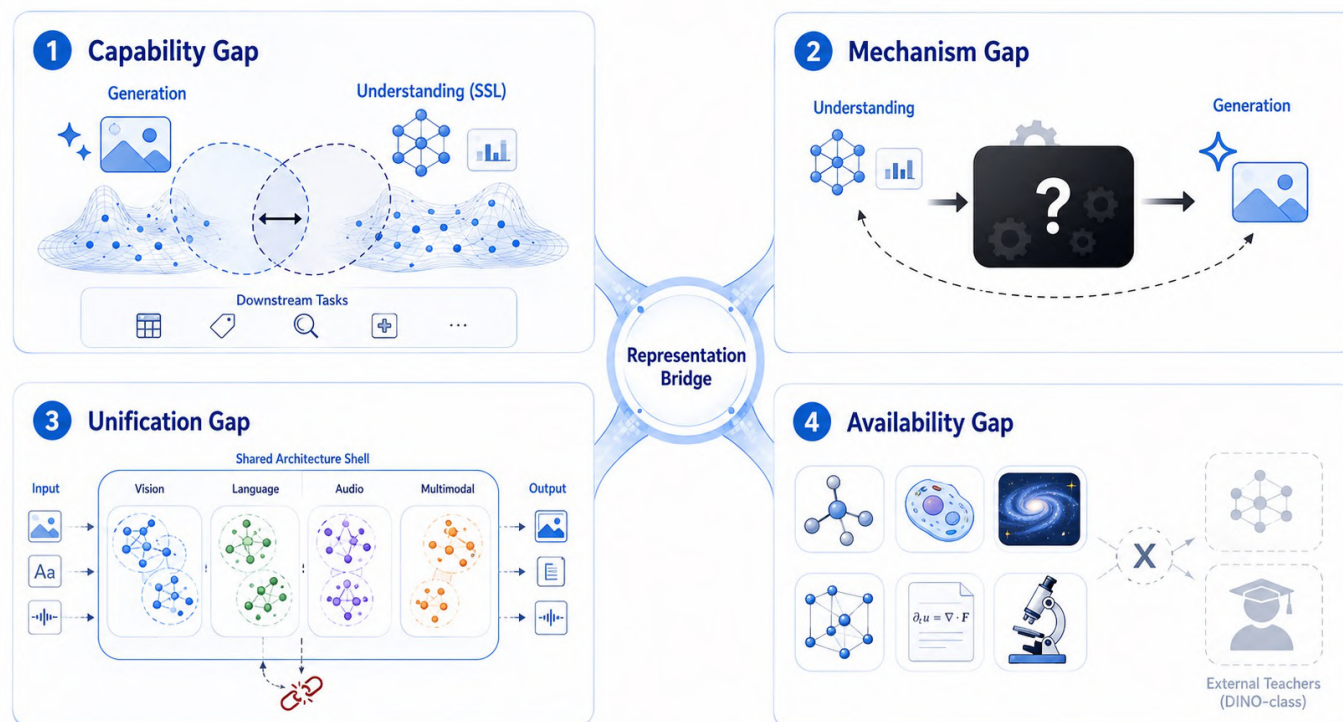
- **A new action space for diffusion improvement:** Beyond optimizing outputs, objectives, or sampling, recent work opens a new axis by directly analyzing and shaping **DiT hidden states**.
- **Old representation-learning ideas, revisited for generation:** Many recent advances are rediscovering and adapting classic ideas from **contrastive learning** and **representation learning**—alignment, self-distillation, masking, relational structure, and regularization.



## §2.1 Background — Several gaps



- **Capability gap** — Generation-assisted understanding remains unverified while generative representations still lag dedicated SSL features on some downstream tasks.
- **Mechanism gap** — Empirically, it is well-known that understanding helps with generation, but why remains unclear.
- **Unification gap** — Current unified models often share architecture, not a jointly optimized representation geometry.
- **Availability gap** — Many scientific domains do not have DINO-class external teachers.

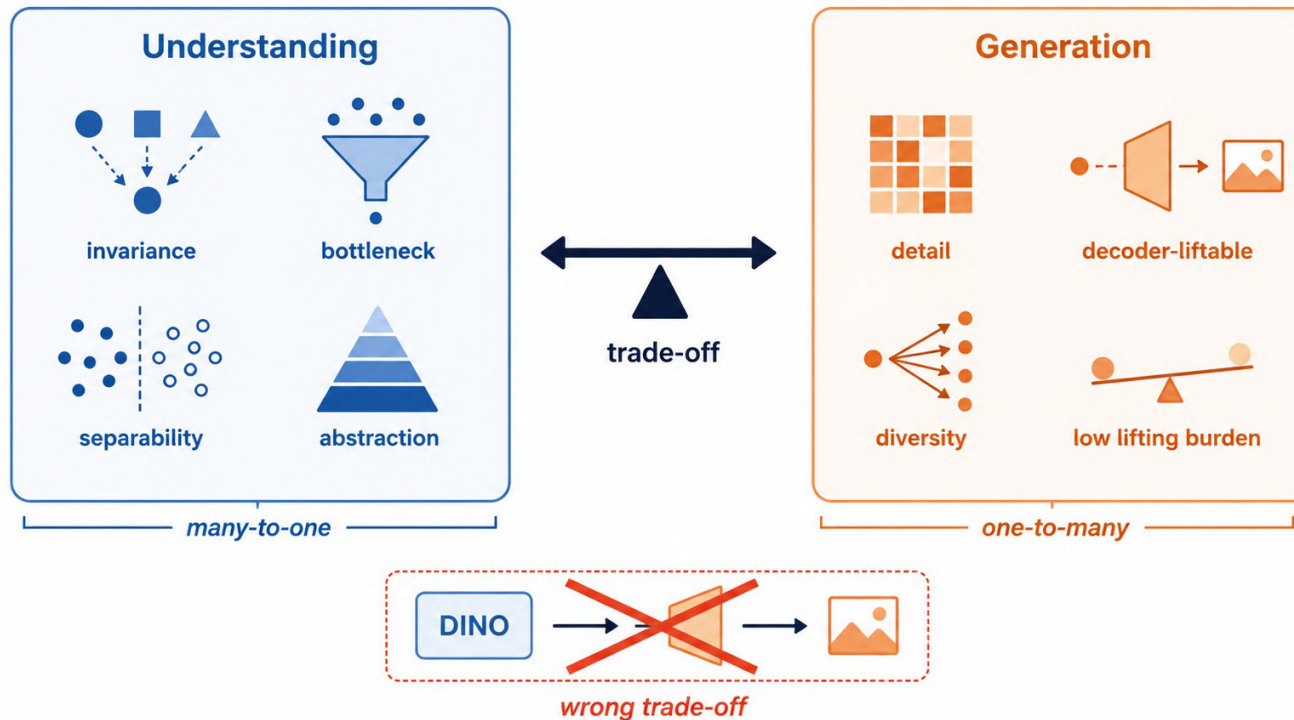


These gaps point at the same missing thing:  
a principled, native fusion.

## §2.2 Why is it hard — the requirements diverge



- **【Why-hard】** A representation good *for understanding* may not be good *for generation* — the two impose different requirements:
  - understanding wants: invariance to nuisance, strong information bottleneck, linear separability, many-to-one abstraction.
  - generation wants: high-frequency detail retained, decoder-liftable, one-to-many diversity, a low generative-lifting burden.
- Naive "just plug DINO into the generator" fails because it optimizes the wrong side of this trade-off.



## §2.4 Prior limits — no native unified model yet; MAE is closest



- **【Prior-limits】** Existing work attacks from the two ends separately — no *natively* unified model exists.
- **MAE is the closest thing to a native unified model** (masked reconstruction is simultaneously a generation task and a representation pretext) — but it has real limitations (weak generative fidelity, pixel-reconstruction target may not define a generation-friendly latent, no explicit information bottleneck control).
- This gap is exactly what §6 revisits.

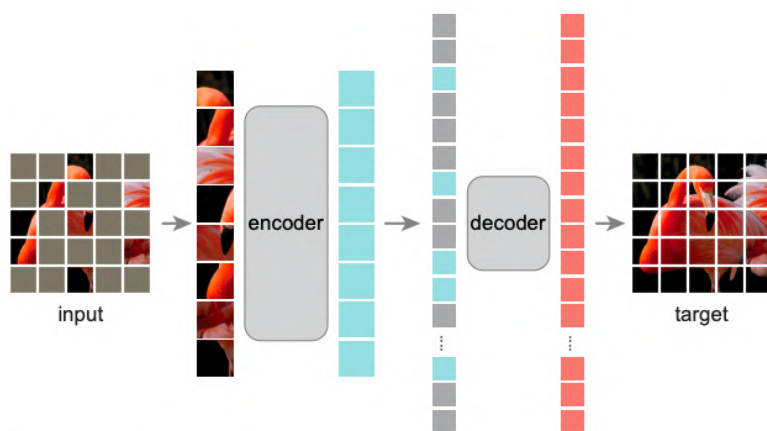
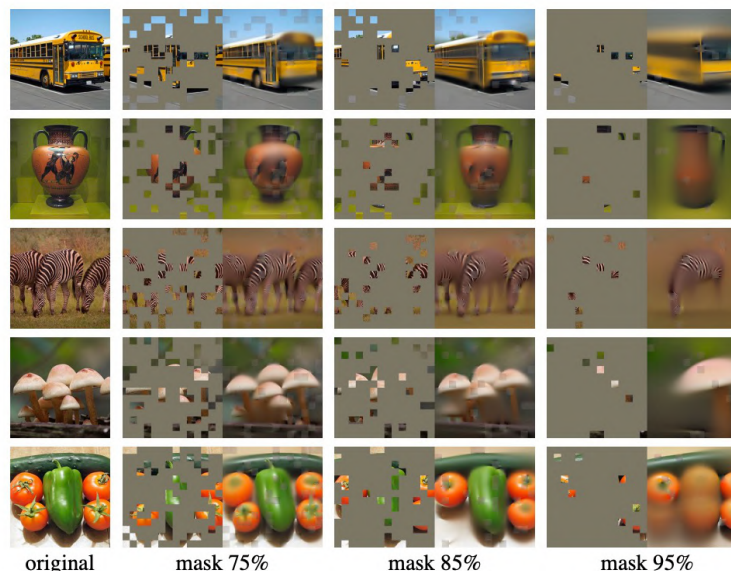


Figure 1. **Our MAE architecture.** During pre-training, a large random subset of image patches (*e.g.*, 75%) is masked out. The encoder is applied to the small subset of *visible patches*. Mask tokens are introduced *after* the encoder, and the full set of encoded patches and mask tokens is processed by a small decoder that reconstructs the original image in pixels. After pre-training, the decoder is discarded and the encoder is applied to uncorrupted images (full sets of patches) for recognition tasks.

| method             | pre-train data | ViT-B       | ViT-L       | ViT-H       | ViT-H <sub>448</sub> |
|--------------------|----------------|-------------|-------------|-------------|----------------------|
| scratch, our impl. | -              | 82.3        | 82.6        | 83.1        | -                    |
| DINO [5]           | IN1K           | 82.8        | -           | -           | -                    |
| MoCo v3 [9]        | IN1K           | 83.2        | 84.1        | -           | -                    |
| BEiT [2]           | IN1K+DALLE     | 83.2        | 85.2        | -           | -                    |
| MAE                | IN1K           | <u>83.6</u> | <u>85.9</u> | <u>86.9</u> | <b>87.8</b>          |

Representation learning

Table 3. Comparisons with previous results on ImageNet-



Generation Task

## §3 Intro — shared vocabulary

Brief, per the survey (A Survey on Self-Supervised Representation Learning, 2023). Build the vocabulary so §4/§5 land.

3.1 representation learning (native survey taxonomy)

3.2 generation (DDPM & some analysis).

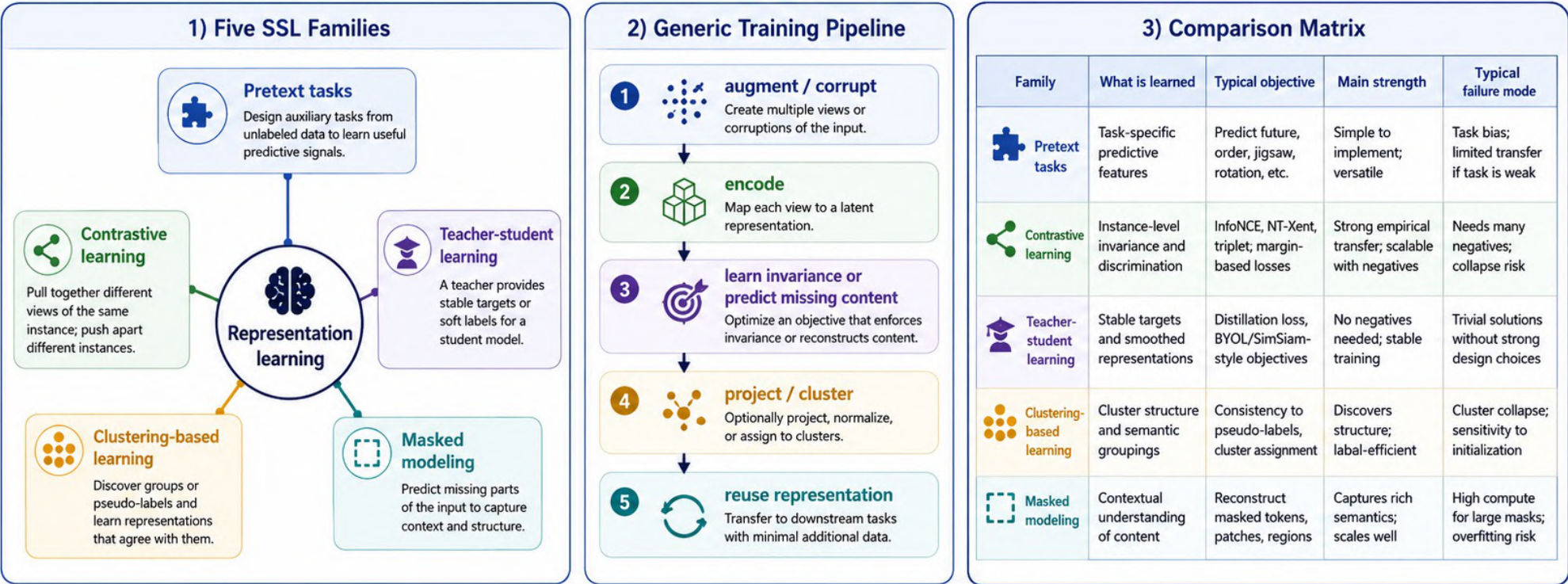
Frame (light): **【Problem】** no shared definition/taxonomy of "representation" and "generation";

**【Components】** the vocabulary = the survey's 5 method families + the diffusion objective.

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# Section 2A Map: Representation Learning

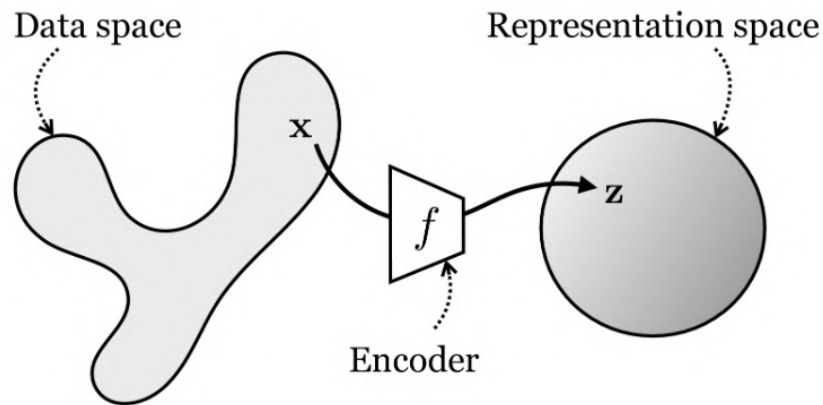
## SECTION MAP



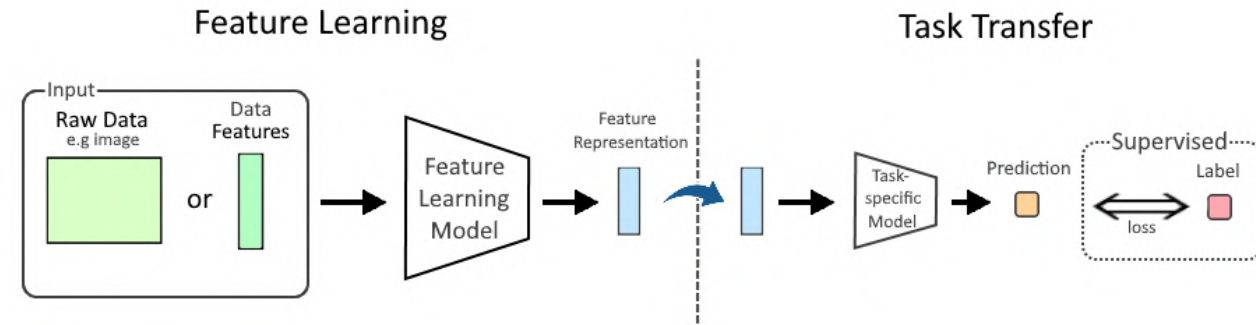
**Main role of this section:** build a shared vocabulary for what a good representation is — transferable, robust, semantically organized, and sufficiently structured for downstream reuse.

# §3.1 What is a representation?

- A representation maps raw data into a structure that is **easier to use**.



From <https://visionbook.mit.edu/>



Goodfellow, Ian, et al. *Deep learning*. Vol. 1. No. 2. Cambridge: MIT press, 2016.

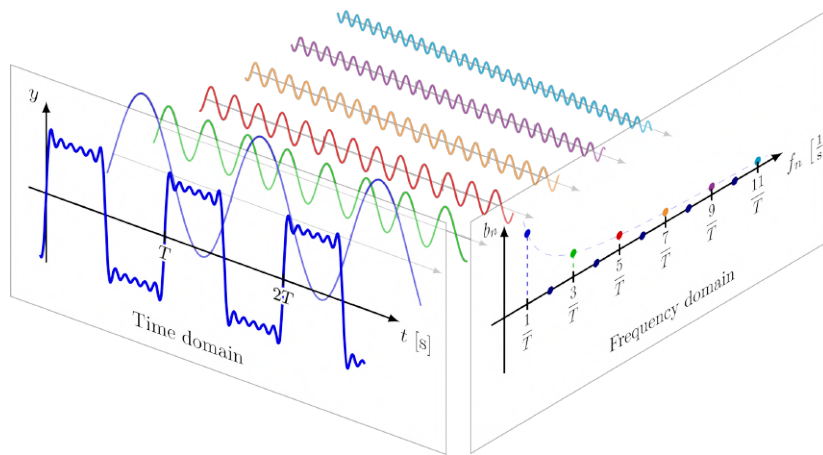
| View        | Representation means                     | Key question                   |
|-------------|------------------------------------------|--------------------------------|
| Statistical | sufficient statistic                     | sufficient for what?           |
| Geometric   | coordinates + metric + neighborhoods     | what does distance mean?       |
| Clustering  | soft sample grouping                     | who should be close?           |
| Generative  | latent state for reconstruction/sampling | can it be decoded and sampled? |

Different perspective for representation learning

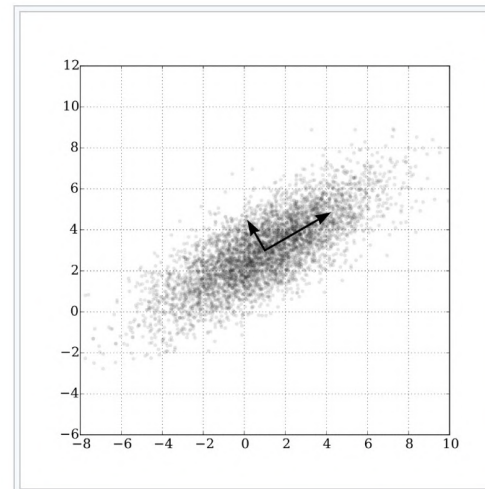
## §3.1.1 How to get representations — the non-AI ancestry



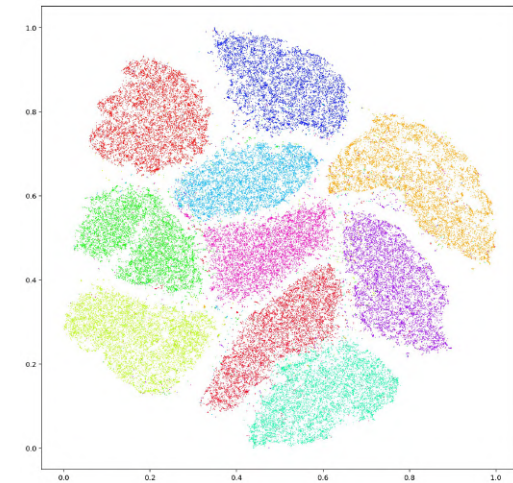
- **Hand-crafted:** SIFT, HOG, Fourier features, wavelet, MFCC, physical descriptors each hard-code a prior (locality, scale, frequency, invariance).
- **Linear dim-reduction:** PCA, ICA, NMF, CCA, LDA — good for theoretical intuition.
- **Manifold learning:** (Isomap / LLE / t-SNE family) — the geometric ancestor of "data lives on a low-dim manifold".



Fourier features



PCA



t-SNE

# §3.1.1 How to get representations — the AI ancestry

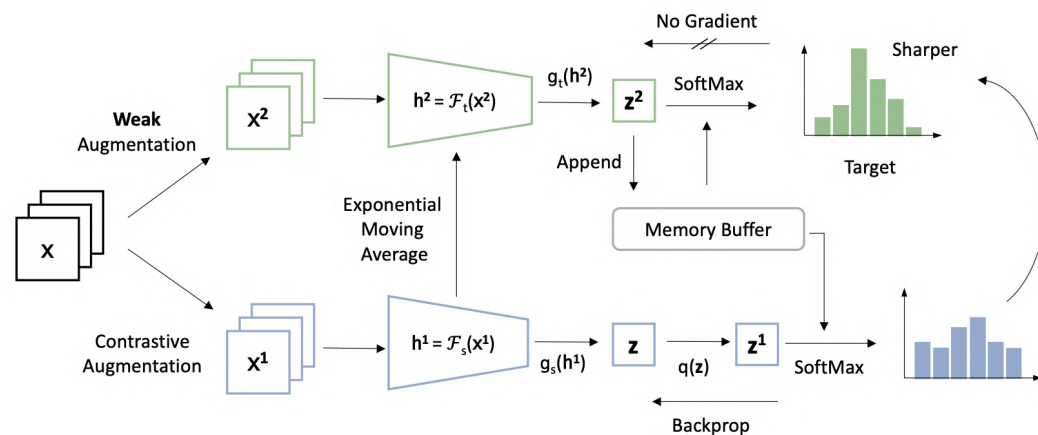


| Method Type                                | Representative Works               | Core Idea                                                      | Strengths / Limitations                                                                                                                   |
|--------------------------------------------|------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pretext Tasks</b>                       | Jigsaw, Colorization, RotNet       | Predict hand-designed transformations or missing attributes.   | Simple and interpretable; but task-specific shortcuts and limited semantic transfer.                                                      |
| <b>Contrastive Learning</b>                | CPC, SimCLR, MoCo, CLIP            | Pull positive views together; push negatives apart.            | Strong generalization and alignment—uniformity trade-off; needs good augmentations, large batches/queues, handles false negatives poorly. |
| <b>Redundancy Reduction / InfoMax</b>      | Deep InfoMax, Barlow Twins, VICReg | Preserve information while decorrelating feature dimensions.   | Avoids explicit negatives; but collapse prevention and feature whitening choices are delicate.                                            |
| <b>Teacher–Student / Self-Distillation</b> | BYOL, SimSiam, DINO, iBOT          | Match a stop-gradient or EMA teacher representation.           | Strong semantics without negatives; but teacher bottleneck and excessive invariance may discard generative details.                       |
| <b>Clustering / Prototype Learning</b>     | DeepCluster, SwAV, SeLa            | Learn features through pseudo-labels or prototype assignments. | Encourages semantic grouping; but sensitive to cluster number, imbalance, and assignment noise.                                           |
| <b>Masked / Generative Prediction</b>      | MAE, BEiT, SimMIM, JEPA            | Recover or predict masked pixels/tokens/latent targets.        | Scalable and spatially dense; but pixel reconstruction may waste capacity, while latent prediction depends heavily on target design.      |

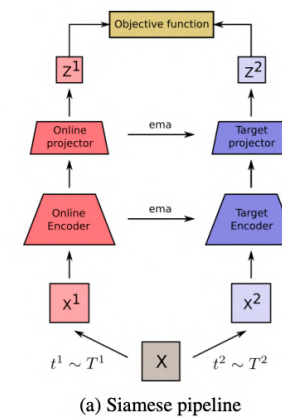
### §3.1.5 Advanced — soft relations (SCE & ReSSL), our ancestors



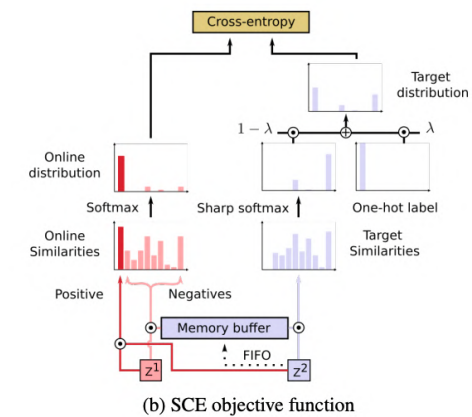
- Binary positive/negative is too crude: it ignores similarity *degree*, hierarchy, inter-class relations, and false negatives.
- **ReSSL**: match a *relation distribution* over the batch (weak-augmentation target) — same-instance views need not be identical, different instances need not be pushed fully apart.
- **SCE**: replace hard classify with a continuous *semantic similarity distribution* that decides push/pull.



ReSSL NeurIPS 2021



(a) Siamese pipeline



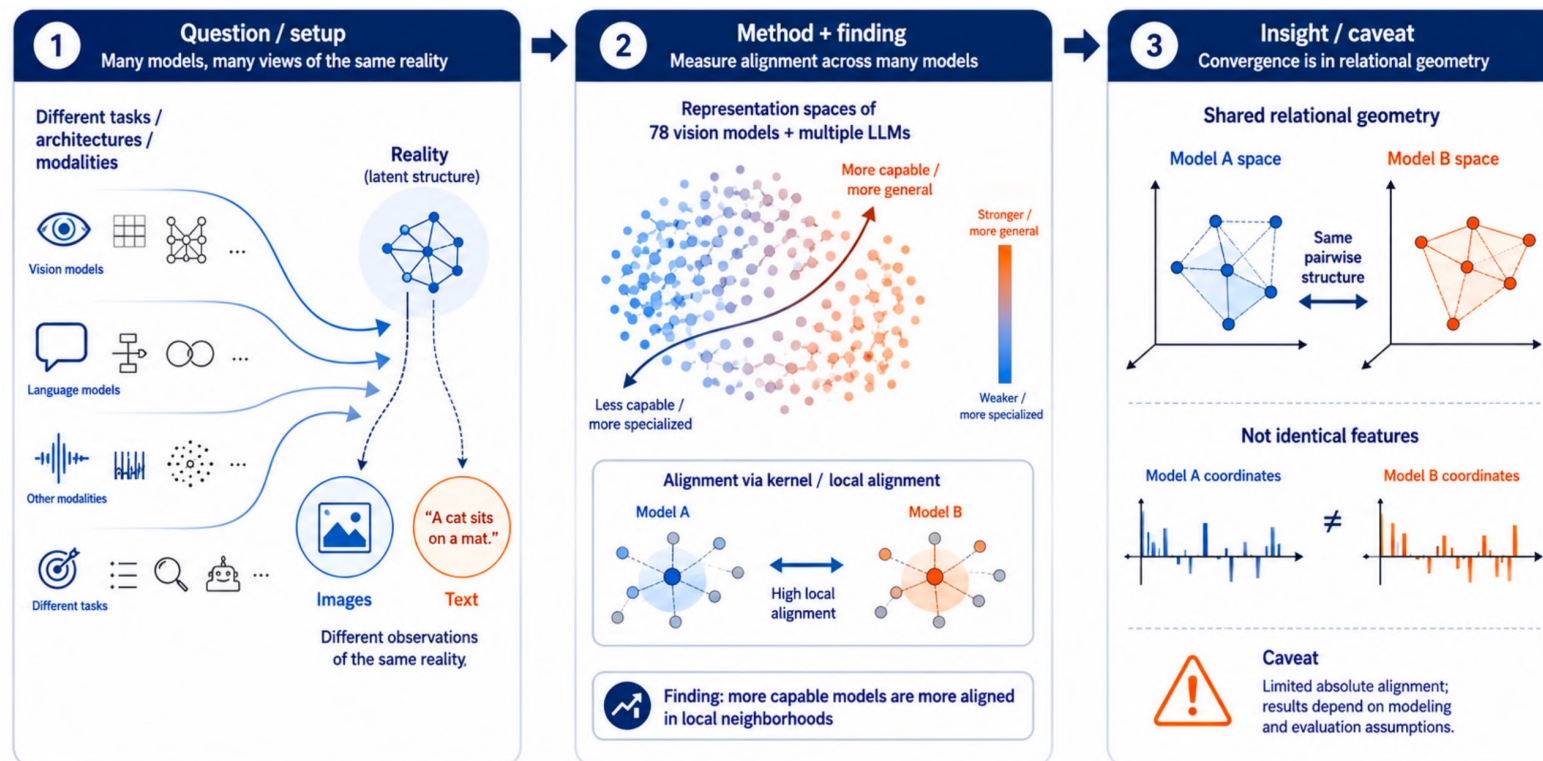
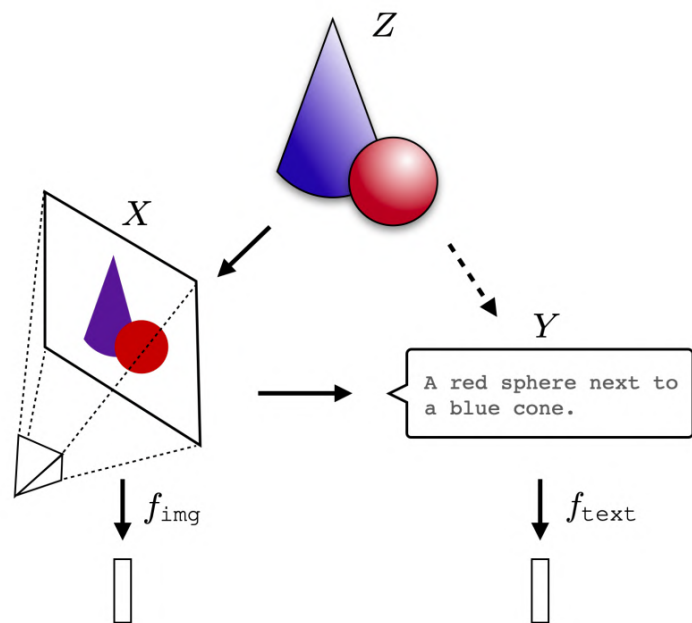
(b) SCE objective function

SCE [2111]

## §3.1.6 some personal view on representation learning

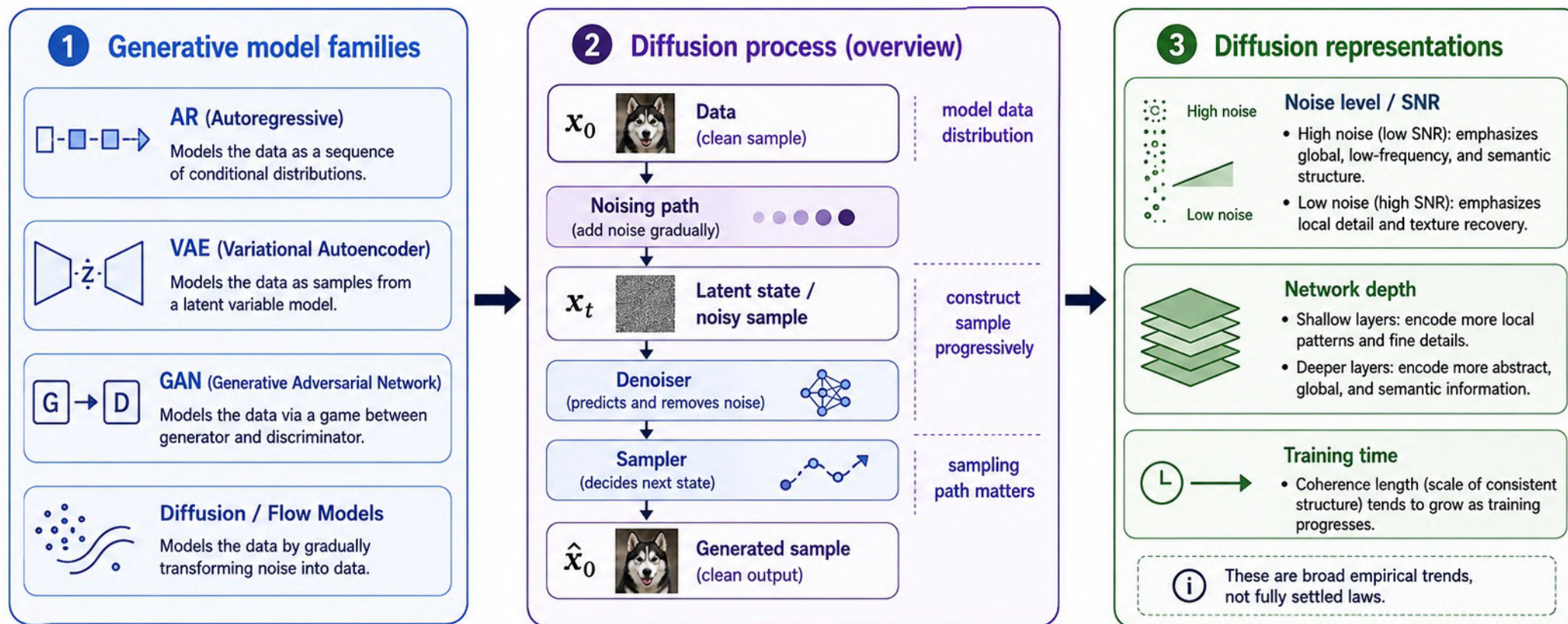


- **Question:** Do stronger models converge to a shared representation despite different tasks, architectures, and modalities?
- **Method:** Measure local/kernel alignment across **78 vision models and multiple LLMs**; derive a shared PMI-kernel view under idealized assumptions.
- **Finding:** More capable/general models become more aligned; shared information, task diversity, capacity, and simplicity bias may drive convergence.
- **Insight / Caveat:** Convergence appears mainly in **shared relational geometry**, not identical features or computation; absolute alignment is still limited and the theory relies on strong assumptions.



## Section 2B Map: Generative Models and Diffusion Representations

### SECTION MAP

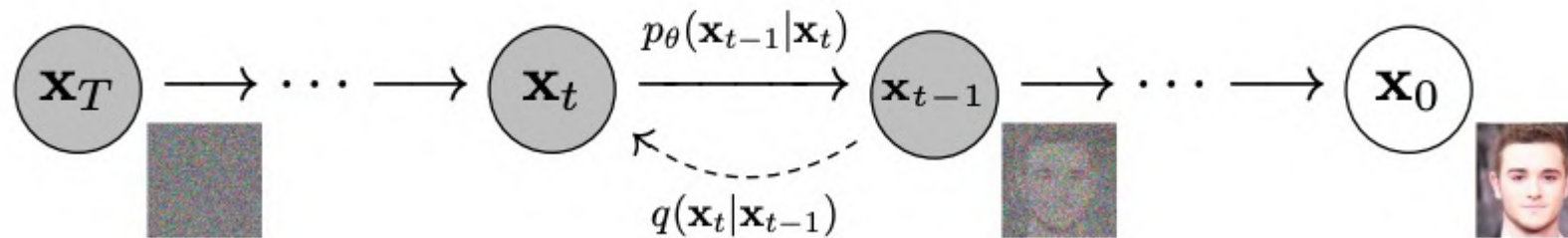


**Takeaway:** Generation is not only about sample synthesis; it also exposes internal structure along time, depth, and noise level.

## §3.2 Generation —DDPM



- Generation: sample / complete / simulate / design from a latent, condition, program, action, or world-state — not just "draw a picture".
- We define one family in depth (**DDPM**) and skip GAN/Flow/VAE/AR detail:



### Algorithm 1 Training

```
1: repeat
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ 
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:   Take gradient descent step on
        $\nabla_\theta \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t)\|^2$ 
6: until converged
```

### Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

## §3.2 Generation —DDPM: some analysis



### Have the Classical Action Spaces Plateaued?

The classical action spaces are not exhausted — but most have been explored for years and are increasingly optimized, coupled, or compute-limited.

| Action space            | Historical leverage | 2026 status                                |
|-------------------------|---------------------|--------------------------------------------|
| Data / Scale            | ★★★★★               | Still scaling, increasingly expensive      |
| Representation Space    | ★★★★★               | Rapidly evolving                           |
| Probability Path        | ★★★★★               | Maturing, still active                     |
| Prediction / Loss       | ★★★★                | Highly optimized; increasingly incremental |
| Architecture            | ★★★★★               | Still scaling; compute-heavy               |
| Guidance                | ★★★★★               | Mature, trade-off limited                  |
| Numerical Solver        | ★★★★★               | Strongly mature                            |
| Distillation / Flow Map | ★★★★★               | Very active for few-step generation        |

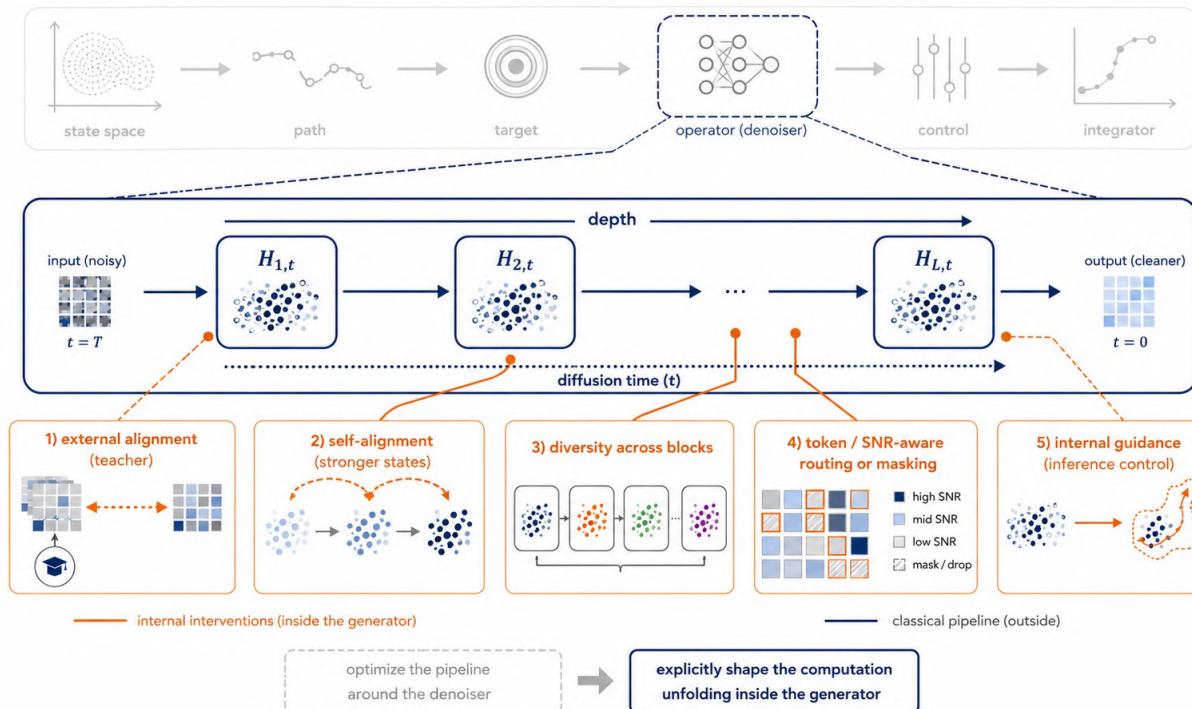
Progress increasingly comes from combining multiple mature knobs rather than discovering a fundamentally new one.

## §3.2 Generation —DDPM: some analysis



### A New Action Space Inside the Generator - A New Frontier: Dynamics of Diffusion Hidden States

- **REPA**: directly shapes intermediate hidden representations using external visual features.
- **SRA / LayerSync**: let stronger internal states supervise weaker ones.
- **HASTE / AHPA / MaskAlign**: show that alignment should depend on training phase, SNR, hierarchy, and token subset.
- **DiverseDiT**: explicitly optimizes representation dynamics across network depth.
- **Internal Guidance**: turns intermediate-layer dynamics into an inference-time control signal.



Take-away: Hidden states turn network depth itself into a new optimization axis.

## §3.2 Generation: DDPM- representation analysis across noise-level



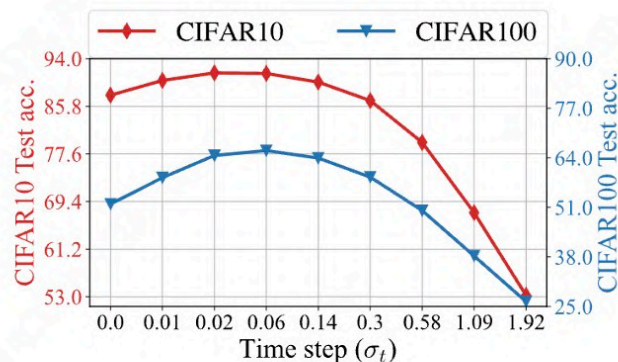
- Problem: How does representation quality change across diffusion noise levels and training stages?
- Extend low-dimensional posterior analysis to study the competition between data denoising and class specification.
- Representation quality is typically unimodal across noise levels. Near the final generation stage, improving reconstruction can conflict with discriminative representation quality.

Low noise-level

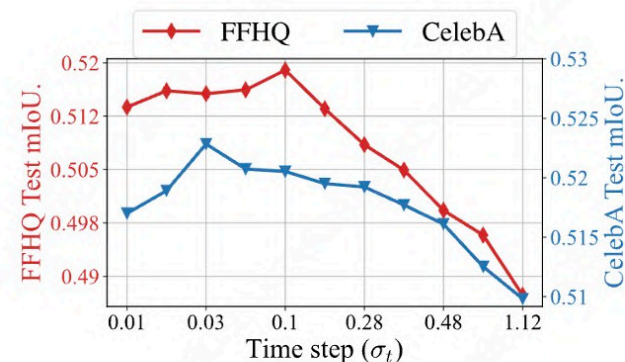
High noise-level

Input is near-clean data, to predict well,  
NN need encode the detailed fiber.

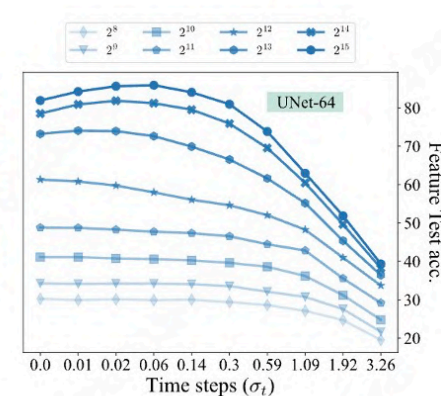
Input is noise, global semantic  
information and global prior help denoise



(a) Classification



(b) Segmentation



## §3.2 Generation: DDPM- representation analysis during training



- Problem: What distinguishes diffusion generalization from memorization inside the learned representation?
- Analyze a two-layer denoising autoencoder and compare its spectral structure with real diffusion models.
- Memorization yields localized, spiky representations dominated by a few directions; generalization yields more balanced representations that capture local data statistics.

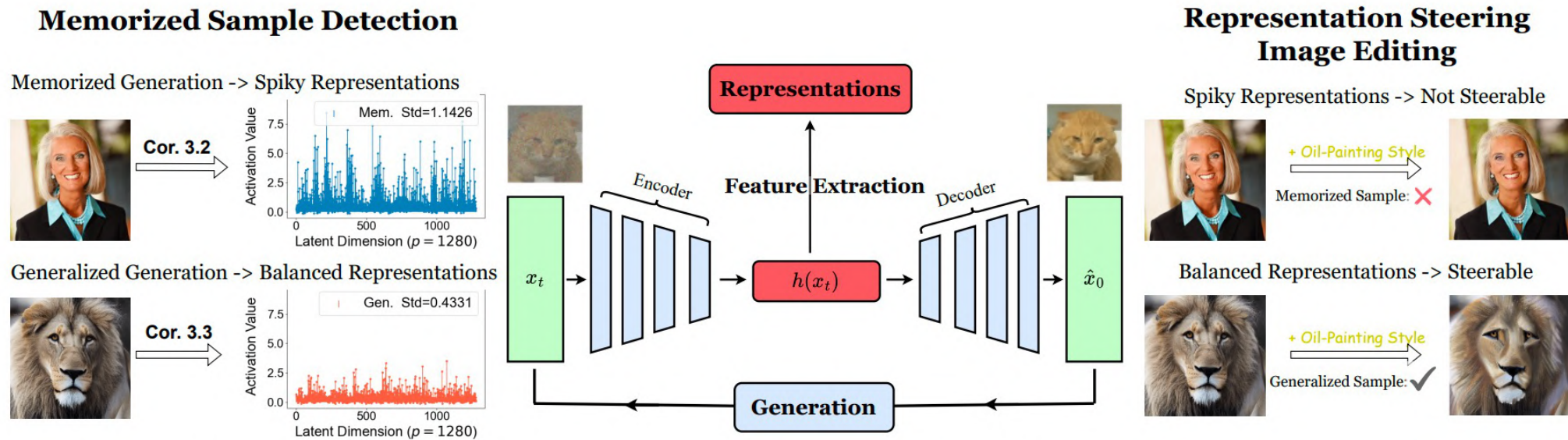
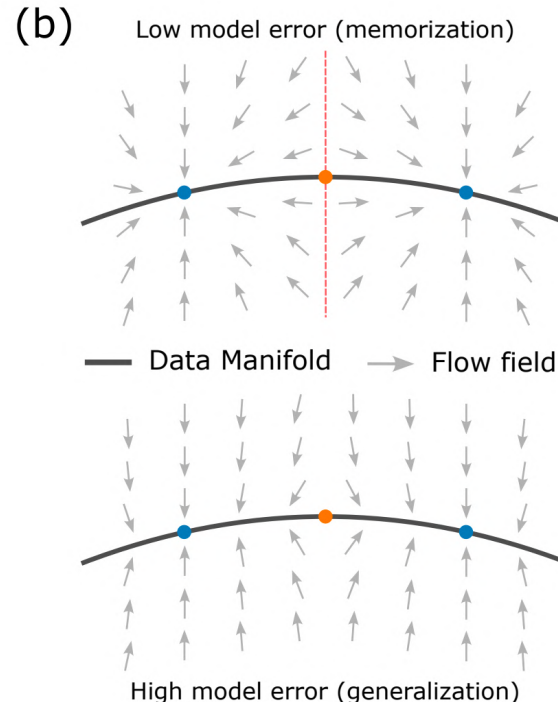


Figure 1: **Diffusion models generalize while learning benign internal representations.** Activa-

## §3.2 Generation: DDPM- representation analysis during training



- Problem: Why do real diffusion models generate novel samples even though an optimal denoiser of the empirical data distribution would reproduce the training set?
- Measure the train–validation gap of the denoising objective across noise levels, model sizes, training durations, and dataset sizes. Use an analytic toy model to relate this gap to model error and flow-field geometry, then compare noisy training samples with actual sampling trajectories.
- Model error can be beneficial: it **smooths the sharply localized empirical flow field** and prevents exact training-sample recall. Diffusion models may therefore **memorize locally during denoising training but generalize along the inference trajectory**.

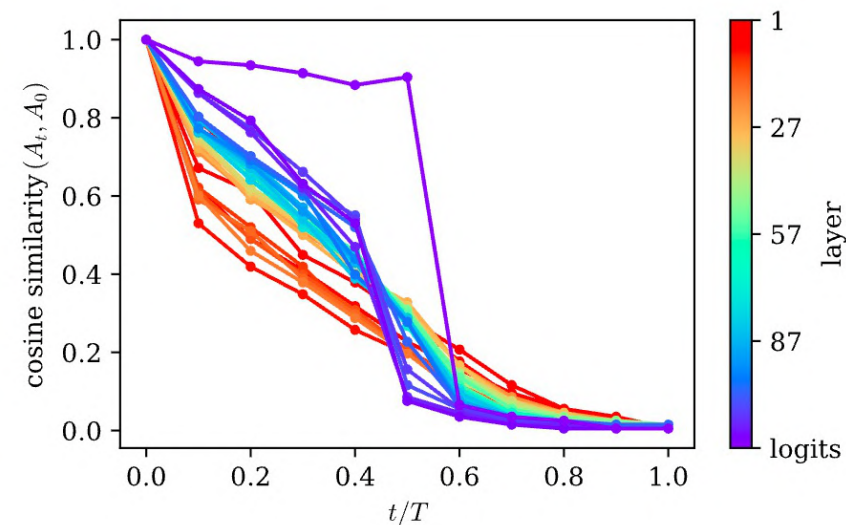


Diffusion Models Memorize in Training — and Generalize in Inference [2603]

## §3.2 Generation: DDPM- representation analysis across noise-level



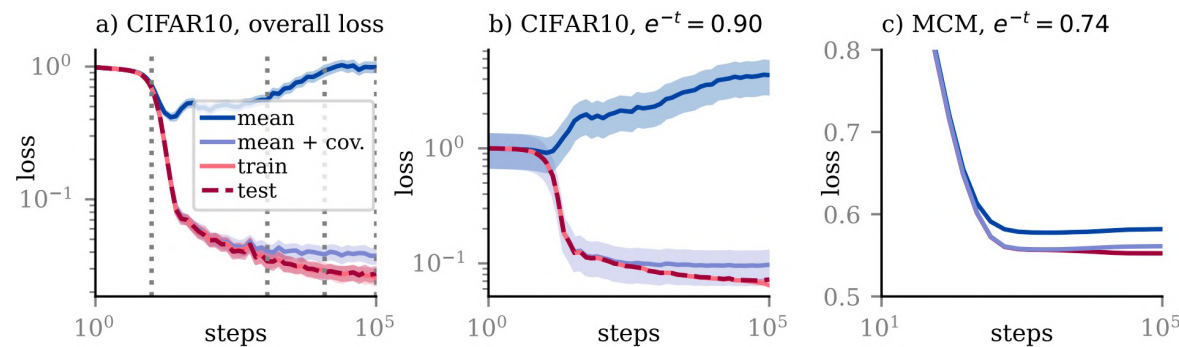
- How does diffusion noise erase different levels of information?
- Analyze forward–backward diffusion in a hierarchical generative model and validate it on ImageNet diffusion models.
- High-level class identity disappears abruptly at a critical noise level, while lower-level components and details decay more smoothly.
- Better view:
  - high-level class information may disappear abruptly near a phase transition;
  - low-level components may decay more smoothly;
  - different tasks may prefer different SNR windows;
  - the same timestep can serve different functions across layers.



## §3.2 Generation: DDPM- representation analysis across layers



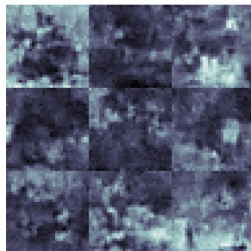
- Which statistics of the data distribution are learned first, and why are some structures much harder to acquire?
- Study natural-image diffusion models and a controlled mixed-cumulant model containing independently adjustable pairwise and higher-order correlations.
- **Main findings.**
  - Pairwise statistics are learned before higher-order correlations.
  - Pairwise structure requires linear sample complexity.
  - Fourth-order statistics generally require at least cubic sample complexity.
  - Higher-order learning becomes much easier when low- and high-order statistics share a correlated latent structure.



d) steps = 10



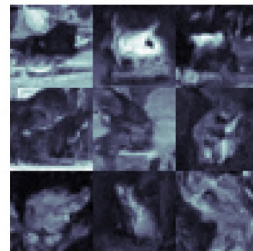
e) steps = 1152



f) steps = 12068



g) steps = 100000



A Theory of Learning Data Statistics in Diffusion Models, from Easy to Hard ICML2026

## §3.2 Generation: DDPM — A Diffusion Representation Atlas



| Axis               | Known pattern                                  | Open question                                    |
|--------------------|------------------------------------------------|--------------------------------------------------|
| Timestep / SNR     | intermediate windows can be most semantic      | are windows task-dependent?                      |
| Layer / resolution | useful features are scattered                  | which layers causally affect generation?         |
| Training time      | low-level statistics learned earlier           | when does generalization turn into memorization? |
| Data scale         | balanced spectra correlate with generalization | cause or consequence?                            |
| Architecture       | conditioning changes feature quality           | which bias is desirable?                         |

The missing object is not a single best feature, but a representation field:  
 $h(\ell, t, \tau, \text{task})$ .

## §4 overview — generation helps understanding (classification + key questions)

Claim: a model trained to generate is secretly a representation learner.

[1 Outline](#)[2 Motiv](#)[3 Intro](#)[4 Gen→Rep](#)[5 Rep→Gen](#)[6 Quick Tour](#)[7 Unified](#)[8 Summary](#)

## §4.2 Generation as Data, Path, and Supervision



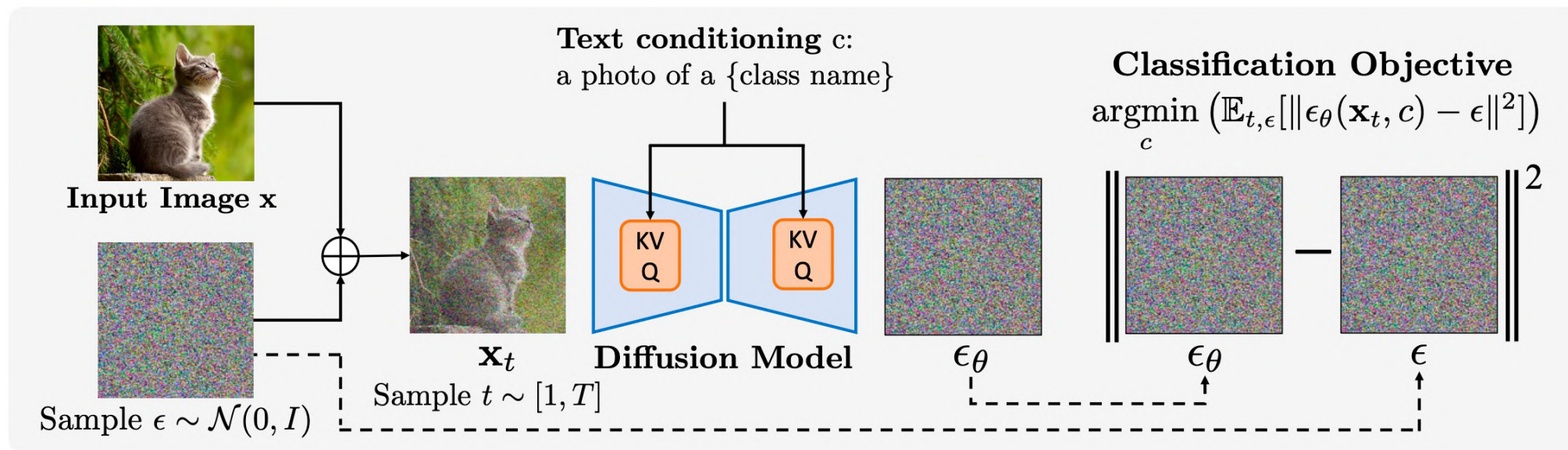
- Generation helps representation in three different ways:

| Mode             | Mechanism                         | Example                    |
|------------------|-----------------------------------|----------------------------|
| Feature source   | read internal features            | DIFT / Hyperfeatures       |
| Data engine      | synthesize training data          | StableRep / SynCLR         |
| Supervision path | corruption creates soft positives | diffusion noise / MAE mask |

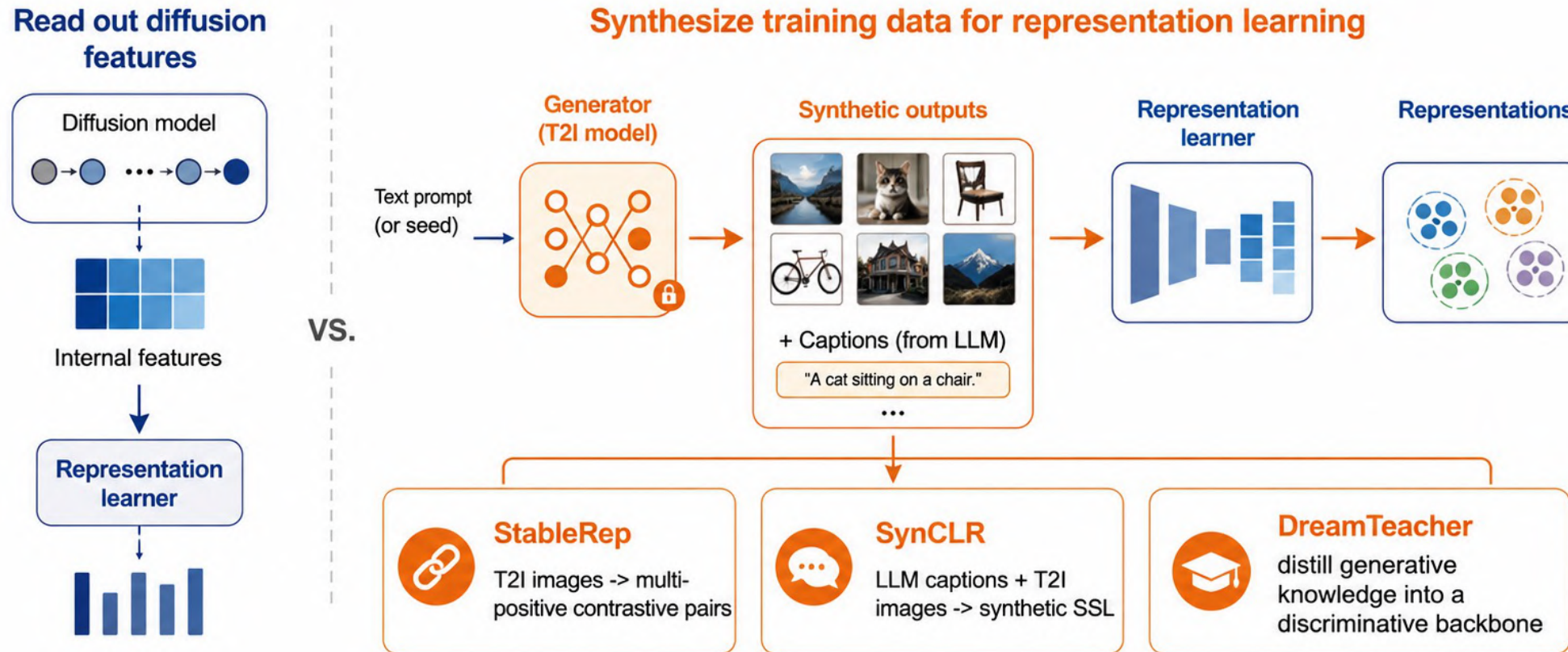
## §4.2 Specific downstream task with representation of Diffusion model



- Can a pretrained conditional diffusion model classify images without training a classifier head?
- For each candidate label, estimate the diffusion denoising or likelihood-related error conditioned on that label and choose the label with the lowest error.
- Text-to-image diffusion models provide competitive zero-shot classification and can classify categories poorly handled by bag-of-words vision-language models.
- Class-conditional generative likelihood can be inverted through Bayes' rule to perform recognition.



## §4.2 Generation as a data engine — the other half of "gen helps rep"

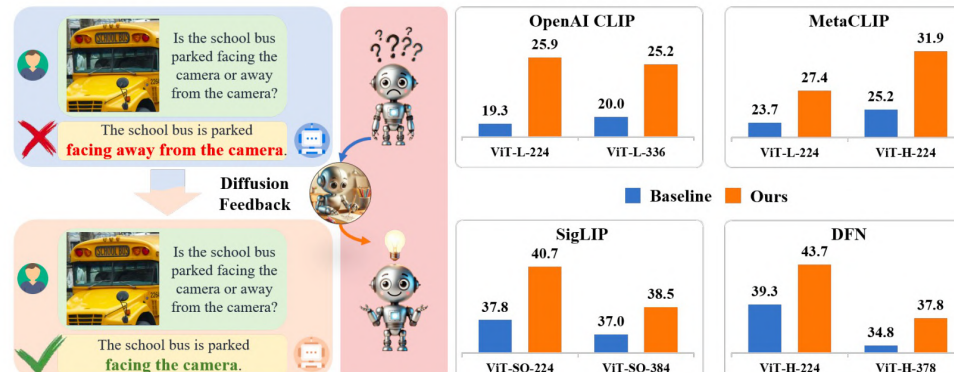


**Takeaway:** Both are generation helps representation; this line **manufactures data** rather than only reading features.

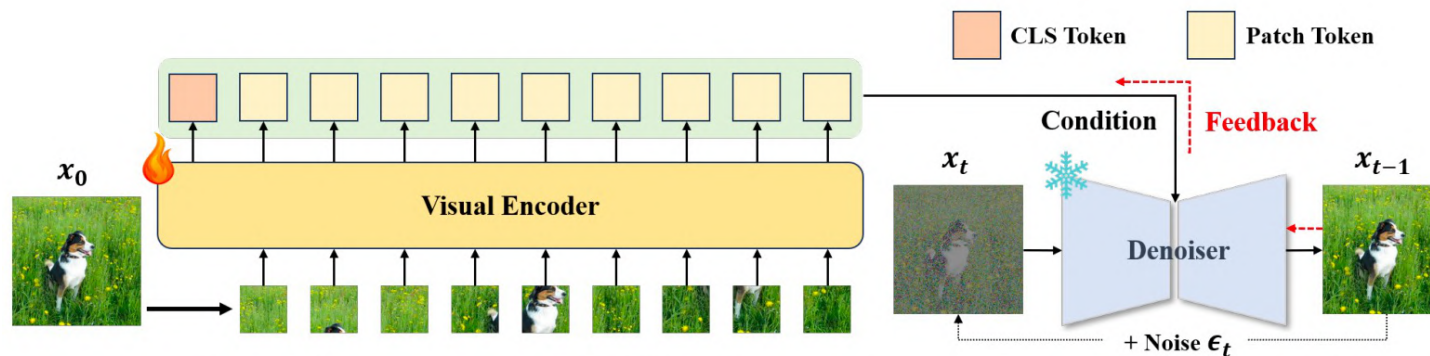
## §4.2 Specific downstream task with representation of Diffusion model



- Problem: CLIP has strong global semantics but often misses fine-grained visual details such as orientation, quantity, color, and local structure.
- Freeze a pretrained text-to-image diffusion model and optimize only the CLIP visual encoder. CLIP class and patch tokens condition the diffusion denoiser, and the diffusion noise-prediction loss provides image-only generative feedback. A moderate subset of local tokens is used as a **visual dense recap**.
- DIVA improves fine-grained perception on MMVP-VLM and downstream multimodal/segmentation tasks while largely preserving CLIP's zero-shot classification and retrieval performance. Partial local-token conditioning works better than using only the class token or all patch tokens.

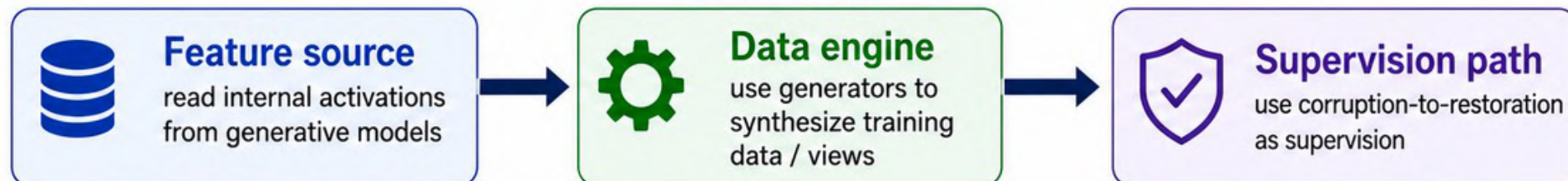


DIVA — Diffusion Feedback  
Helps CLIP See Better [2407]



## Section 4 Map: How Generation Helps Representation

GEN -> REP



| Mode                    | Representative works (selected)                                                                                                                                                                     | Main lesson                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Feature source</b>   | <ul style="list-style-type: none"> <li>Zero-shot Diffusion Classifier</li> <li>DiFT (Diffusion Features as Targets)</li> <li>Hyperfeatures</li> </ul>                                               | Denoising activations carry semantic and dense information. |
| <b>Data engine</b>      | <ul style="list-style-type: none"> <li>StableRep</li> <li>SynCLR</li> <li>DreamTeacher</li> </ul>                                                                                                   | Generators can manufacture representation data.             |
| <b>Supervision path</b> | <ul style="list-style-type: none"> <li>MAE (Masked Autoencoders)</li> <li>Deconstructing DDM for SSL</li> <li>I-DAE (Implicit Denoising Autoencoders)</li> <li>Diffusion-Noise-Based SSL</li> </ul> | Corruption strength defines soft supervision relations.     |

### ? Open questions

- Where does the gain come from (capacity, inductive bias, priors, or something else)?
- How should we aggregate information across time (diffusion steps) and layers?
- Which invariances are preserved or broken by generation and corruption?
- What is the compute cost at scale, and how can it be reduced?



**Key claim:** generative models are not only sample factories; they also provide feature sources, data engines, and supervision paths for representation learning.

## §5 overview — understanding helps generation (five categories + key questions)

Representation-helps-generation is *proven*; the frontier is which/where/what-granularity/why. Five injection categories:

(1) alignment-based (REPA family), (2) generate directly in representation space (RAE/RCG), (3) inversion — understanding-as-generation, (4) condition-based (DiffAE/SODA), (5) representation-learning loss injected (Dispersive loss), (6) others.

**【Why-hard】** naive add-ons fail: bigger/longer = slow-growth tax; blind dispersion is marginal.

**【Prior-limits】** external teacher bias + endpoint feature; pointwise/uniform ignores relation; "good latent" undefined.

1 Outline

2 Motiv

3 Intro

4 Gen→Rep

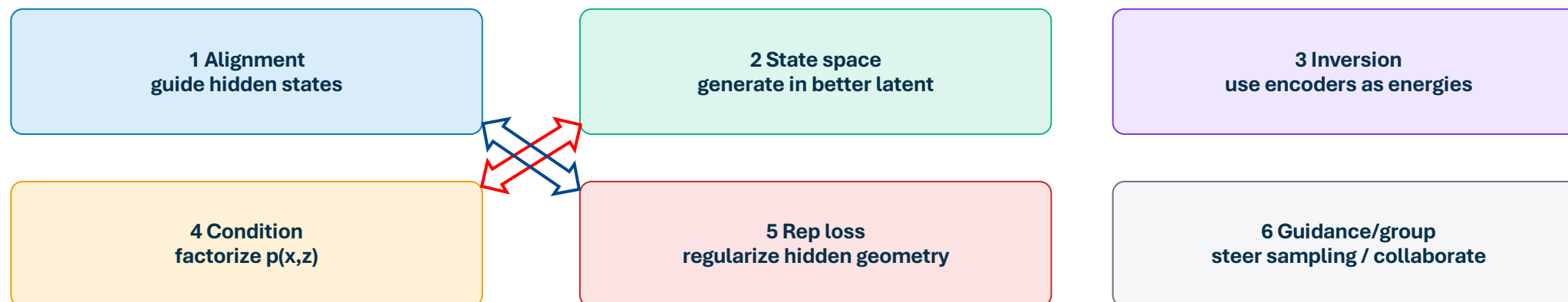
5 Rep→Gen

6 Quick Tour

7 Unified

8 Summary

## §5.0 A Better Taxonomy of Rep → Gen



| subsection | Category                              | Core question                                                   |
|------------|---------------------------------------|-----------------------------------------------------------------|
| §5.1       | Training-time representation transfer | how to guide denoiser hidden states?                            |
| §5.2       | Generative state-space design         | what latent/manifold should be generated?                       |
| §5.4       | Representation condition              | should representation be fixed, generated, or jointly denoised? |
| §5.5       | Internal geometry regularization      | can hidden structure be improved without teachers?              |
| §5.4       | Inference-time representation control | can features steer sampling?                                    |
| §5.3       | Discriminative inversion              | can classifiers/encoders be used as energies?                   |
| §5.6       | Group / multi-trajectory generation   | should samples collaborate?                                     |

## §5.1 Cat-1 alignment — REPA family with external model



- DiT slowly re-invents a DINOv2-like representation while training — so just tell it the answer.
- **REPA**:  $L = L_{\text{diffusion}} + \lambda \cdot (-\text{sim}(h_{\phi}(h_{\ell}(x_t)), f_{\text{SSL}}(x)))$  — a small MLP maps a DiT middle layer to frozen DINOv2; train-time only; near-zero cost, large convergence speedup.
- Fixes prior limitation: DiT grows its representation too slowly → a train-time scaffold to a frozen teacher.

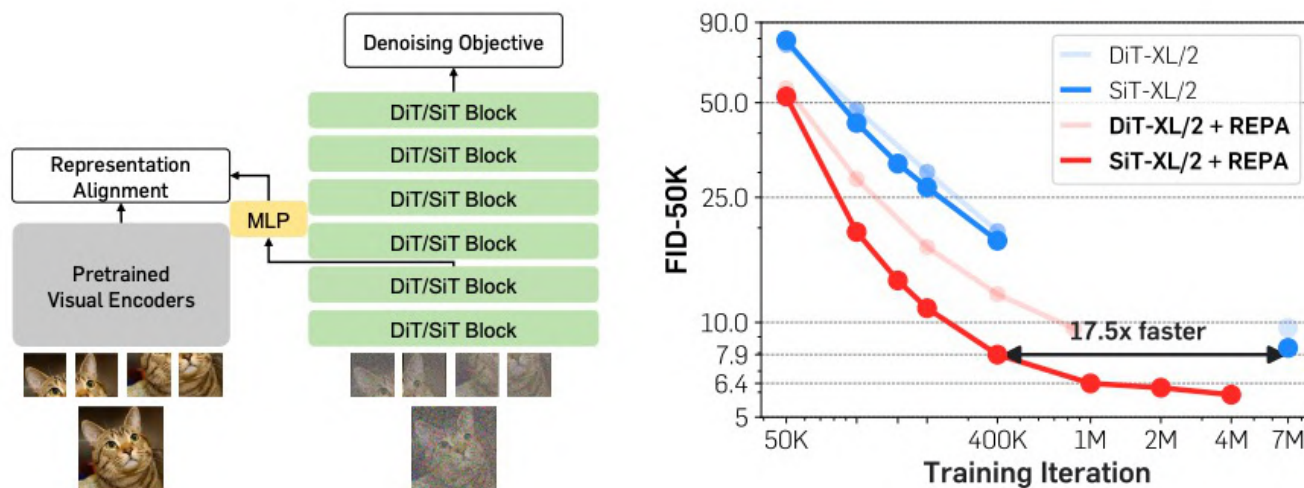


Figure 1: **Representation alignment makes diffusion transformer training significantly easier.** Our framework, REPA, explicitly aligns the diffusion model representation with powerful pretrained visual representation through a simple regularization. Notably, model training becomes significantly more efficient and effective, and achieves  $>17.5\times$  faster convergence than the vanilla model.

## §5.1 Cat-1 alignment — REPA family with external model



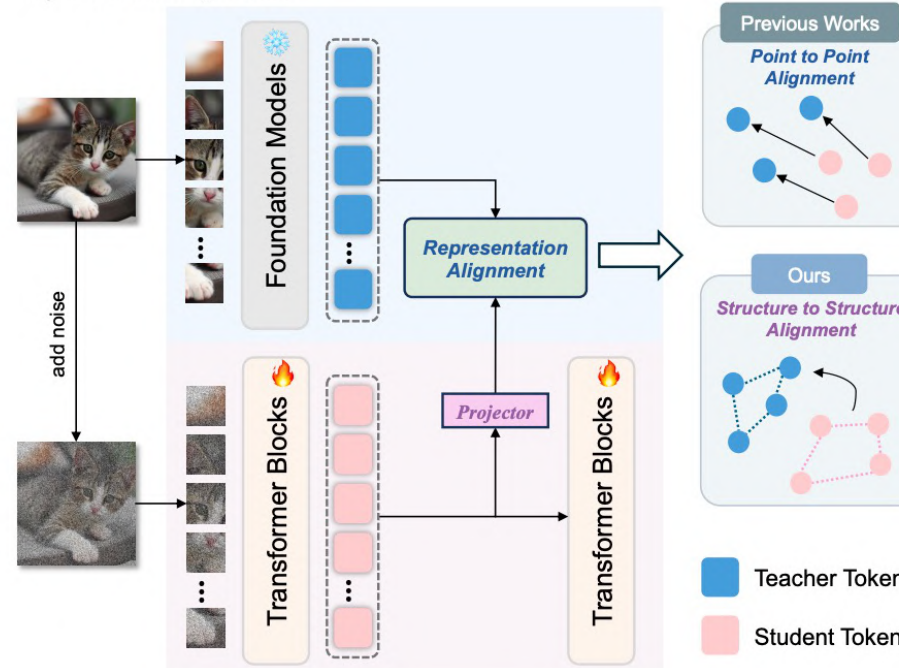
| Question                 | Related Papers                                                          |
|--------------------------|-------------------------------------------------------------------------|
| How to align?            | SARA (2025); iREPA (2025); HASTE (2025); sREPA (2026); MaskAlign (2026) |
| What to align?           | REPA-E (2025); SRA-2 (2026); Send-VAE (2026); AHPA (2026)               |
| Why does alignment work? | iREPA (2025); HASTE (2025); AG-REPA (2026)                              |
| Other extensions         | REPA-G (2026); Don't Retrain—Align (2026)                               |

## §5.1 Cat-1 alignment — the REPA family tree: how to align?



- sREPA:
- Problem: REPA aligns corresponding teacher and student tokens independently, but point-wise matching does not explicitly transfer the spatial relationships and global topology among tokens.
- sREPA matches the **pairwise cosine-similarity structure of teacher and student feature maps**. It proposes two variants: direct Gram-matrix regression and row-wise relational-distribution matching, used together with the original denoising and point-wise alignment losses.
- The useful information in a vision teacher is not only stored in individual token coordinates, but in the **relational geometry that organizes image regions**.

a) Overall Pipeline



sREPA [2605]

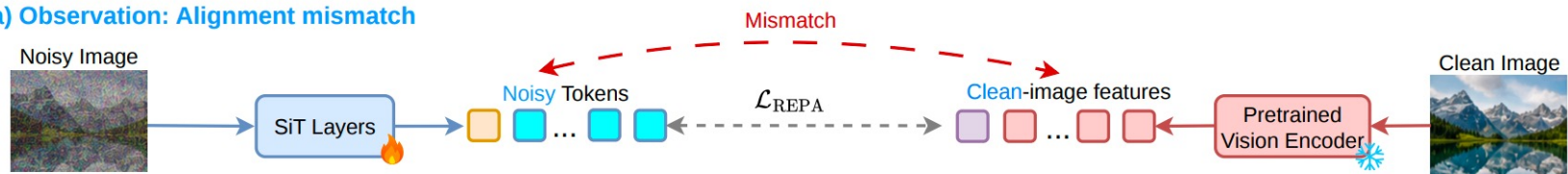
# §5.1 Cat-1 alignment — the REPA family tree: how to align?



## MaskAlign:

- **Problem:** Full-token alignment forces noisy diffusion features to match every clean-image teacher token. Large alignment gradients concentrate repeatedly at particular spatial positions, potentially creating dependence on the complete clean-token set.
- **Method:** Randomly align only a subset of patch tokens during each training step. Before masking, a lightweight token-mixing block shares information across tokens to reduce information loss; all tokens are restored at inference.
- **Core insight:** Alignment itself requires augmentation: supervising every fixed token correspondence can create co-adaptation and feature-fitting shortcuts.

### a) Observation: Alignment mismatch



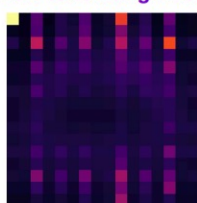
Wild ideas: Random masking is too crude.  
Can we make it smarter and adaptive?  
MAE also shows that mask ratio/policy matters—can we leverage this insight here?

### b) Token-level observation

High-gradient tokens **concentrate** at specific positions

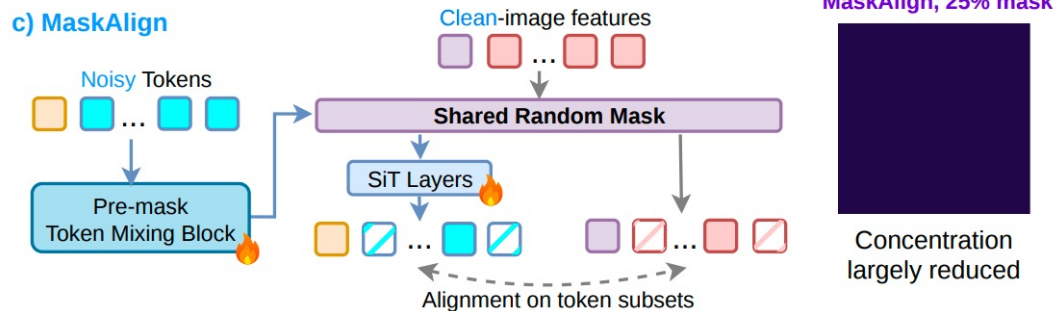


### Full-token alignment



Stable spatial preference

### c) MaskAlign



MaskAlign [2606]

# §5.1 Cat-1 alignment — the REPA family tree: how to align?



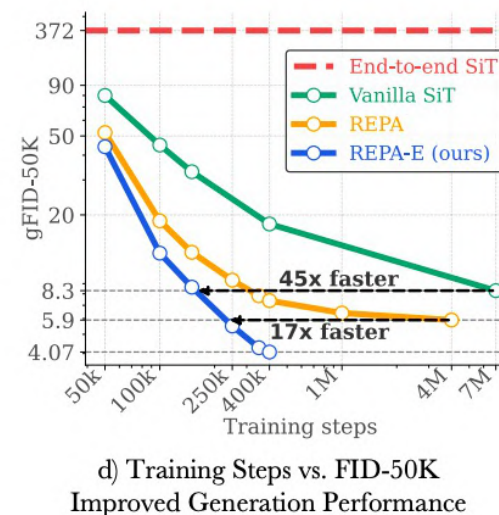
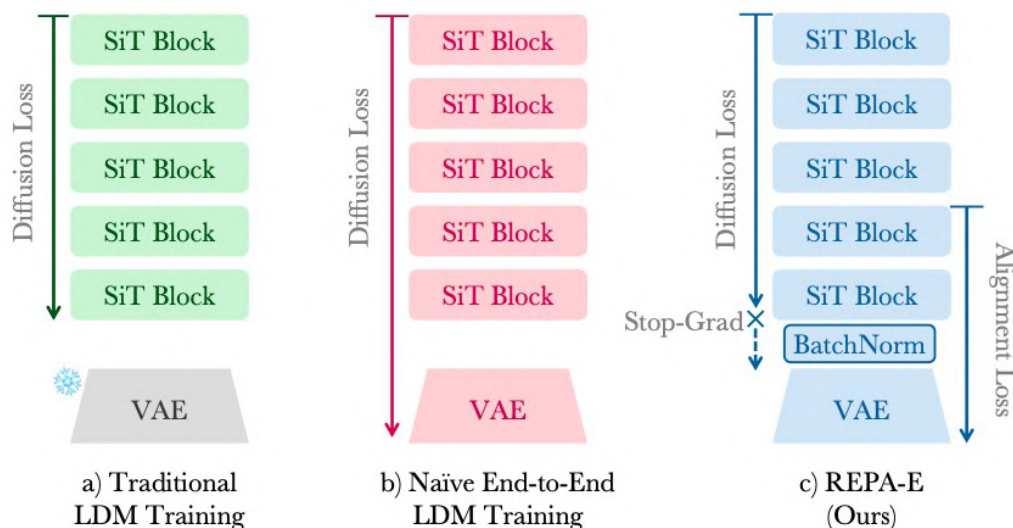
REPA-E: How can the VAE tokenizer and diffusion transformer be jointly tuned without degrading the latent space?

## Method

- Stop diffusion-loss gradients at the VAE output.
- Use REPA loss to update both VAE and DiT.
- Retain reconstruction/perceptual/adversarial VAE losses.
- Stabilize the evolving latent distribution with differentiable BatchNorm.

## Key Findings

- Naive end-to-end diffusion training collapses latent spatial variation: denoising becomes easier, but generation worsens.
- REPA-E improves latent semantics and structure while preserving reconstructability.
- The tuned tokenizer accelerates training and transfers across diffusion architectures.

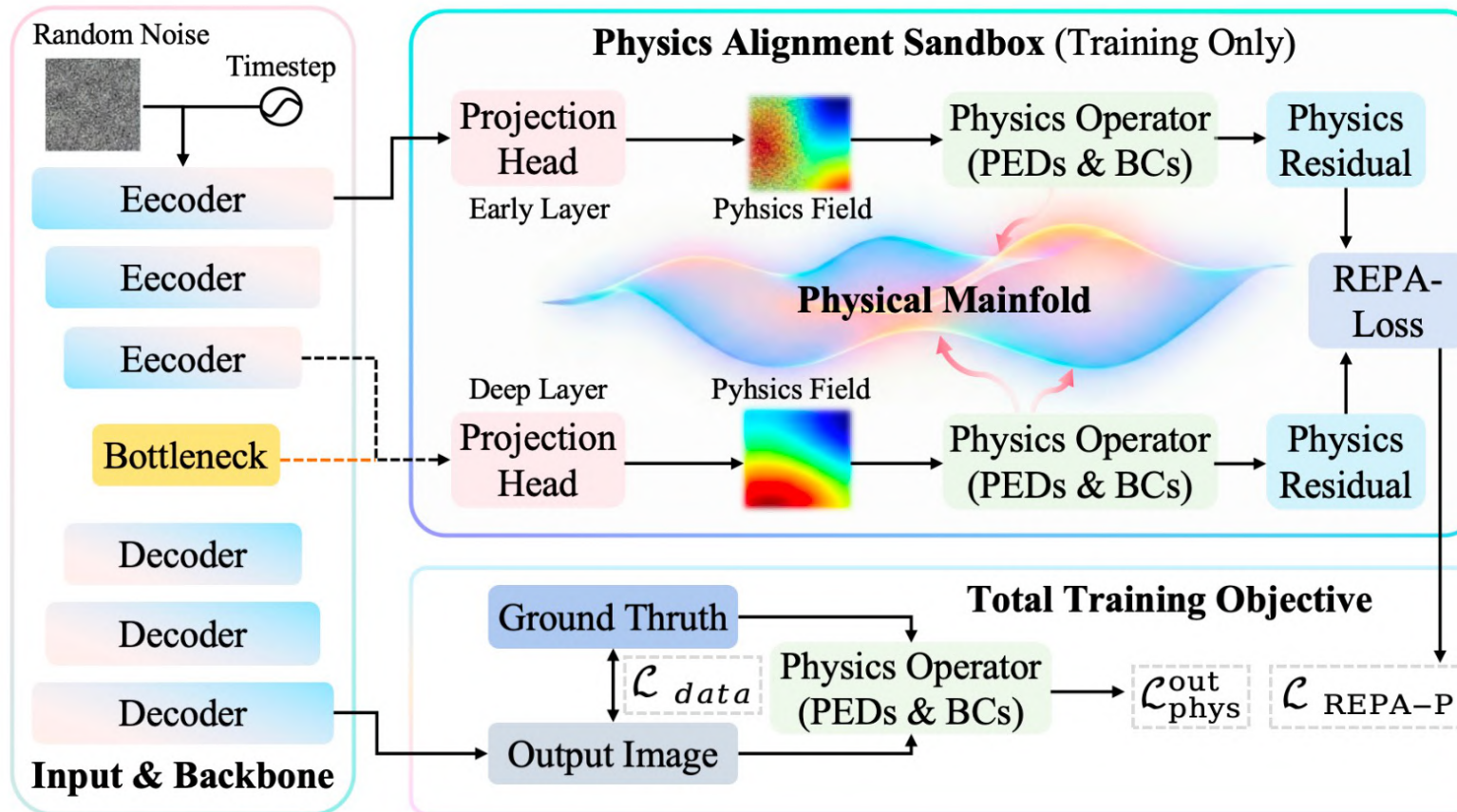


REPA-E [2504]

## §5.1 Cat-1 alignment — the REPA family tree: align what?



- **REPA-P**: Uses physical constraints to shape hidden states, replacing external feature targets with functional supervision.



REPA-P ICML2026

## §5.1 Cat-1 alignment — the REPA family tree: others



- **Co-adapt the state space:** REPA-E uses alignment to jointly tune the tokenizer and generator while preventing destructive diffusion gradients from collapsing latent structure.
- **Adapt across architectures and paradigms:** U-REPA redesigns alignment for U-Nets; Don't Retrain—Align preserves AR language geometry during AR-to-diffusion conversion.
- **Extend to inference:** REPA-G turns the learned representation space into a test-time potential for conditioning and trajectory steering.
- **Make alignment selective:** HASTE, AHPA, MaskAlign, and AG-REPA respectively select supervision by training stage, SNR/granularity, token subset, and causal layer contribution.
- **Unified design space:** target×relation×layer×timestep×training stage×deployment.
- **Takeaway:** REPA is evolving from a fixed auxiliary loss into a broader framework for deciding **what information should shape generation, where it should enter, and when it should be released.**

## §5.1 Cat-1 alignment — REPA family with external model



| Question                 | Related Papers                                                           |
|--------------------------|--------------------------------------------------------------------------|
| How to align?            | SARA (2025); iREPA (2025); HASTE (2025); sREPA (2026); MaskAlign (2026); |
| What to align?           | REPA-E (2025); SRA-2 (2026); Send-VAE (2026); AHPA (2026); SRA(2025)     |
| Why does alignment work? | iREPA (2025); HASTE (2025); AG-REPA (2026)                               |
| Other extensions         | REPA-G (2026); Don't Retrain—Align (2026)                                |

## §5.1 Cat-1 alignment — self-teacher (drop the external model)

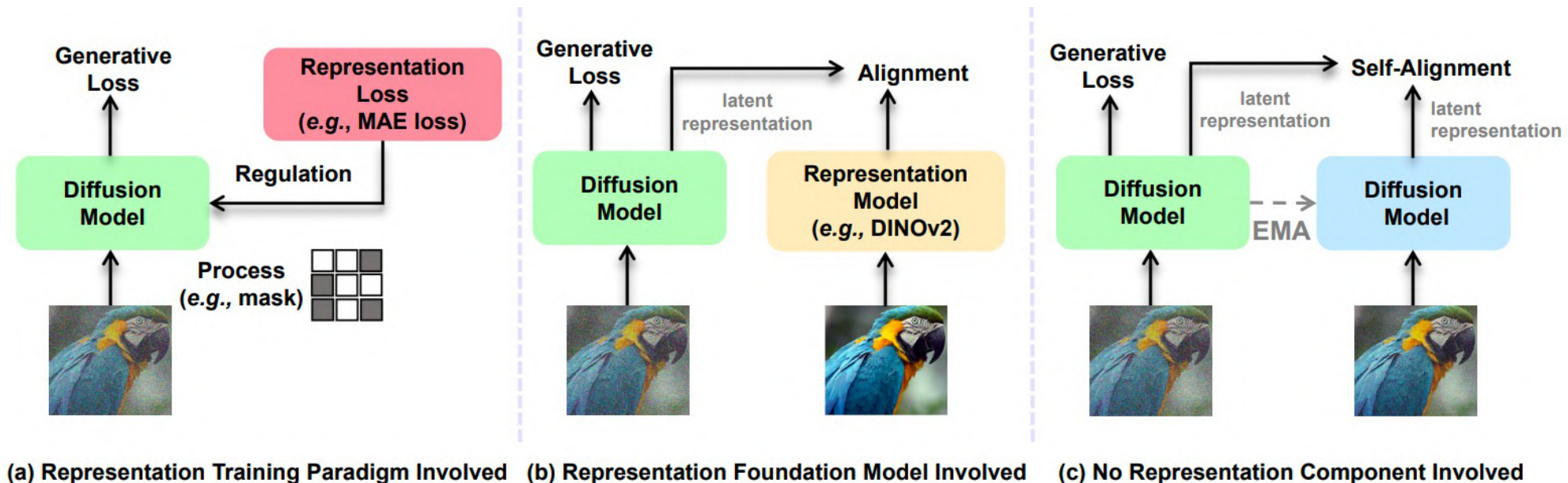


**SRA:** Can a diffusion transformer provide its own representation guidance without an external teacher?

**Method:** Align weak representations from early layers and high-noise states to stronger representations from later layers and lower-noise states.

### Key Findings

- Diffusion representations vary substantially across layers and noise levels.
- Internal weak-to-strong alignment can improve training without external models.
- Lower-noise and later-layer features often provide stronger semantic supervision.



## §5.1 Cat-1 alignment — self-teacher (drop the external model)



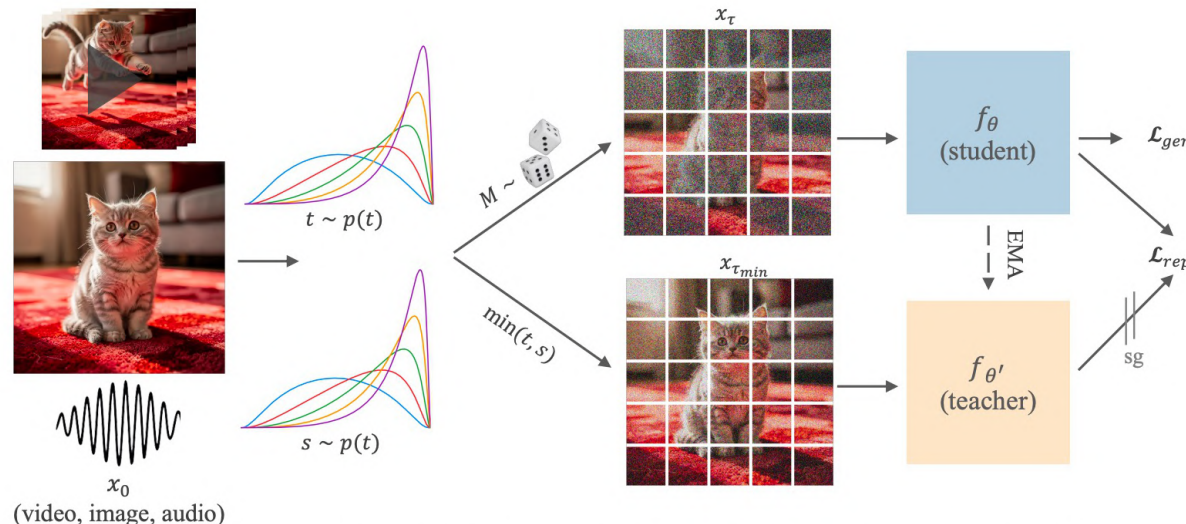
| Category                                          | Representative Papers                                                                                                                                      |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Training-time self-alignment                      | SRA (ICLR 2026); LayerSync (ICLR 2026); Self-Flow (2026)                                                                                                   |
| Inference-time internal / counterfactual guidance | SAG (CVPR 2023); PAG (CVPR 2024); AutoGuidance (NeurIPS 2024); Internal Guidance (CVPR 2026); S <sup>2</sup> -Guidance (2025); Sparse Guidance (CVPR 2026) |
| Training-sampling bridge / self-distillation      | Self Step-Distillation (CVPR 2025); SGG (CVPR 2026); Synthetic Self-Guidance (2026)                                                                        |

## §5.1 Cat-1 alignment — self-teacher (drop the external model)



Self-Flow:

- Self-Flow attributes its gain to interactions between tokens at different noise levels, where cleaner tokens help infer noisier ones.
- **Dual-Timestep Scheduling assigns heterogeneous noise levels to different tokens, creating information asymmetry.**
- Learns stronger representations and improves image, video, and audio generation without external supervision.
- Different-noise tokens create an internal self-supervised prediction task.



Self-Flow [2603]

**Figure 3. Illustration of our method.** Given a clean input  $x_0$ , we draw two timesteps  $t, s$ , and a random mask  $M$ , then noise each token according to its assigned timestep. The teacher input is noised with  $\tau_{min} = \min\{t, s\}$ , creating an information asymmetry compared to the student. The student is trained to simultaneously denoise the input and reconstruct the teacher's features given its mixed-noised view.

## §5.1 Cat-1 alignment — self-teacher (drop the external model)



### From SRA to Self-Flow:

- Self-Flow claims cleaner tokens teach noisier tokens; is this really the mechanism?
- Introduces Attention Separation: **preserve dual-timestep inputs but block attention between different-noise token groups.**
- Blocking cross-noise interaction does not hurt and can even improve performance.
- Self-Flow's gain may come mainly from **noise-dimension data augmentation**, not cross-token self-supervision.

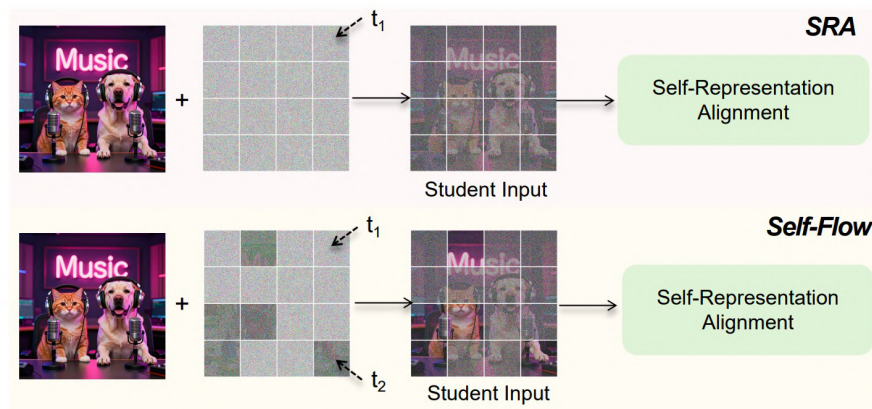


Figure 1. **Difference Between Self-Flow and SRA.** Self-Flow adopts SRA's Self-Representation Alignment method while differs in the student input processing: SRA utilizes a single noise level ( $t_1$ ) for all student input tokens, whereas Self-Flow employs a dual-timestep scheduler where tokens at two distinct noise levels ( $t_1$  and  $t_2$ ) coexist in the same image.

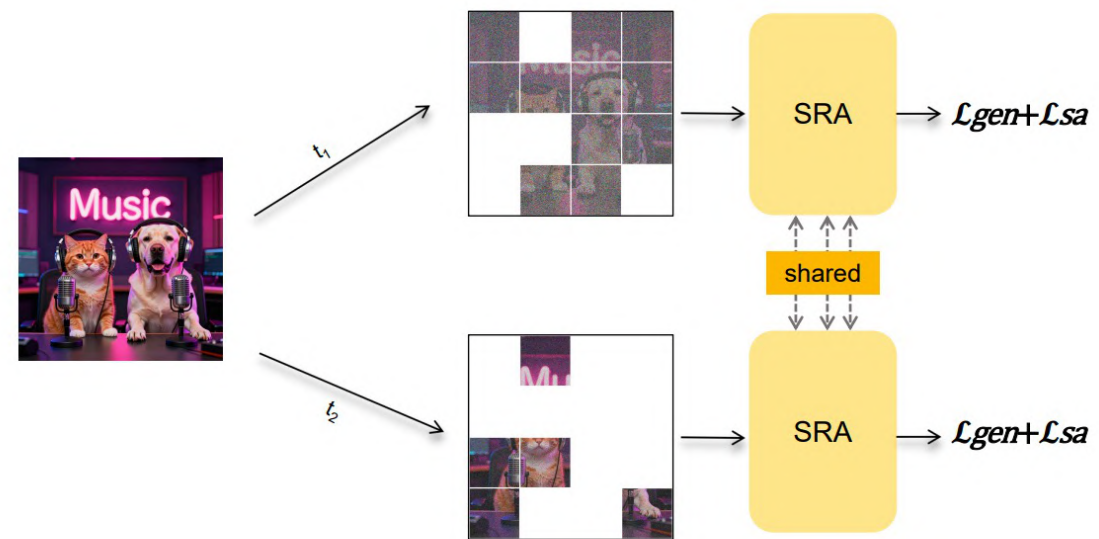


Figure 2. **Attention Separation disentangles self-supervision from augmentation.** Given a sample with dual-timestep schedul-

## §5.1 Cat-1 alignment — self-teacher (drop the external model)

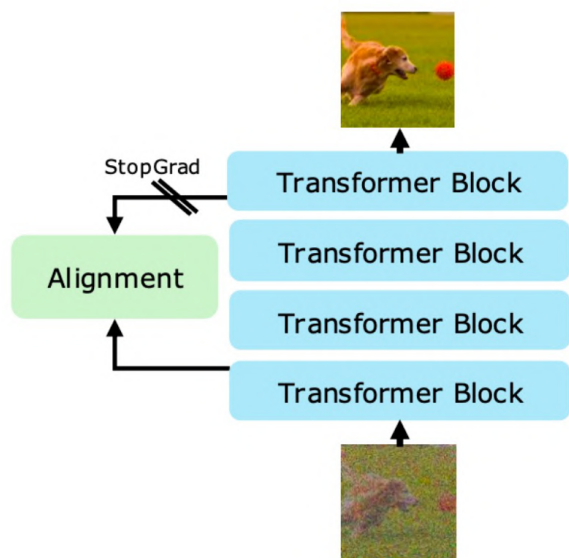


**LayerSync:** Can layer-wise representation imbalance be corrected within **a single forward pass?**

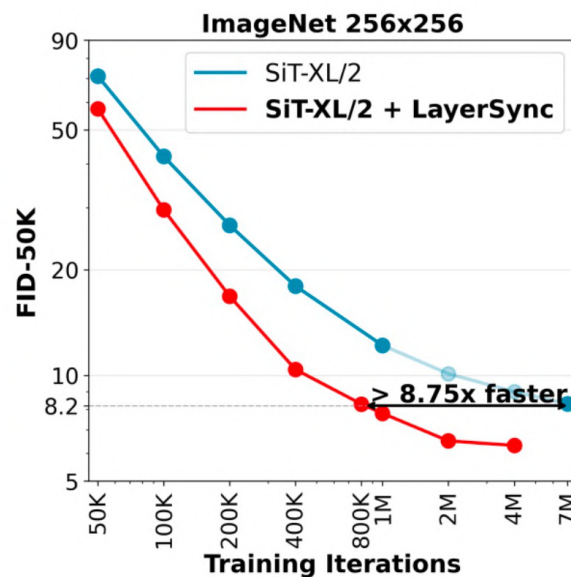
**Method:** Identify semantically stronger intermediate layers and use them to guide weaker layers from the same forward pass.

### • Key Findings

- DiT layers naturally specialize into local, semantic, and decoding stages.
- Aligning selected weak layers to strong internal layers improves generation.
- Extra teacher models, data, and forward passes are unnecessary.



(a)



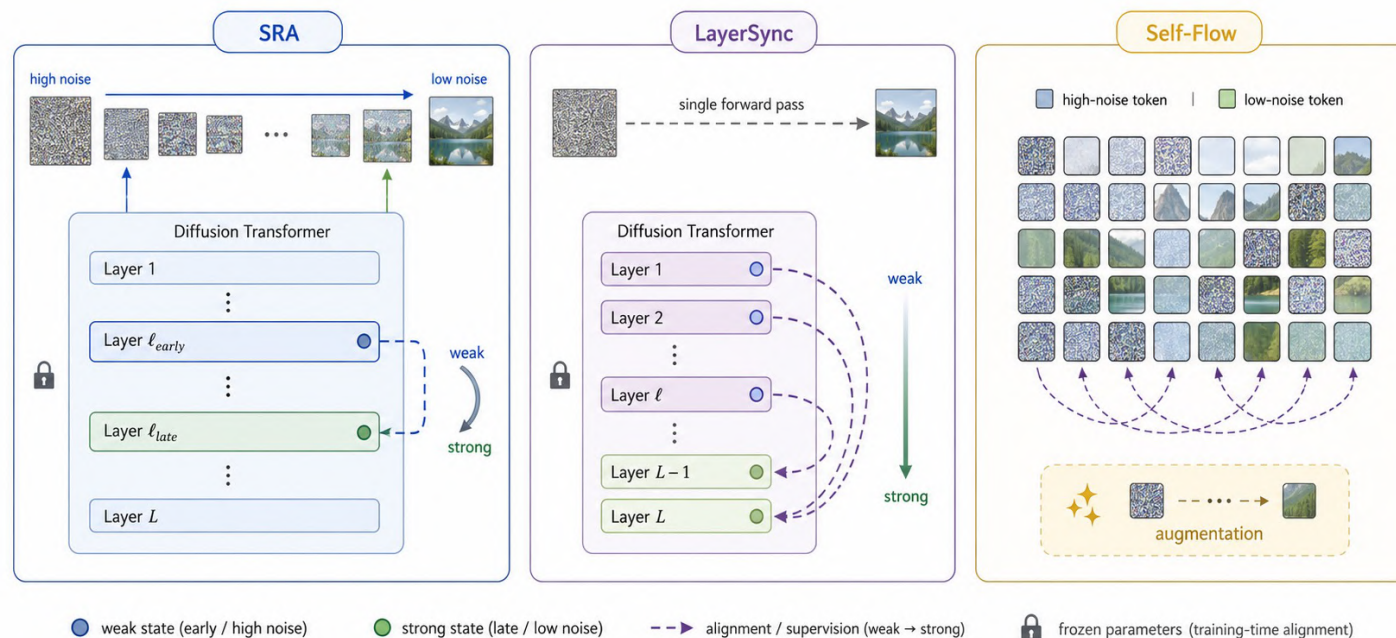
(b)

LayerSync [ICLR 2026]

## §5.1 Cat-1 alignment — self-teacher (drop the external model)



- **SRA** aligns weak high-noise and early-layer representations with stronger low-noise, deep EMA features.
- **LayerSync** removes the external teacher and performs weak-to-strong alignment across layers within the same forward pass.
- **Self-Flow** introduces heterogeneous noise levels across tokens, but later analysis suggests that much of its gain may come from noise-state augmentation rather than cross-noise information transfer.
- **Takeaway:** self-alignment accelerates representation formation and optimization, while its gains mix privileged supervision, deep supervision, and augmentation effects.



## §5.1 Cat-1 alignment — self-teacher (drop the external model)

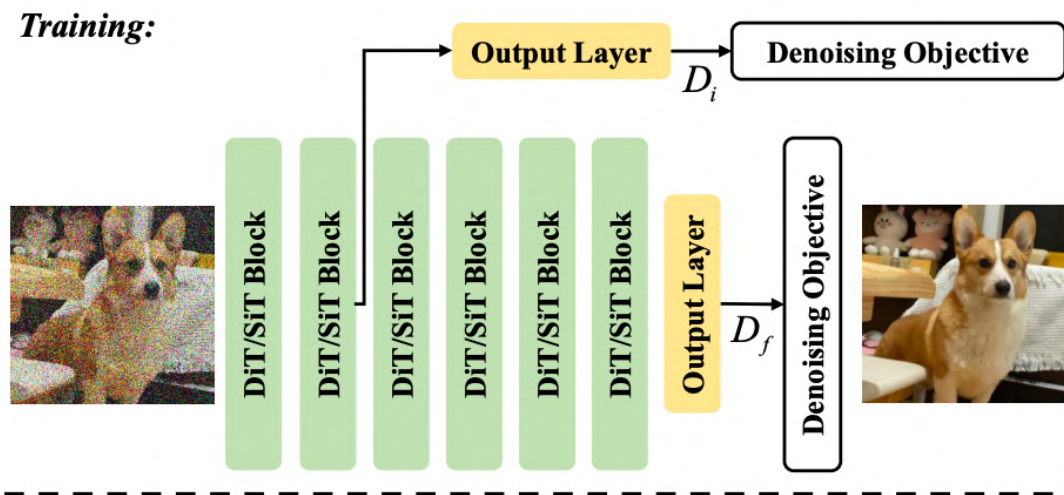


**Internal Guidance:** Can intermediate diffusion predictions improve both training and sampling?

**Method:** Add auxiliary denoising supervision to a middle layer during training, then extrapolate between middle-layer and final-layer predictions during sampling

### Key Findings

- Intermediate supervision improves gradient flow and makes middle layers predictive.
- The mid-to-deep prediction difference provides an effective self-guidance direction.
- Sampling quality improves without an external classifier or guidance model.



[Internal Guidance](#) [CVPR 2026]

*Sampling:*

$$D_w(x; c) = D_i(x; c) + w(D_f(x; c) - D_i(x; c))$$

## §5.1 Cat-1 alignment — self-teacher (drop the external model)

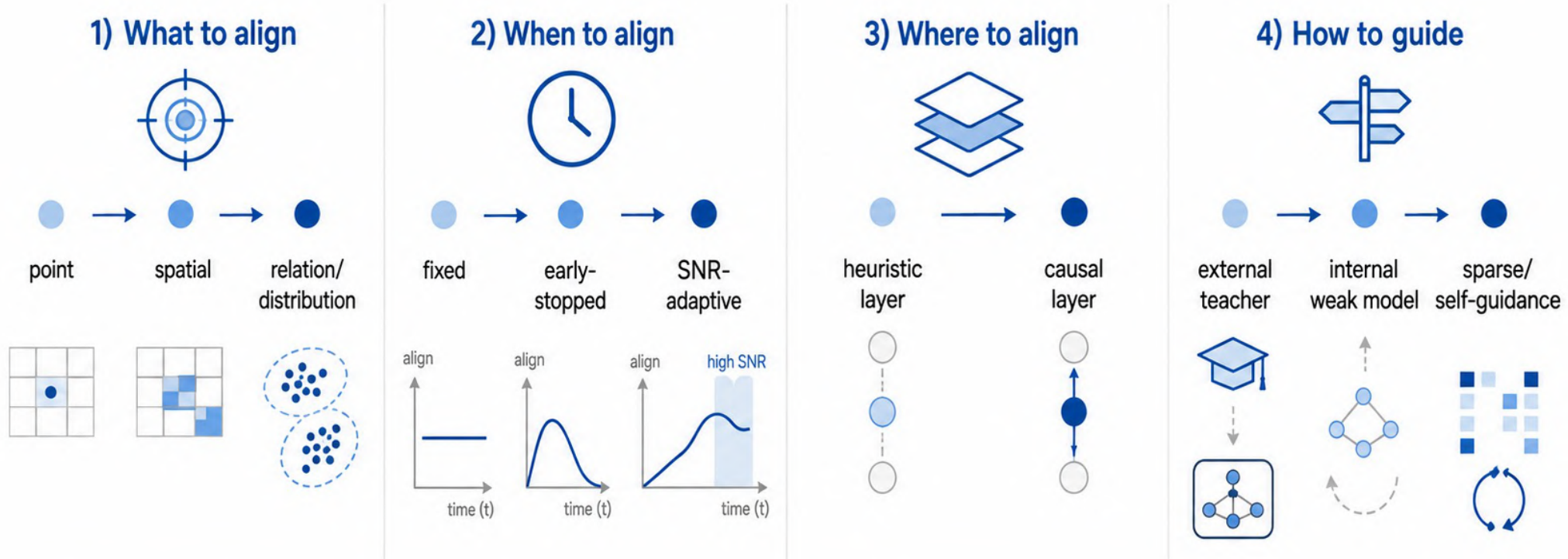


| Category                                          | Representative Papers                                                                                                                                      |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Training-time self-alignment                      | SRA (ICLR 2026); LayerSync (ICLR 2026); Self-Flow (2026)                                                                                                   |
| Inference-time internal / counterfactual guidance | SAG (CVPR 2023); PAG (CVPR 2024); AutoGuidance (NeurIPS 2024); Internal Guidance (CVPR 2026); S <sup>2</sup> -Guidance (2025); Sparse Guidance (CVPR 2026) |
| Training-sampling bridge / self-distillation      | Self Step-Distillation (CVPR 2025); SGG (CVPR 2026); Synthetic Self-Guidance (2026)                                                                        |

## §5.1 Cat-1 alignment — Unifying message.



These works are not just “more alignment tricks.” They explore four different ways to reshape the optimization problem of generative models:



**Open problem:** disentangle representation content, deep supervision, gradient conditioning, augmentation, and inference-time correction.

## §5.1 Cat-1 alignment — why alignment works?



| Candidate Mechanism              | Explanation                                                         |
|----------------------------------|---------------------------------------------------------------------|
| <b>Privileged information</b>    | Clean, low-noise, or deep states provide richer target information. |
| <b>Credit assignment</b>         | Auxiliary objectives shorten gradient paths for shallow layers.     |
| <b>Representation geometry</b>   | Meaningful topology enhances main task performance.                 |
| <b>Implicit regularization</b>   | Alignment narrows the hypothesis space.                             |
| <b>Counterfactual correction</b> | Strong-weak discrepancies identify error directions.                |
| <b>Task augmentation</b>         | Novel corruptions increase training sample diversity.               |

## §5.1 Cat-1 alignment —open questions?

---



- **Relation to OT / GW**  
Should alignment preserve pointwise coordinates, pairwise geometry, or cross-domain structural correspondences?
- **What part of the teacher matters?**  
Semantic category, spatial layout, local texture, frequency statistics, or relational structure?
- **Does alignment help the main denoising objective?**  
Does it reduce denoising loss, improve gradient conditioning, or only improve final FID?
- **Is the comparison fair?**  
External VFMs carry extra data, compute, and inductive bias that are often not counted in training cost.

## §5.2 Cat-2 generate in representation space — RAE



- **RAE**: don't keep DiT's own space — make the frozen SSL encoder's output the diffusion target space. Train (a) a decoder  $z \rightarrow \text{pixel}$  and (b) a DiT modeling  $p(z)$ .
- Because the latent is already semantic, diffusion on it converges faster — you save learning-semantics-from-scratch, not training-the-diffusion.
- RAEv2: don't use only the last layer; RAE & REPA are complementary.

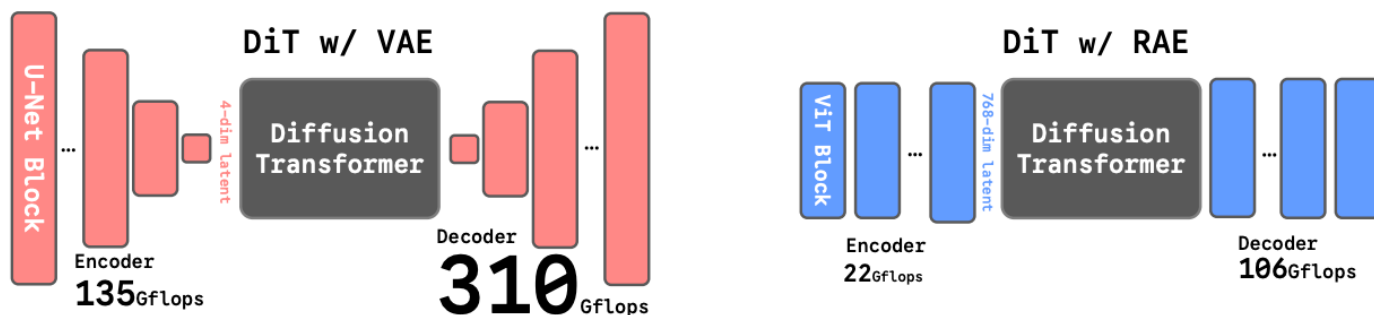


Figure 2: **Comparison of SD-VAE and RAE (DINOv2-B)**. The VAE relies on convolutional backbones with aggressive down- and up-sampling, while the RAE uses a ViT architecture *without* compression. SD-VAE is also more computationally expensive, requiring about 6× and 3× more GFLOPs than RAE for the encoder and decoder, respectively. GFlops are evaluated on one  $256 \times 256$  image.

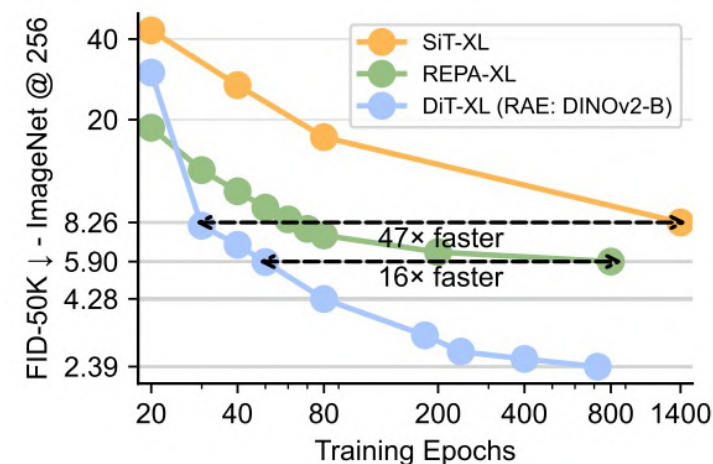


Figure 4: **DiT w/ RAE reaches much faster convergence and better FID than SiT or REPA.**

[RAE ICLR2026](#)

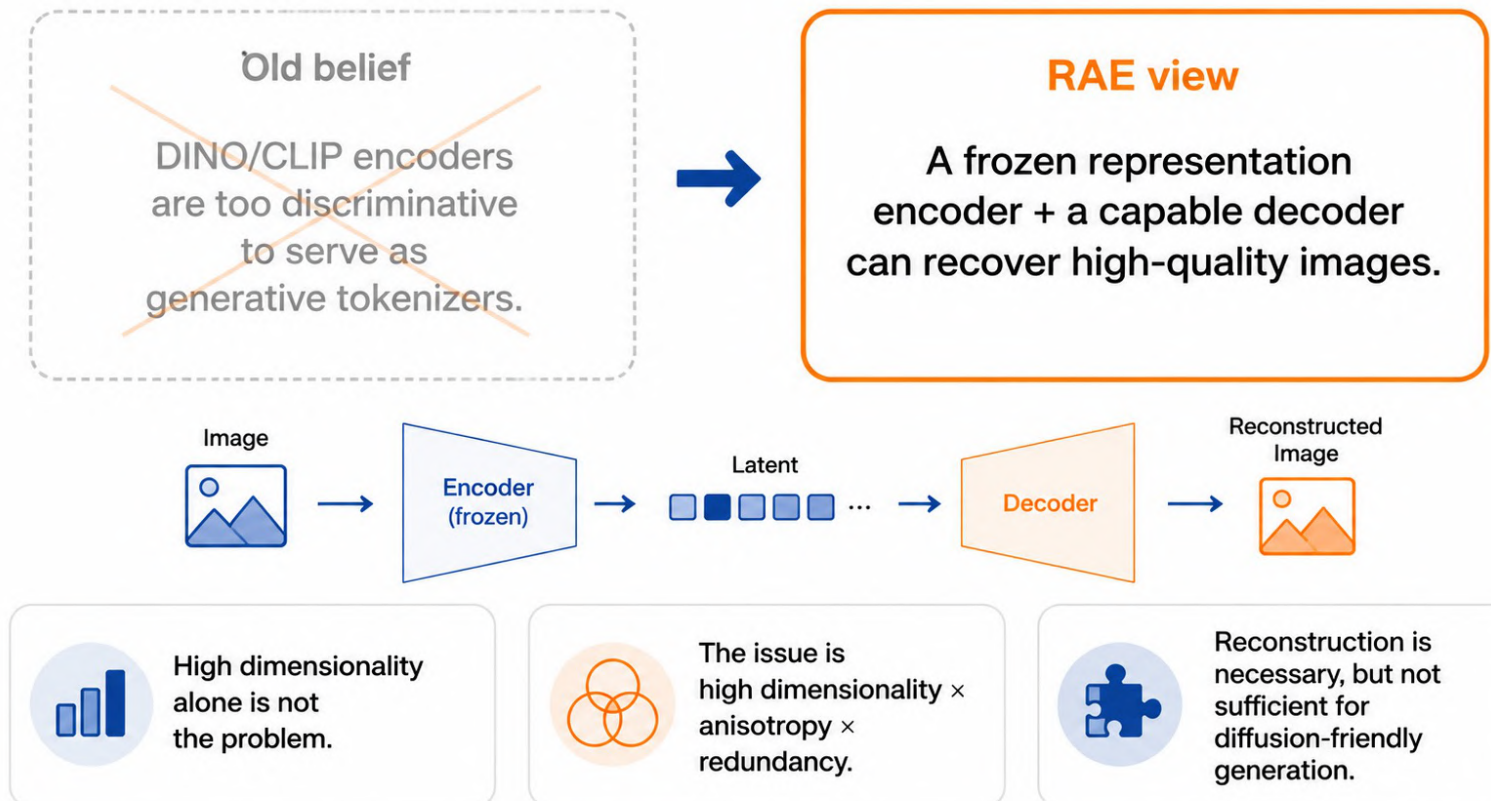
## §5.2 Cat-2 generate in representation space — some analysis



### RAE Changes the Old Assumption

- A common belief was: DINO / CLIP-like encoders are too discriminative and discard low-level details, so they cannot serve as generative tokenizers.
- RAE shows this view is too strong. A frozen representation encoder, paired with a sufficiently capable decoder, can still recover high-quality images.

### • Key takeaways

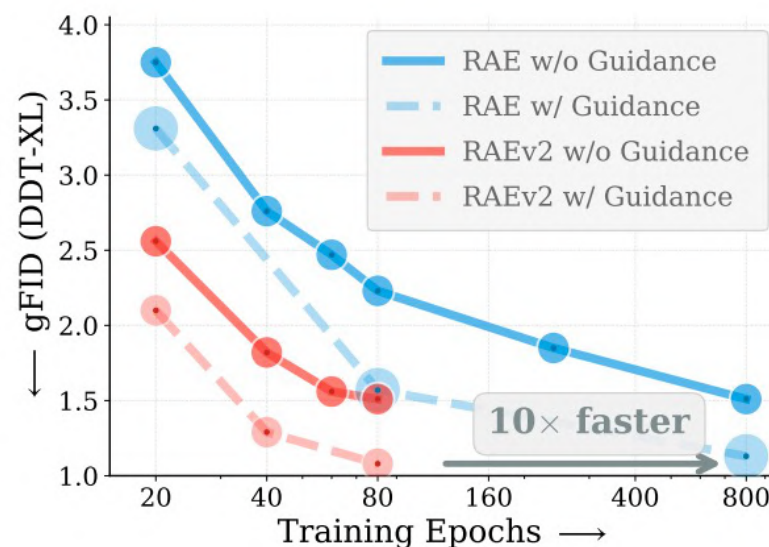
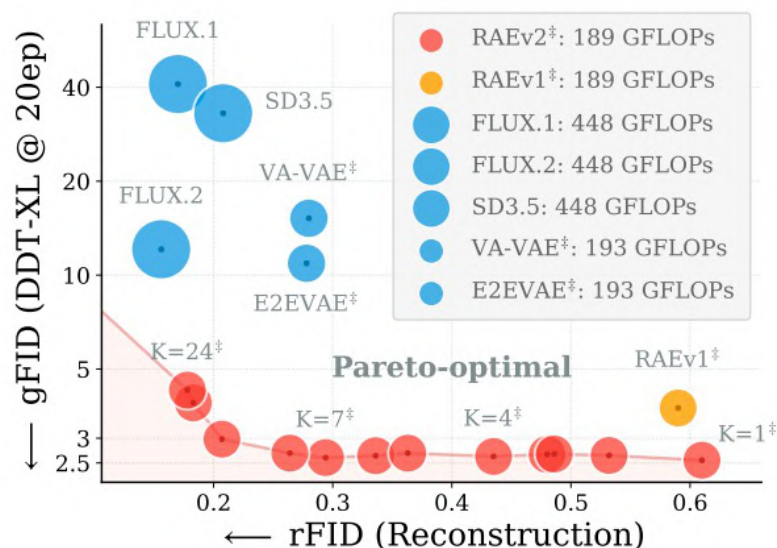


## §5.2 Cat-2 generate in representation space — RAEv2



RAEv2:

- The original RAE uses only the final encoder layer and treats RAE and REPA as competing approaches, potentially underestimating the RAE baseline.
- Aggregate several late representation-encoder layers rather than only the final layer; combine RAE latent modeling with REPA-style intermediate supervision; reinterpret REPA in the RAE space as a form of latent  $\mathcal{X}$ -prediction and derive a low-cost guidance mechanism.
- RAE chooses **where generation occurs**, while REPA changes **how the generator is optimized**. These are orthogonal design axes.



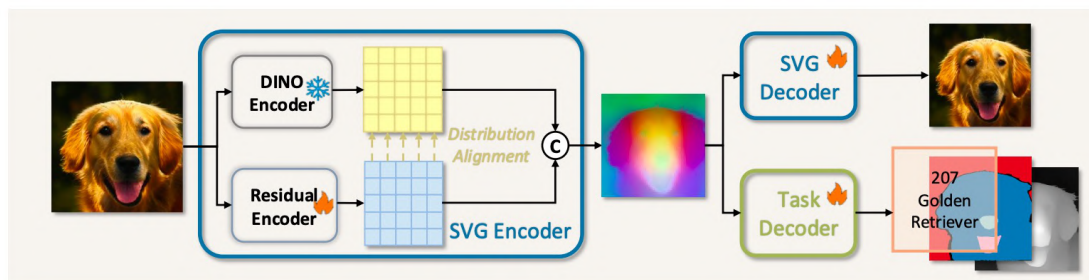
[RAEv2](#) [2605]

## §5.2 Cat-2 generate in representation space — SVG

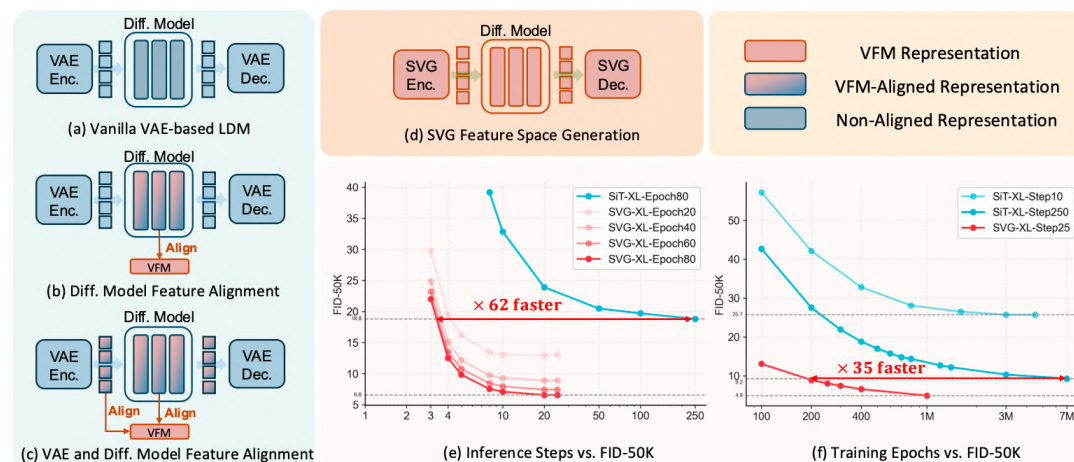


SVG:

- VAE latents are reconstructive but often lack explicit semantic separation, while frozen discriminative features contain semantics but discard fine visual details.
- Use frozen DINO features as the semantic latent trunk and introduce a lightweight residual branch to preserve information needed for high-fidelity reconstruction. Train diffusion directly over the combined semantic-plus-residual representation.
- A generation-ready representation should be decomposed into a **semantic trunk** and a **detail residual**, rather than forcing one latent to simultaneously provide invariance and pixel sufficiency.



SVG ICLR2026

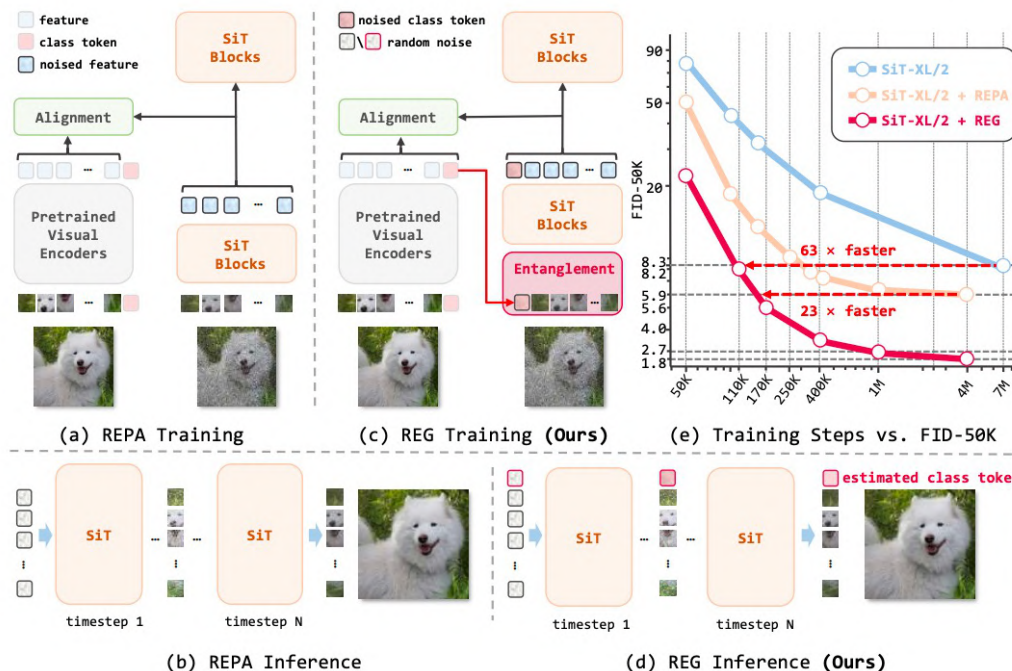


## §5.2 Cat-2 generate in representation space — RAE



### REG:

- REPA uses visual representations only as training-time supervision; the representation disappears during inference and cannot directly guide the denoising trajectory.
- REG treats **a DINOv2 class token as an additional generative state**. The semantic token and image latents are independently noised, concatenated, and jointly denoised within the same Diffusion Transformer. REPA-style intermediate alignment is retained during training.
- Representation can be more than an auxiliary loss: it can serve as a persistent semantic plan that co-evolves with the image throughout denoising.

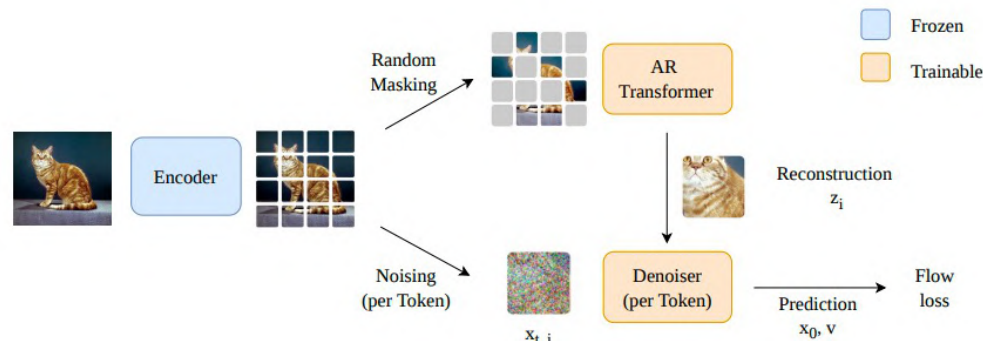


REG [2507]

## §5.2 Cat-2 generate in representation space — analysis

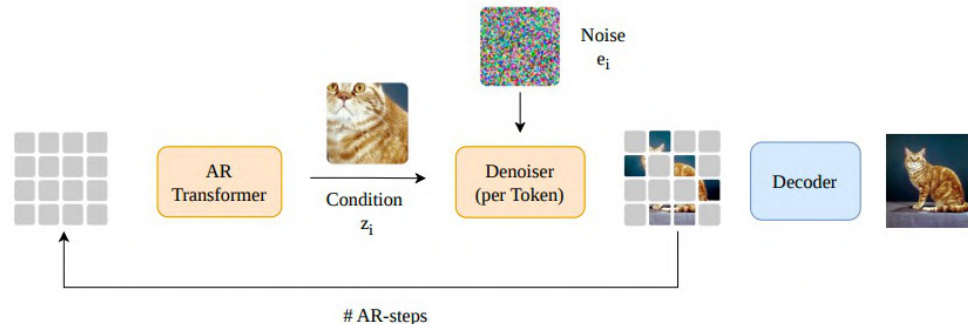


- **Question:** Does a better representation truly reduce generative difficulty, or merely redistribute it?
- Compare **Pixels, SD-VAE, DINOv2, and MAE** in a **unified masked-AR + per-token flow framework**.
- **DINOv2 is much easier to train**, while MAE reconstructs better but generates worse.
- **Key finding:** representation choice reallocates difficulty across **context modeling, local denoising, and transport**.



(a) Training pipeline.

Recall REPA-E, we need **co-design** generation and representation.

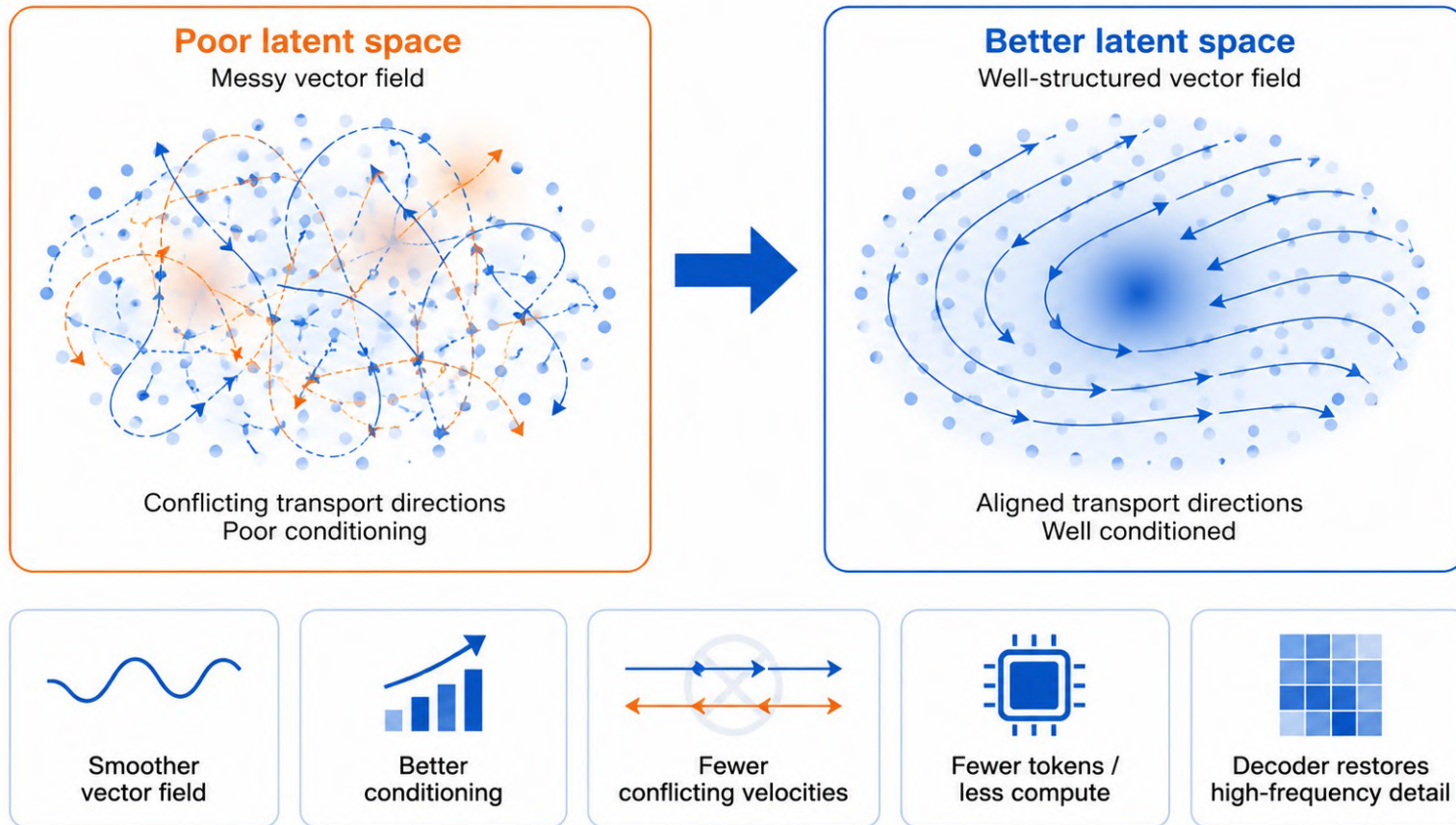


(b) Inference pipeline.

## §5.2 Cat-2 generate in representation space — why works?



A better latent space may make the generative vector field easier to model:



Representation-space generation works when it reduces the generator's modeling burden more than it increases the decoder's lifting burden.

## §5.2 Real Goal, Proxy Loss, and Evaluation Metric Are Not the Same



- Many failures come from optimizing the wrong proxy.

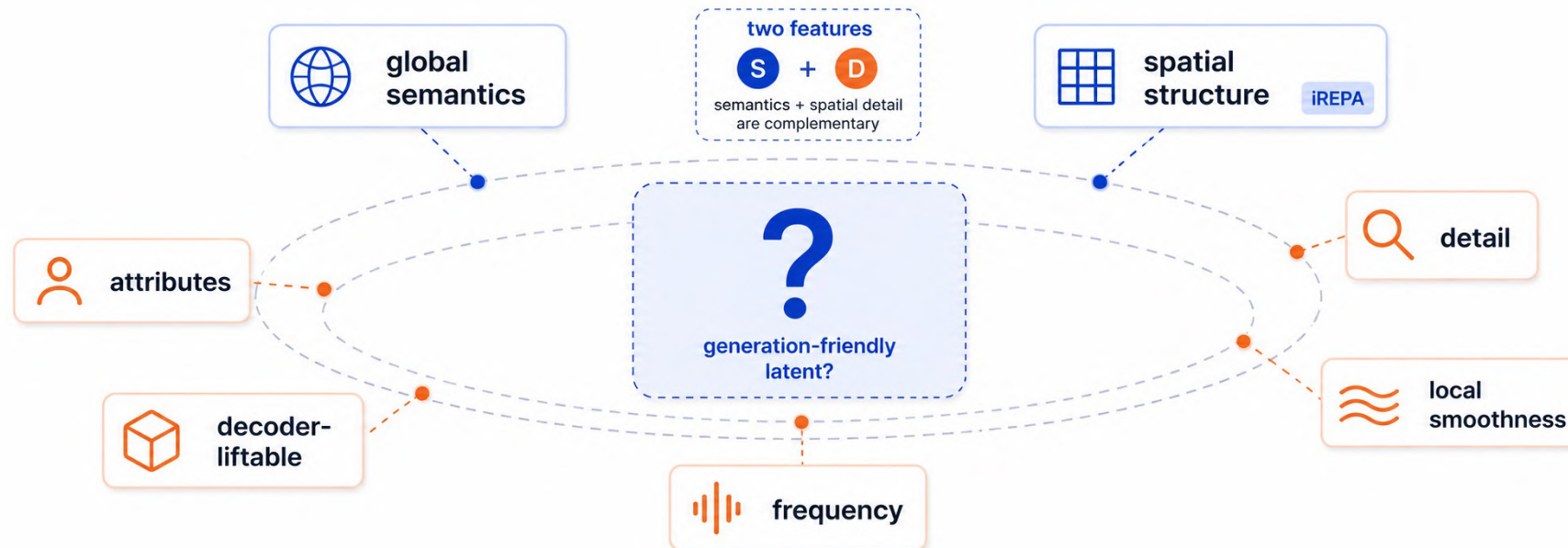
| Level      | Example                                        | Failure mode                         |
|------------|------------------------------------------------|--------------------------------------|
| Real goal  | High-quality, diverse, controllable generation | Hard to optimize directly            |
| Proxy loss | Denoising MSE, alignment, InfoNCE, dispersion  | May improve while generation worsens |
| Metric     | FID, linear probe, CKA, precision/recall       | Measures only part of the goal       |

A better alignment loss does not necessarily mean better generation.

## §5.2 what is a generation-friendly latents?



- Not yet settled — the field's live debate:
  - iREPA: **spatial structure** may matter more than global semantic accuracy.
  - A Tale of Two Features: semantics (DINO) + spatial detail (SD) are *complementary* — for generation you may want both.
  - High-level semantic latent may **drop high-frequency detail** needed for generation.
- Open: global semantic? spatial structure? attribute disentanglement? local smoothness? frequency? fiber variables? decoder liftability?

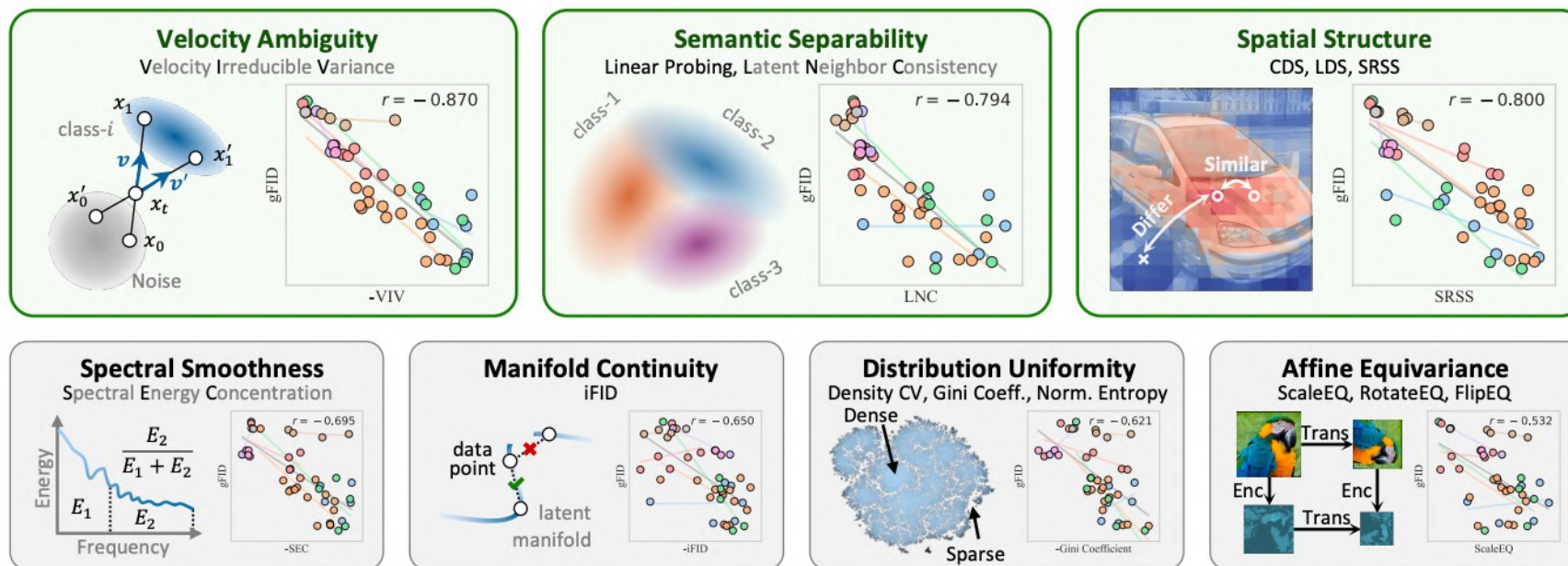


## §5.2 what is the generation-friendly latent space?



- Many works propose different explanations for diffusion-friendly latents—semantics, spatial structure, smoothness, continuity, uniformity, or equivariance—but their conclusions are often based on a small, method-specific set of tokenizers.
- Systematically evaluate seven latent properties across 86 tokenizers, three tokenizer families, and four diffusion backbones. Introduce **Velocity Irreducible Variance (VIV)** to measure the intrinsic ambiguity caused by different transport velocities crossing at the same intermediate state.

Recall ICR metric from [Evaluating the Representation Space of Diffusion Models via Self-Supervised Principles](#) [2606]



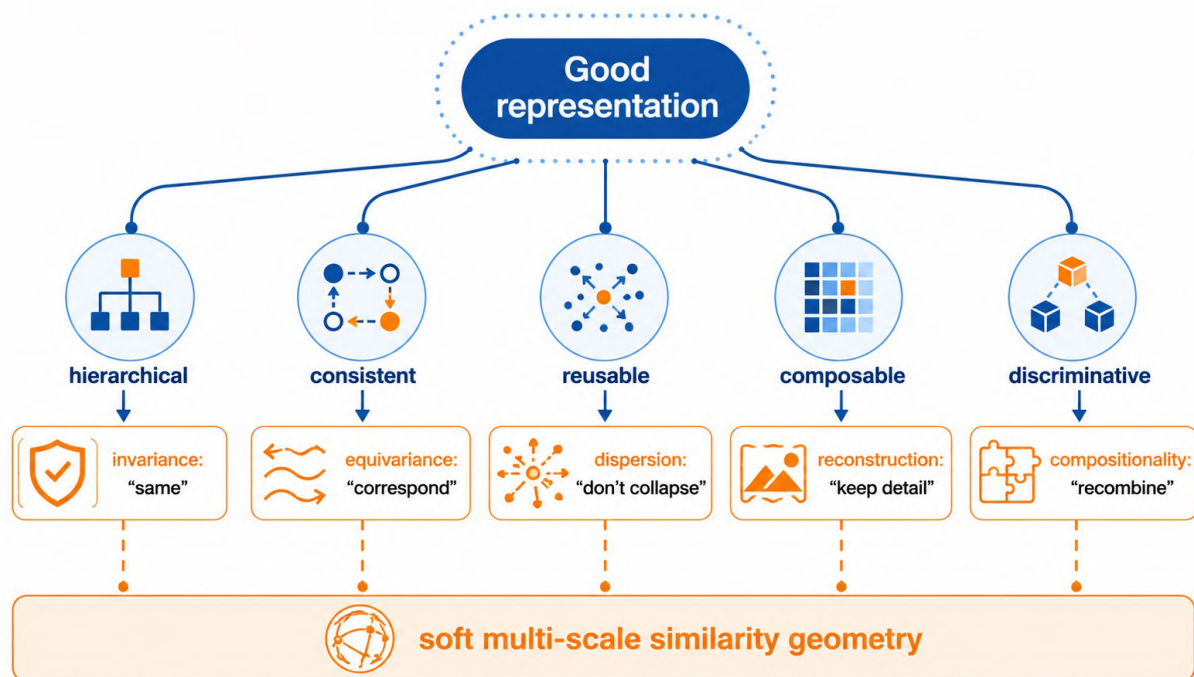
Not all “good representation” metrics are equally predictive. A key robust predictor is transport ambiguity: **good latents make the velocity field less intrinsically multi-valued (VIV).**

[Diffusing in the Right Space: A Systematic Study of Latent Diffusability](#) [2606]

## §5.2 My view on a good representation



- Personal thesis: the properties a representation must have are **hierarchical** · **consistent** · **reusable** · **composable** · **discriminative**.
- Operationalized as a family of losses that each *define* one property:
  - invariance loss — "what should be the same"
  - equivariance loss — "how change should correspond"
  - dispersion/uniformity loss — "don't collapse"
  - reconstruction/detail loss — "what must not be lost"
  - compositionality loss — "can factors recombine"



So many insights from <https://jchencxh.github.io/blog/>

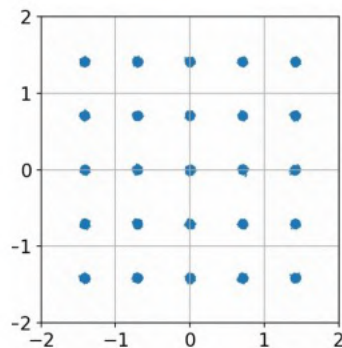
## §5.2 what is a generation-friendly latents?



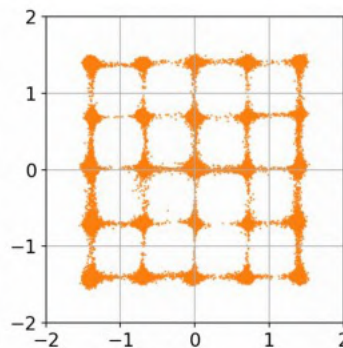
- Reconstruction FID measures whether individual latents can reconstruct training images, but it poorly predicts whether diffusion can generate valid samples between those latents.
- Propose **interpolated FID (iFID)**:
  - find the nearest neighbor of each image in latent space;
  - interpolate their latent representations;
  - decode the interpolated latent;
  - compute FID between the decoded interpolations and real images.

$$\text{rFID} = d_{\text{FID}}(x^{(1:N)}, g(z^{(1:N)})), \text{gFID} = d_{\text{FID}}(x^{(1:N)}, g(\Phi(\epsilon^{(1:M)}, T))). \quad (3)$$

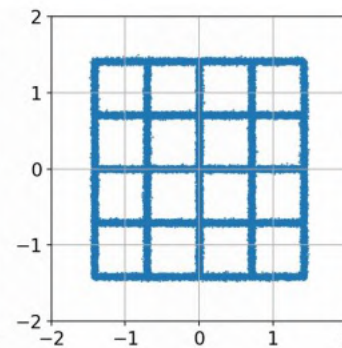
$$\text{iFID} := d_{\text{FID}}(x^{(1:N)}, g(\hat{z}^{(1:N)})), \text{ where } \hat{z}^{(i)} = \frac{z^{(i)} + z^{(i^*)}}{2}, z^{(i^*)} := \arg \min_{j \neq i} \|z^{(j)} - z^{(i)}\|. \quad (4)$$



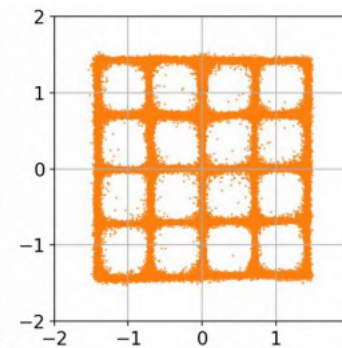
Source  $p(z)$  with **bad iFID**,  
interpolated  $\hat{z}$  are not in data  
manifold



Diffusion samples that  
interpolate modes are **not in**  
data manifold



Source  $p(z)$  with **good iFID**,  
interpolated  $\hat{z}$  are in data  
manifold



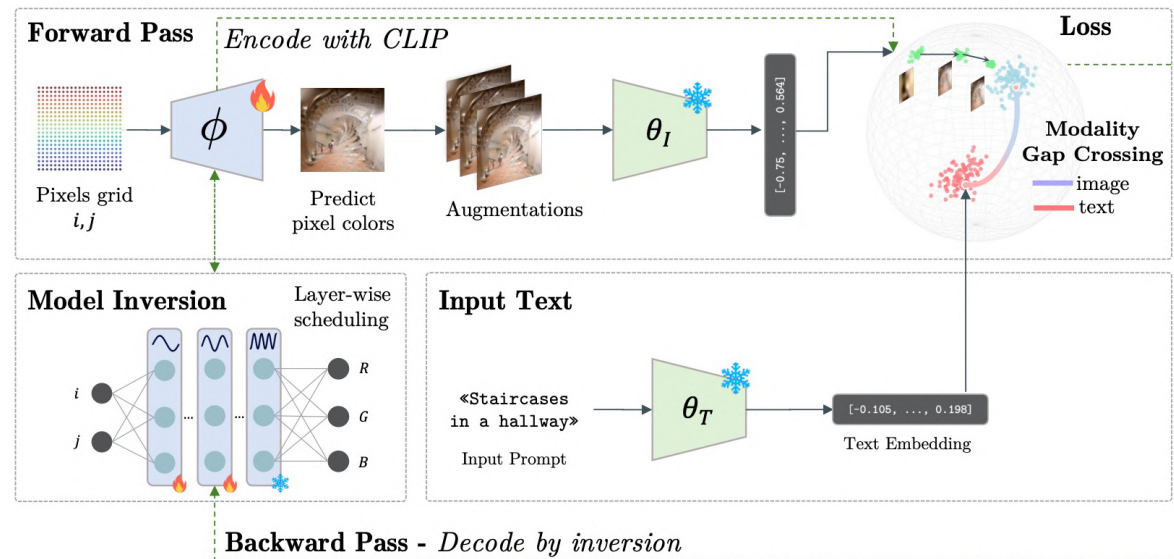
Diffusion samples that  
interpolate modes are **in** data  
manifold

## §5.3 Cat-3 inversion — understanding-as-generation



CLIP-1:

- The inverse of understanding can be a generation process: invert a discriminative model to synthesize.
- Implicit Inversion turns CLIP into a Decoder:** CLIP gives an energy-like score  $\rightarrow$  backprop directly to generate.



[CLIP-1](#) ICLR2026

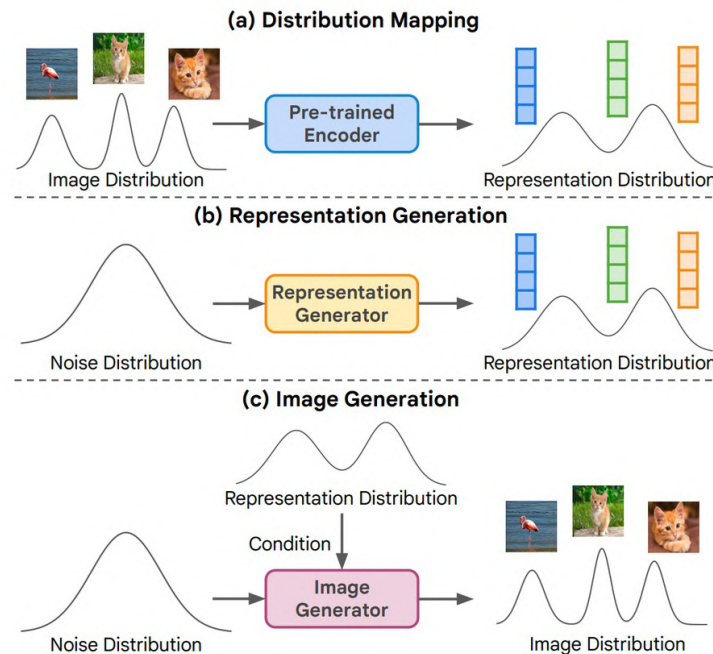
Figure 2: **CLIP<sup>-1</sup> text-to-image inversion pipeline.** The image is initially represented using an Implicit Neural Representation (INR)  $f_\phi(i, j)$ , optimizing the INR weights to match an input text prompt. The starting point for the inversion is a robust INR trained with Adversarial Weight Perturbation (AWP). The optimization updates the INR layer-wise, so that the embedding of its rendering aligns with the input text prompt embedding.

Recap: **PRG ICCV2025**: pretrained reversible generation as unsupervised representation learning — the reverse of generation is understanding.

## §5.4 Cat-4 condition-based



- **RCG:**
- Problem: Unconditional image generation lacks the semantic organization supplied by class labels, and therefore often underperforms conditional generation.
- Methods: First learn a prior over self-supervised image representations; sample a representation; then condition an image generator on that sampled representation.



RCG [2312]

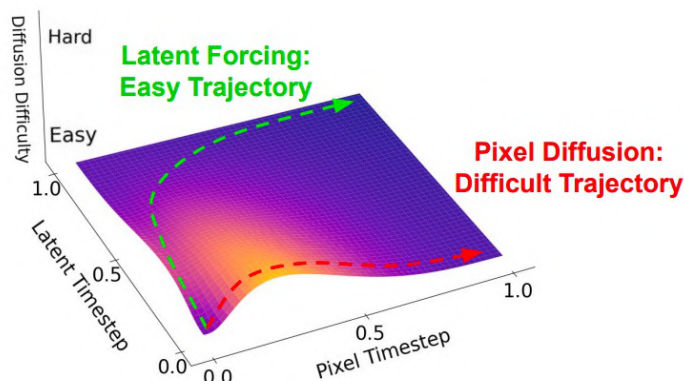
Figure 1: **The Representation-Conditioned Generation (RCG) framework** for unconditional generation. RCG consists of three parts: (a) it uses a pre-trained self-supervised encoder to map the image distribution to a representation distribution; (b) it learns a representation generator that samples from a noise distribution and generates a representation subject to the representation distribution; (c) it learns an image generator (e.g., which can be ADM [18], DiT [50], or MAGE [41]) that maps a noise distribution to the image distribution conditioned on the representation distribution.

## §5.4 Cat-4 condition-based

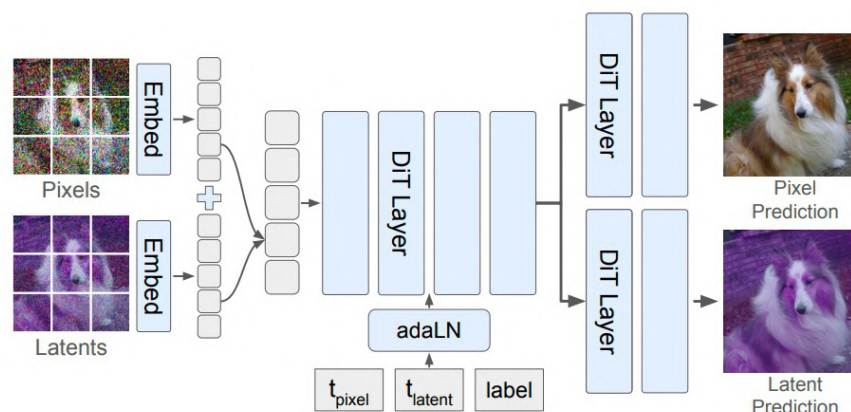


**Latent forcing:** model representation and pixel sample jointly while denoising representation faster to help pixel generation.

- Pixel diffusion is end-to-end but expensive; latent diffusion is efficient but loses information and requires a separate decoder.
- Jointly model pixels and a self-supervised latent using separate time variables and noise schedules. Recover the high-level latent earlier so that it acts as a scratchpad for subsequent pixel denoising.
- Changing the order in which information becomes available improves pixel-space generation: semantic structure is formed before expensive high-frequency refinement.



*Figure 1.* Conceptual Diagram. We diffuse both latents and pixels together, each with their own time variable. This allows us to denoise an easier trajectory than pure pixel diffusion by diffusing latents first, leading to improved performance.



Latent forcing [2602]

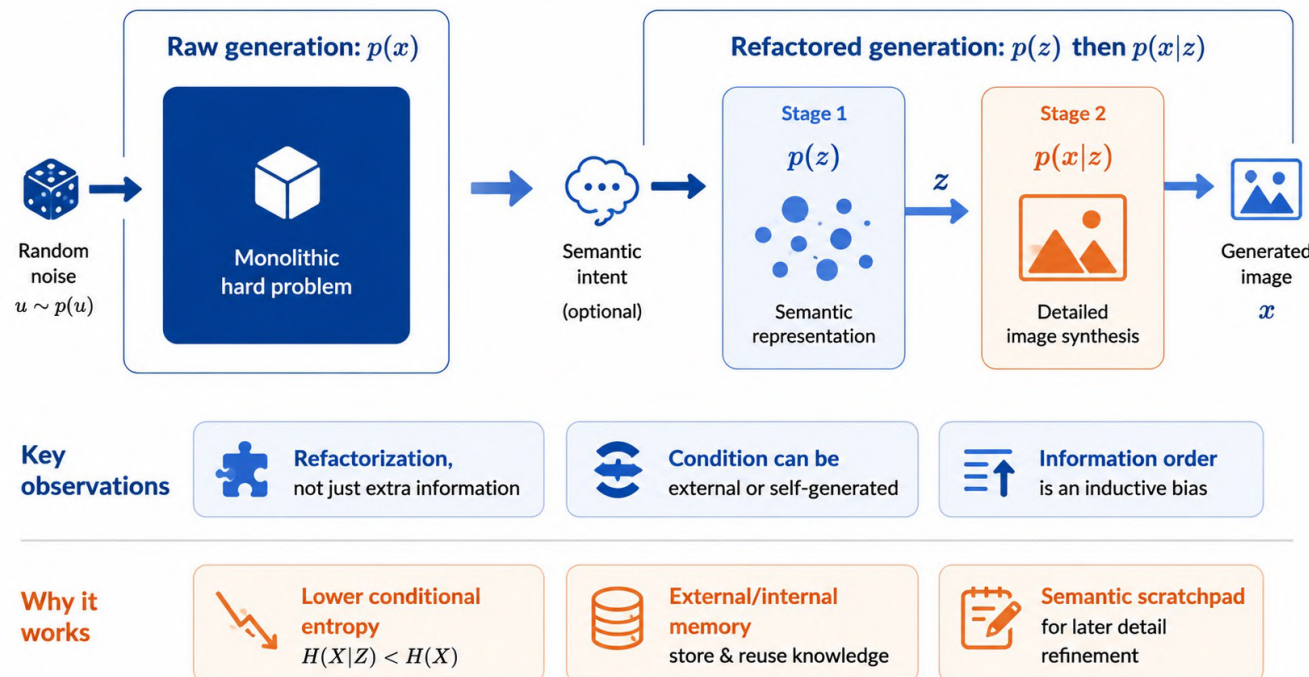
## §5.4 Cat-4 condition-based: Representation Condition Re-factorizes Generation



Representation conditioning changes the problem from  $p(x)$  to  $p(z)p(x|z)$ .

### Three observations

1. The value of representation condition is not just extra information; it re-factorizes the generative problem.
2. The condition does not have to come from an external model; it can be self-generated.
3. The order in which information becomes available is itself an inductive bias.

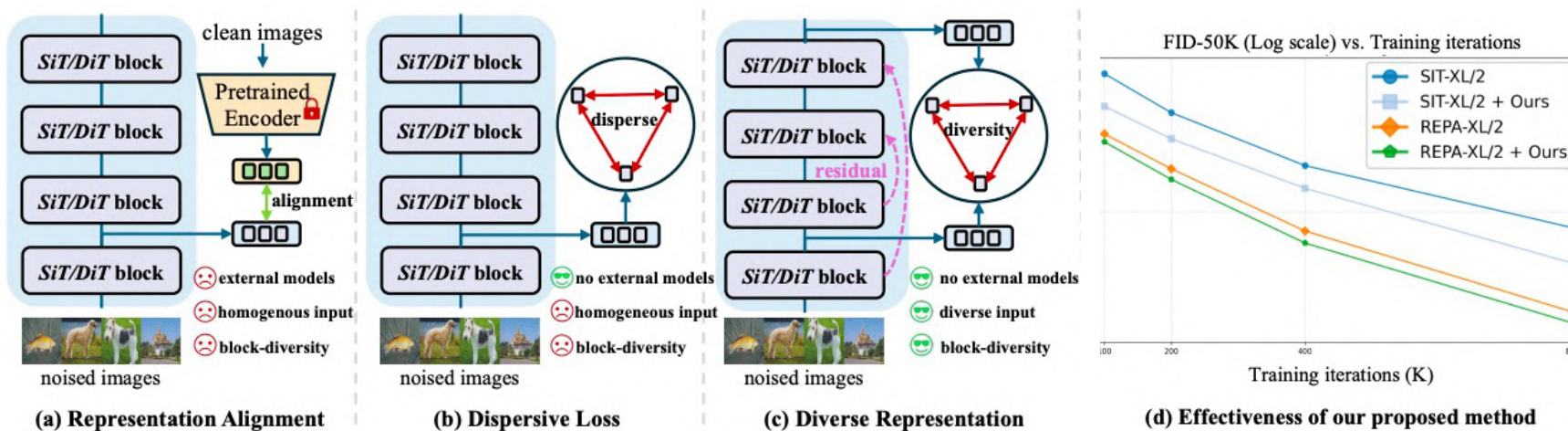


## §5.5 Cat-5 representation-learning loss



- **DiverseDiT**: block-wise representation diversity is key; long residual + diversity loss makes different blocks learn different features.
- Problem: Different DiT blocks can learn highly similar representations, so model depth does not necessarily produce complementary computation.
- Analyze cross-block similarity during training; introduce **long residual connections** and a **block-wise representation-diversity loss** to encourage functional specialization.

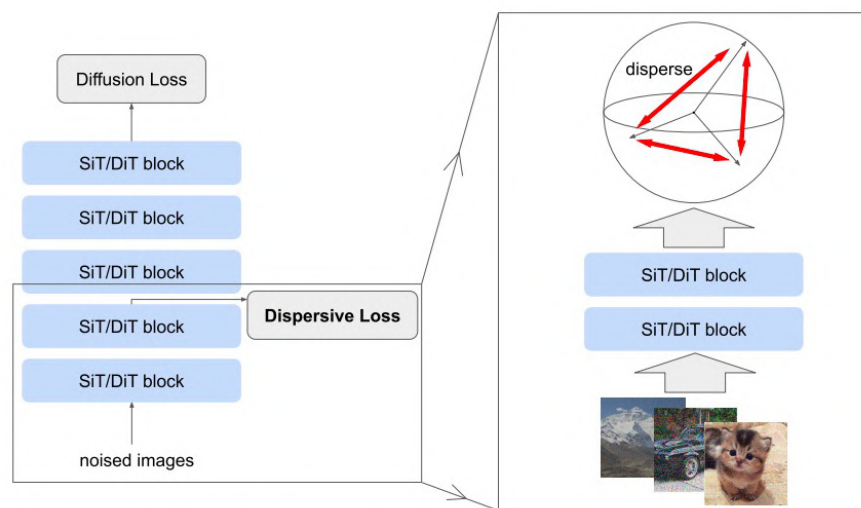
**Wild ideas:** Do DiverseDiT and LayerSync have conflicting motivations? Could this reveal deeper connections to architecture design, especially skip connections?



## §5.5 Cat-5 representation-learning loss



- **Dispersive Loss** (Diffuse and Disperse): add dispersion on internal representation, no external encoder — just prevent collapse.
- Diffusion models are mainly trained with pointwise regression and do not explicitly prevent internal representation concentration or collapse.
- Add a dispersive loss that pushes apart hidden representations of different batch samples, without positive pairs, an external encoder, new inference modules, or extra model parameters.



Wild idea: Uniformly pulling together or pushing apart representations across samples is too crude. How can we make this process adaptive, or choose better objects for attraction and repulsion?

Figure 1: **Dispersive Loss for Generative Modeling.** **Left:** A standard diffusion-based model (*e.g.*, SiT [26] or DiT [31]) with a regression-driven diffusion loss, augmented by a *Dispersive Loss* applied to an intermediate block. **Right:** A zoom-in on the first few blocks with *Dispersive Loss*. Our loss encourages intermediate representations to disperse in the hidden space. It operates on the same input batch of noised images, shares the existing network blocks and computations, introducing negligible overhead, no additional parameters, and no reliance on external data.

[Dispersive Loss](#) [2506]

# §5.5 Cat-5 representation-learning loss



## Perceptual Flow Matching:

- Problem: Standard flow matching uses Euclidean velocity regression, which can be poorly aligned with perceptual quality and accumulates visible errors under coarse few-step integration.
- Compare predicted and target flow outcomes in a pretrained perceptual feature space rather than relying only on raw latent or pixel velocity MSE.
- Perceptual supervision enables high-quality generation with substantially fewer integration steps, without requiring teacher distillation or an auxiliary score model.
- The training metric changes the statistical estimator: Euclidean regression tends toward conditional averaging, whereas perceptual objectives favor more on-manifold, mode-like solutions under limited sampling steps.

**Algorithm 1** Perceptual Flow Matching (PFM) with optional CFG baking

```
# =====
# PFM
# =====
v_pred = model(x_t, t, c)
x0_pred = velocity_to_x0(v_pred, x_t, t)

loss = perceptual_loss(
    decode(x0_pred),
    decode(x0)
)

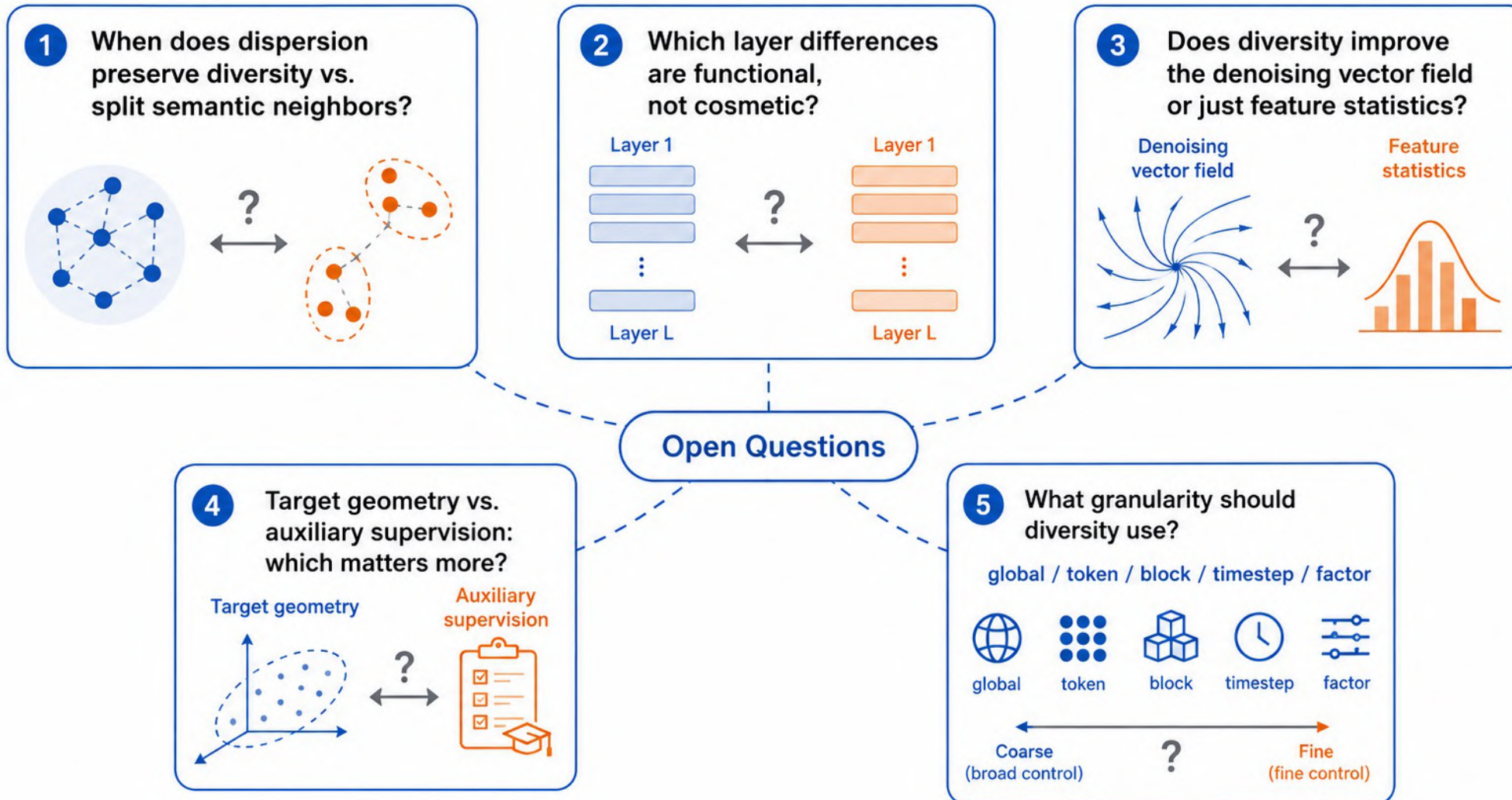
# =====
# PFM with prediction side CFG baking
# =====
v_pred = model(x_t, t, c)
v_uncond = model(x_t, t, null)
v_pred = beta * v_pred + (1 - beta) * stopgrad(v_uncond)
x0_pred = velocity_to_x0(v_pred, x_t, t)

loss = perceptual_loss(
    decode(x0_pred),
    decode(x0)
)
```

[Perceptual Flow Matching \[2607\]](#)

**Wild ideas:** Better representations for few-step generation—  
serial vs. parallel complexity?

# §5.5 Cat-5 representation-learning loss: open question?

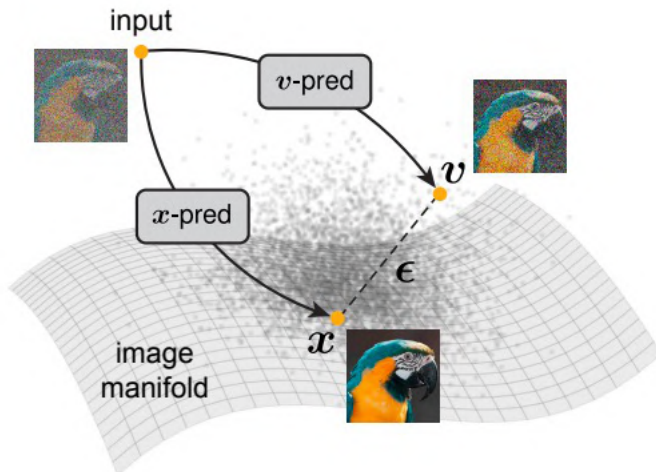


## §5.6 more analysis: The manifold lever — x-prediction (JiT)



### JiT:

- **Problem:** Pixel-space diffusion is hard due to extremely high dimensionality, motivating VAEs/tokenizers and noise/velocity prediction.
- Removes tokenizers, pretraining, and auxiliary losses; a large-patch Transformer directly predicts the clean image  $x_0$  from raw pixels.
- **Finding:**  $x_0$ -prediction remains effective even with patch sizes 16/32 at 256 or 512, while noisy-target prediction can fail in the same setting.
- **Insight:** Natural images lie near a low-dimensional manifold, whereas noise/velocity fill the ambient space; predicting  $x_0$  therefore provides a more structured, lower-effective-dimensional target.



|                                                                                               | (a) $x$ -pred                                                              | (b) $\epsilon$ -pred                                                       | (c) $v$ -pred                                                      |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------|
|                                                                                               | $\mathbf{x}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$                  | $\boldsymbol{\epsilon}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$       | $\mathbf{v}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$          |
| (1) $x$ -loss: $\mathbb{E} \ \mathbf{x}_\theta - \mathbf{x}\ ^2$                              | $\mathbf{x}_\theta$                                                        | $\mathbf{x}_\theta = (\mathbf{z}_t - (1-t)\boldsymbol{\epsilon}_\theta)/t$ | $\mathbf{x}_\theta = (1-t)\mathbf{v}_\theta + \mathbf{z}_t$        |
| (2) $\epsilon$ -loss: $\mathbb{E} \ \boldsymbol{\epsilon}_\theta - \boldsymbol{\epsilon}\ ^2$ | $\boldsymbol{\epsilon}_\theta = (\mathbf{z}_t - t\mathbf{x}_\theta)/(1-t)$ | $\boldsymbol{\epsilon}_\theta$                                             | $\boldsymbol{\epsilon}_\theta = \mathbf{z}_t - t\mathbf{v}_\theta$ |
| (3) $v$ -loss: $\mathbb{E} \ \mathbf{v}_\theta - \mathbf{v}\ ^2$                              | $\mathbf{v}_\theta = (\mathbf{x}_\theta - \mathbf{z}_t)/(1-t)$             | $\mathbf{v}_\theta = (\mathbf{z}_t - \boldsymbol{\epsilon}_\theta)/t$      | $\mathbf{v}_\theta$                                                |

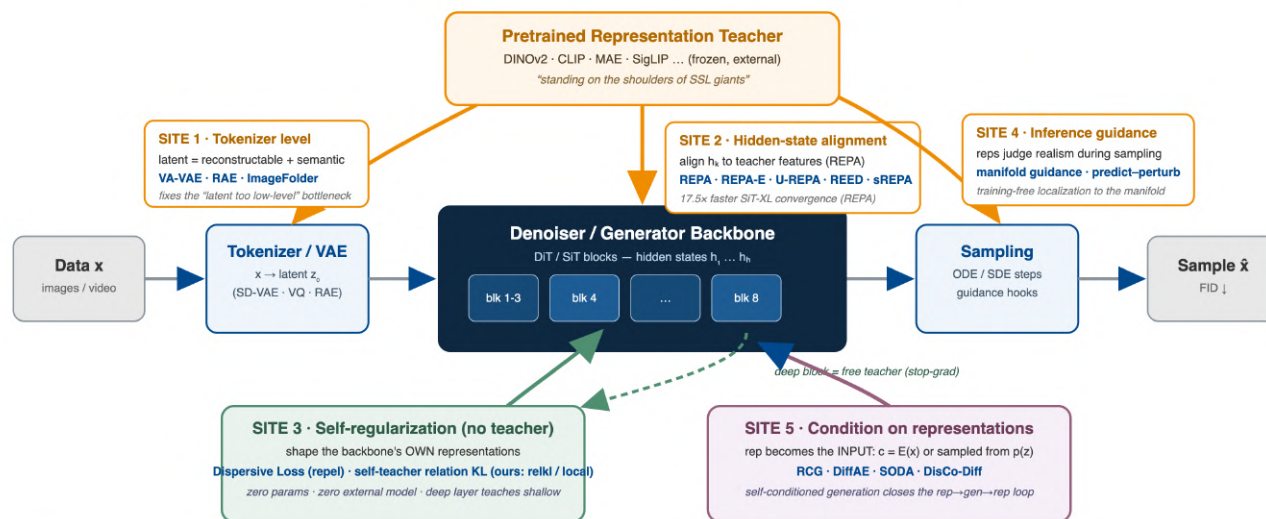
# §5.6 the whole landscape (five categories at a glance)



| Category                 | Representatives                                         | Injection site      |
|--------------------------|---------------------------------------------------------|---------------------|
| (1) alignment            | REPA / U-REPA / SoftREPA / SRA / LayerSync / VA-VAE     | middle / VAE / self |
| (2) rep-space generation | RAE / RAEv2 / RCG                                       | latent space        |
| (3) inversion            | CLIP-inversion / PRG                                    | model inverse       |
| (4) condition            | DiffAE / SODA / self-condition                          | condition           |
| (5) rep-learning loss    | Dispersive / Repulsor / DiverseDiT / LSEP / <b>ours</b> | internal loss       |
| (+) tokenizer / manifold | MAGVIT-v2 / TiTok / MAETok / JiT                        | encoder / target    |

## Where Can Representations Help a Generative Model? Five Injection Sites

taxonomy following the training pipeline — every method in §4 lives on exactly one of these arrows



### WHY IT WORKS (three recurring mechanisms)

M1 Denoising alone learns semantics SLOWLY — alignment shortcuts the semantic part so capacity goes to generation details.

M2 Structure > instance: relations / affinity carry the transferable signal (class-relations, token topology). M3 Better-conditioned latent manifold → easier score field → faster convergence, better FID.

# Roadmap

§1 Outline

§2 Motivation — why a unified model, why now, the several gaps, and my personal views on what a good representation needs.

§3 Intro — understanding (survey taxonomy) + generation (DDPM).

§4 Generation helps Understanding — diffusion's own representations: where, when good, when broken.

§5 Understanding helps Generation — five ways representation is injected into generation.

§6 The Story of Hiddens of Generative Model: A Quick Tour

§7 Building a Unified Model — existing attempts and their limits.

§8 Open Question & Wild Ideas & Summary — answer the root question.

1 Outline

2 Motiv

3 Intro

4 Gen→Rep

5 Rep→Gen

6 Quick Tour

7 Unified

8 Summary

## §6 The Story of Hiddens of Generative Model: A Quick Tour

§6.1 Does it matter?

§6.2 What is the target?

§6.3 How do we reach it?

1 Outline

2 Motiv

3 Intro

4 Gen→Rep

5 Rep→Gen

6 Quick Tour

7 Unified

8 Summary

Note: This section is primarily a synthesis.

# §6.0 Outline

---



**How should DiT hidden states evolve so that generation becomes:**

- easier to optimize,
- easier to approximate with finite network depth,
- easier to integrate with few sampling steps,
- and more robust in inference rollouts?

**Three questions**

- **Does it matter?**  
Do hidden-state properties causally affect training and generation?
- **What is the target?**  
What should each layer learn at each SNR and training stage?
- **How do we reach it?**  
How can we accelerate useful structure without suppressing generative capacity?

Root question: **How can we reach the target hidden state faster and more accurately, while avoiding the capacity loss, bias, and train–inference mismatch introduced by external or self-alignment?**

## §6.1 Does it matter?

---



### Hidden-State Interventions Can Strongly Change DiT Training

- **REPA:** aligns projected noisy DiT states with clean SSL features and reports large convergence gains.
- **SRA / LayerSync:** remove the external vision encoder and use stronger internal states as supervision.
- **DiverseDiT:** improves learning by increasing useful representation differentiation across blocks.
- **Residual-routing studies:** changing skip strength alone changes both internal representations and generation.
- **PixelREPA:** shows that the same alignment can fail when the state space and teacher compression are mismatched.

Hidden states are causal intervention points—but no universal hidden-state quality score has been established.

## §6.1 Does it matter?

---



### Why Can Better Hidden States Improve Final Generation?

#### Lower suffix complexity

A well-organized state makes the remaining map

$$h_{l,t} \rightarrow v^*(x_t, t)$$

simpler for a finite-depth suffix to approximate.

#### Lower velocity ambiguity

If similar hidden states require conflicting velocities, MSE regression averages modes. A better state separates these trajectories.

#### Better spatial computation

Object support, part relations, and layout can make attention routing and global integration easier.

#### Better numerical behavior

A state producing smoother, more on-manifold predictions can tolerate larger solver steps.

#### Better generalization

Balanced internal statistics may avoid both low-rank collapse and sample-specific spikes.

## §6.1 Does it matter?

---



A More Operational Definition: Remaining Generative Risk

- A hidden state is better when it enables:
  - lower error with the **same suffix depth**,
  - the same error with a **smaller suffix**,
  - better performance under **layer skipping**,
  - or better robustness under **few-step sampling**.

## §6.1 Does it matter?

---



### Why the Simple “Better Representation → Better Generation” Claim Fails

- **Recognition–generation mismatch:** invariance helps recognition but can delete texture, pose, background, and instance identity.
- **Early gain ≠ final optimum:** a target can speed optimization yet bias the final solution.
- **Storage ≠ use:** a layer may encode semantics but contribute little to the predicted velocity.
- **Geometry ≠ function:** attractive global geometry does not guarantee correct object–attribute binding.
- **Better FID ≠ better diversity:** a tighter semantic target may improve precision while reducing recall.
- **Steps-to-FID ≠ actual efficiency:** teacher pretraining, extra forward passes, memory, and decoder cost must be counted.

The defensible claim is conditional: hidden-state shaping helps only when it reduces task difficulty without deleting required generative degrees of freedom.

## §6.2 What Should the Target Hidden State Be?



### Question:

Is there a generation-useful hidden-state metric that consistently predicts FID, recall, and few-step generation error?

**Answer:** Not yet — no single metric has been shown to reliably predict all three.

| Metric                                                         | Strength                                                                                                                                             | Limitation                                                                                                            |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Linear-Probe Accuracy (LPA) / Teacher-Feature Similarity (TFS) | Measures <b>semantic readability / stored information</b> ; simple for layer- and stage-wise analysis.                                               | <b>Stored ≠ used</b> ; teacher/task dependent, and semantic compression may discard generative detail.                |
| Centered Kernel Alignment (CKA)                                | Compares representation geometry across <b>layers, models, and training stages</b> ; useful for redundancy/specialization analysis.                  | Similarity is not quality: it does not tell whether differences are <b>causal or generation-useful</b> .              |
| Weighted Diversity Score (WDS)                                 | Summarizes <b>inter-block specialization</b> ; strongly correlates with synthesis quality across diverse DiT settings. ( <a href="#">arXiv</a> )     | Measures <b>difference, not useful innovation</b> ; correlation is not causality and includes a depth-distance prior. |
| Invariant Contamination Ratio (ICR)                            | Label-free measure of <b>invariant signal vs. residual variation</b> ; tracks semantic windows and memorization onset. ( <a href="#">arXiv</a> )     | Depends on the chosen perturbation/invariance definition; desirable behavior is <b>SNR- and task-dependent</b> .      |
| Participation Ratio / Effective Rank (PR / eRank)              | Cheap measure of <b>spectral balance, dimensional collapse, and concentration</b> ; useful for generalization/memorization diagnosis.                | Global spectrum ignores <b>binding, functional use, and output relevance</b> .                                        |
| Forward-only Gate Ablation (FoG-A)                             | Intervenes on layers and measures induced velocity change; directly probes <b>what the network actually uses</b> . ( <a href="#">arXiv</a> )         | Layer-level and coarse; <b>used ≠ beneficial to align or regularize</b> .                                             |
| Jacobian Effective Rank (JER)                                  | Measures <b>functional / response geometry</b> rather than static point-cloud geometry; informative for compositional binding.                       | Local and more costly; not yet established as a general predictor of DiT generation quality.                          |
| Velocity Irreducible Variance (VIV)                            | Generation-native measure of <b>velocity ambiguity / trajectory crossing</b> ; a strong predictor of latent diffusability. ( <a href="#">arXiv</a> ) | Mainly a <b>latent-space</b> metric; does not measure internal layer specialization or suffix computation.            |
| Remaining Generative Risk (RGR, proposed)                      | Directly measures <b>how much generative work remains under fixed suffix compute</b> ; links representation to depth/few-step difficulty.            | Not yet established; requires controlled suffix probes and cross-model / rollout validation.                          |

## §6.2 What Should the Target Hidden State Be?



### Question:

Is there a generation-useful hidden-state metric that consistently predicts FID, recall, and few-step generation error?

**Answer:** Not yet — no single metric has been shown to reliably predict all three.

| Metric & Formula                                                                                                                                            | Symbols & Physical / Functional Meaning                                                                                                                                                                                                                                           | Strength                                                                                                                                                | Limitation                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Linear-Probe Accuracy (LPA) / Teacher-Feature Cosine Similarity (TFCS)</b><br>$\text{LPA} = \text{Acc}(W^*H, y), \quad \text{TFCS} = \cos(P(H), T(x_0))$ | $H$ : hidden representation; $W^*$ : trained linear probe; $y$ : semantic label; $P$ : student projector; $T(x_0)$ : clean teacher feature. <b>Meaning:</b> asks whether semantic/teacher information is easily <b>readable or stored</b> in $H$ .                                | Simple and interpretable; useful for tracking <b>where and when semantic information emerges</b> across layers/SNR/training.                            | <b>Stored <math>\neq</math> used <math>\neq</math> useful.</b> High readability does not show that the generator uses the feature; teacher semantics may also suppress detail, diversity, or other generation-relevant information. |
| <b>Centered Kernel Alignment (CKA)</b><br>$\text{CKA}(X, Y) = \frac{\text{HSIC}(X, Y)}{\sqrt{\text{HSIC}(X, X)\text{HSIC}(Y, Y)}}$                          | $X, Y$ : representation matrices from the same samples; <b>HSIC</b> : kernel dependence measure. <b>Meaning:</b> high CKA means two representations preserve similar pairwise/kernel geometry, even if their coordinates differ.                                                  | Robust for comparing <b>layers, models, checkpoints, and alignment effects</b> ; reveals redundancy, specialization, and representation dynamics.       | Measures <b>similarity, not quality</b> . High/low CKA alone cannot tell whether a representation difference is causally used or beneficial for generation.                                                                         |
| <b>Weighted Diversity Score (WDS)</b><br>$\text{WDS} = 1 - \frac{\sum_{i < j}  j - i  \text{CKA}(H_i, H_j)}{\sum_{i < j}  j - i }$                          | $H_i, H_j$ : feature matrices of blocks $i, j$ ; $ j - i $ : depth distance; CKA measures their similarity. <b>Meaning:</b> larger WDS means especially <b>distant blocks become less redundant and more differentiated</b> ; $\text{WDS} = 0$ corresponds to maximal similarity. | Converts a full CKA matrix into a scalar <b>block-specialization diagnostic</b> ; strongly correlates with generation quality across many DiT settings. | <b>Diversity <math>\neq</math> useful innovation.</b> Random/nuisance differences can also raise diversity; depth weighting assumes distant blocks should differ, and WDS–FID evidence is primarily correlational.                  |

## §6.2 What Should the Target Hidden State Be?



### Question:

Is there a generation-useful hidden-state metric that consistently predicts FID, recall, and few-step generation error?

**Answer:** Not yet — no single metric has been shown to reliably predict all three.

|                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                 |                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Invariant Contamination Ratio (ICR)</b><br>$\text{ICR} = \frac{1}{1 + \frac{1}{d} \sum_i \lambda_i}, \quad \Sigma_s v_i = \lambda_i \Sigma_\xi v_i$                                                     | $h = s + \xi$ : feature = invariant component $s$ + perturbation-sensitive residual $\xi$ ; $\Sigma_s, \Sigma_\xi$ : their covariances; $v_i$ : Fisher direction; $\lambda_i$ : invariant-to-residual variance ratio along $v_i$ ; $d$ : feature dimension. <b>Meaning:</b> low ICR means stable/invariant structure dominates residual variation; high ICR means stronger contamination by view/noise-specific variation. | Label-free and intrinsic; captures the <b>invariance-detail balance</b> , identifies intermediate-noise semantic windows, and can signal memorization dynamics. | The desired value is <b>SNR- and task-dependent</b> : low-noise generation legitimately needs residual detail. It also depends on how perturbations/augmentations define “invariance.”                                |
| <b>Global Participation Ratio / Global Isotropy Score (G.PR / G.Iso)</b><br>$\text{G.PR} = \frac{(\sum_i \lambda_i)^2}{D \sum_i \lambda_i^2}, \quad \text{G.Iso} = 1 - \frac{\lambda_1}{\sum_i \lambda_i}$ | Here $\lambda_i$ : eigenvalues of the embedding covariance (different from ICR's generalized eigenvalues); $D$ : feature dimension; $\lambda_1$ : largest covariance eigenvalue. <b>Meaning:</b> high G.PR means variance occupies many dimensions; high G.Iso means no single principal direction dominates.                                                                                                              | Cheap intrinsic diagnostics of <b>effective dimensionality, anisotropy, spectral concentration, and representation collapse</b> .                               | Global spectral geometry says little about <b>what information is encoded, how factors are bound, or whether direction affect the generator</b> ; high-rank/isotropic representations can still be functionally poor. |
| <b>Forward-only Gate Ablation (FoG-A)</b><br>$\text{FoG-A}_k = \mathbb{E}_{x_t, t, c} \left[ \frac{\ v_\theta^{\setminus k} - v_\theta\ _2}{\ v_\theta\ _2 + \epsilon} \right]$                            | $k$ : ablated block; $v_\theta$ : original predicted velocity; $v_\theta^{\setminus k}$ : velocity after disabling block $k$ ; $x_t$ : noisy state; $t$ : diffusion/flow time; $c$ : condition; $\epsilon$ : numerical stabilizer. <b>Meaning:</b> high FoG-A means removing block $k$ strongly changes the velocity field, so that block is <b>causally important to the current computation</b> .                        | Moves from correlation to <b>interventional usage</b> ; directly exposes the Storage-Use gap and identifies functionally critical layers.                       | Coarse at layer level and conditional on the current trained network. <b>Used ≠ useful to align/regulate</b> : a necessary block may already be optimal, and intervention ignores finer causal subspaces/interaction  |

## §6.2 What Should the Target Hidden State Be?



### Question:

Is there a generation-useful hidden-state metric that consistently predicts FID, recall, and few-step generation error?

**Answer:** Not yet — no single metric has been shown to reliably predict all three.

|                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Jacobian Effective Rank (JER)</b><br>$\text{JER} = \mathbb{E}_x \left[ \frac{(\sum_i \sigma_i(J_f(x)))^2}{\sum_i \sigma_i^2(J_f(x))} \right], \quad J_f(x) = \frac{\partial f(x)}{\partial x}$ | $f$ : representation mapping; $J_f(x)$ : local input-to-feature Jacobian; $\sigma_i$ : Jacobian singular values; $x$ : input sample. <b>Meaning:</b> high JER means many independent input directions can meaningfully change the representation; low JER indicates <b>functional sensitivity collapse</b> into a few directions.                                                                                                   | Measures <b>functional/response geometry</b> , rather than static point-cloud geometry; substantially more relevant to compositional sensitivity and binding.                                                | Local first-order diagnostic; depends on input distribution and scaling and requires Jacobian-vector computations. Its ability to predict DiT FID/recall/few-step generation is still unestablished.                                               |
| <b>Velocity Irreducible Variance (VIV)</b><br>$\text{VIV} = \mathbb{E} [\ v - v^*(x_t, t, y)\ _2^2], \quad v^* = \mathbb{E}[v \mid x_t, t, y]$                                                    | $x_0$ : source noise; $x_1$ : target latent; $x_t = (1 - t)x_0 + tx_1$ : interpolated state; $v = x_1 - x_0$ : target transport velocity; $y$ : condition; $v^*$ : Bayes-optimal conditional mean velocity. <b>Meaning:</b> high VIV means the same intermediate state admits conflicting transport directions—strong <b>trajectory crossing / velocity ambiguity</b> .                                                             | Directly tied to the generative regression objective; gives a principled measure of <b>intrinsic diffusability / transport ambiguity</b> and strongly predicts latent generation difficulty.                 | Primarily a <b>tokenizer/latent-space property</b> , not an internal DiT hidden-state diagnostic. It does not measure layer specialization, causal usage, or finite-suffix computation; analytic versions also require distributional assumptions. |
| <b>Remaining Generative Risk (RGR, proposed)</b><br>$R_k(H_{\ell,t}) = \inf_{g \in \mathcal{G}_k} \mathbb{E} [\ Y_t - g(H_{\ell,t}, t, c)\ _2^2]$                                                 | $H_{\ell,t}$ : hidden state at layer $\ell$ and noise level $t$ ; $Y_t$ : true generation target (e.g., velocity); $\mathcal{G}_k$ : suffix/probe class under fixed compute budget $k$ ; $g$ : downstream predictor; $\inf$ : best achievable risk within that budget. <b>Meaning:</b> lower $R_k$ means the hidden state has already transformed the problem into one that requires <b>less remaining downstream computation</b> . | Directly targets <b>generation usefulness</b> , not semantics or geometry; naturally connects hidden-state quality to suffix depth, compute efficiency, layer skipping, and potentially few-step robustness. | A proposed diagnostic, not yet an established metric. Results depend on the chosen suffix class/compute budget and require cross-model, cross-SNR, cross-NFE, and rollout-state validation to establish generality.                                |

## §6.2 What Should the Target Hidden State Be?



| Axis        | Desired property                                                                                                                                               | Why it matters                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Information | <b>SNR-calibrated functional sufficiency</b> — retain what is useful and predictable from the current ( $x_t$ ), without fabricating unavailable clean details | Avoid false certainty and clean-teacher mismatch                                            |
| Geometry    | <b>Spatial, relational, and compositional organization</b> with local continuity and low velocity ambiguity                                                    | Make attention routing and the remaining vector-field prediction easier                     |
| Function    | <b>Output-accessible shared state + useful layer-specific innovation</b>                                                                                       | Stored semantics are insufficient; every block should reduce the remaining generative error |
| Capacity    | <b>Detail/fiber liftability + balanced spectrum</b>                                                                                                            | Preserve texture, geometry, instance diversity, and avoid collapse or memorization spikes   |
| Dynamics    | <b>Layer/SNR/stage adaptivity + rollout robustness</b>                                                                                                         | A useful training state must remain useful on model-generated sampling trajectories         |

A good hidden state should make the remaining generation problem easier—while preserving uncertainty, detail, and useful depth-wise computation.

## §6.3 Gap to target state, the idea source



| Category             | Current Gap                                                                                                                                                                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Information          | <b>Information asymmetry:</b> clean/lower-noise teachers contain information unavailable to the current noisy state, causing averaging, shortcuts, or teacher bottlenecks under strong alignment; Non-stationarity, Collective Bias, and Erroneous Self-Reinforcement in Self-Teacher                            |
| Task mismatch        | <b>Understanding <math>\neq</math> generation:</b> discriminative representations favor invariance, while generation requires fine detail, diversity, and decoder liftability.                                                                                                                                   |
| Geometry             | <b>Insufficient relational order:</b> point-wise matching and pairwise/Gram geometry mainly capture low-order structure, but may miss object binding, attributes, and higher-order composition; high-order loss, more than MSE, to share the geometry; more losses to share representation directly are in need. |
| Dynamics             | <b>Highly non-uniform representation field:</b> hidden-state quality and function vary strongly across layers, SNR, and training stage; <b>slow emergence does not necessarily mean low usefulness.</b>                                                                                                          |
| Depth specialization | <b>Redundancy vs. specialization:</b> blocks can be redundant, but excessive alignment can also erase useful layer-wise functional differences; information stored in the layer may not affect generation.                                                                                                       |
| Architecture         | <b>Residual echo:</b> residual streams can repeatedly preserve shallow information, reducing the pressure for deeper layers to form genuinely new abstractions.                                                                                                                                                  |

The desired hidden state is not simply “more semantic,” but information-calibrated, higher-order, stage-aware, and functionally specialized for generation.

## §6.3 Several research gaps



| Meta Gap                                                      | What is Missing                                                                                                                                                                        | What Should We Do?                                                                                                                                                                                                                  |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>G1. Scientific-Object Gap</b>                              | “Good hidden state” is still defined by semantic/readability proxies, not by what makes generation easier.                                                                             | Define it by <b>remaining generative risk / suffix complexity</b> : task-sufficient, uncertainty-calibrated, structured, progressive, and rollout-robust.                                                                           |
| <b>G2. Measurement Gap</b>                                    | WDS, ICR, VIV, CKA, probes, etc. each capture one property; no metric robustly spans <b>models, SNRs, FID/recall, and few-step error</b> .                                             | Build <b>functional diagnostics</b> : suffix risk, causal accessibility, ambiguity, and rollout robustness—not another single geometric scalar.                                                                                     |
| <b>G3. Intervention / Controllability Gap</b>                 | We increasingly know <i>what</i> to measure, but current actuators—alignment, masking, EMA, diversity losses, routing, residuals—are mostly <b>coarse, indirect, and open-loop</b> .   | Develop <b>diagnosis-conditioned closed-loop shaping</b> over layer/token/subspace/SNR/training phase, with adaptive constraint strength.                                                                                           |
| <b>G4. Causality / Mechanism Gap</b>                          | <b>Readable / stored <math>\neq</math> used <math>\neq</math> useful</b> ; gains may instead come from deep supervision, projector capacity, gradient scaling, or regularization.      | Establish a <b>Storage <math>\rightarrow</math> Use <math>\rightarrow</math> Utility</b> causal chain with ablation, matched-compute/placebo supervision, and intervention.                                                         |
| <b>G5. Analysis <math>\rightarrow</math> Intervention Gap</b> | Phase transitions, easy-to-hard learning, and slow/uneven emergence remain mostly descriptive; point-MSE/Gram objectives may miss higher-order binding.                                | Turn diagnostics into <b>phase-aware objectives and response/structural supervision</b> , rather than naïve uniform or higher-moment matching.                                                                                      |
| <b>G6. Formulation / Design-Space Gap</b>                     | REPA/SRA/LayerSync/MaskAlign/sREPA/HASTE/guidance look like separate tricks, while increasingly reusing mature SSL ideas: EMA, stop-grad, masking, relations, weak-to-strong learning. | Unify <b>target <math>\times</math> object <math>\times</math> metric <math>\times</math> routing <math>\times</math> gradient topology <math>\times</math> controller <math>\times</math> inference role</b> into one formulation. |

The missing piece is not another “better feature,” but a principled way to define, diagnose, and control a generation-useful hidden-state field.

- **The Missing Scientific Links**

- Is there a metric that predicts **training speed, final FID, recall, and few-step error** simultaneously?
- Which subspace of  $T(x_0)$  is actually predictable from  $x_t$ ?
- How should shared information and layer-specific innovation be separated?
- How can higher-order binding be measured beyond pairwise Gram matrices?
- How are suffix risk, VIV, Jacobian structure, and causal layer contribution related?
- Can network depth absorb part of the computation currently assigned to diffusion-time NFE?
- When is a training-only teacher truly a catalyst rather than a hidden inference dependency?

- **Final thesis**

- A better DiT is not one whose hidden states are merely more semantic.  
It is one whose hidden trajectory makes the remaining generation problem progressively easier.

## §7 Building a unified model — back to the root question

§4 (gen→rep) and §5 (rep→gen) are the two halves;

§7 asks the root question directly: can one model do both, natively?

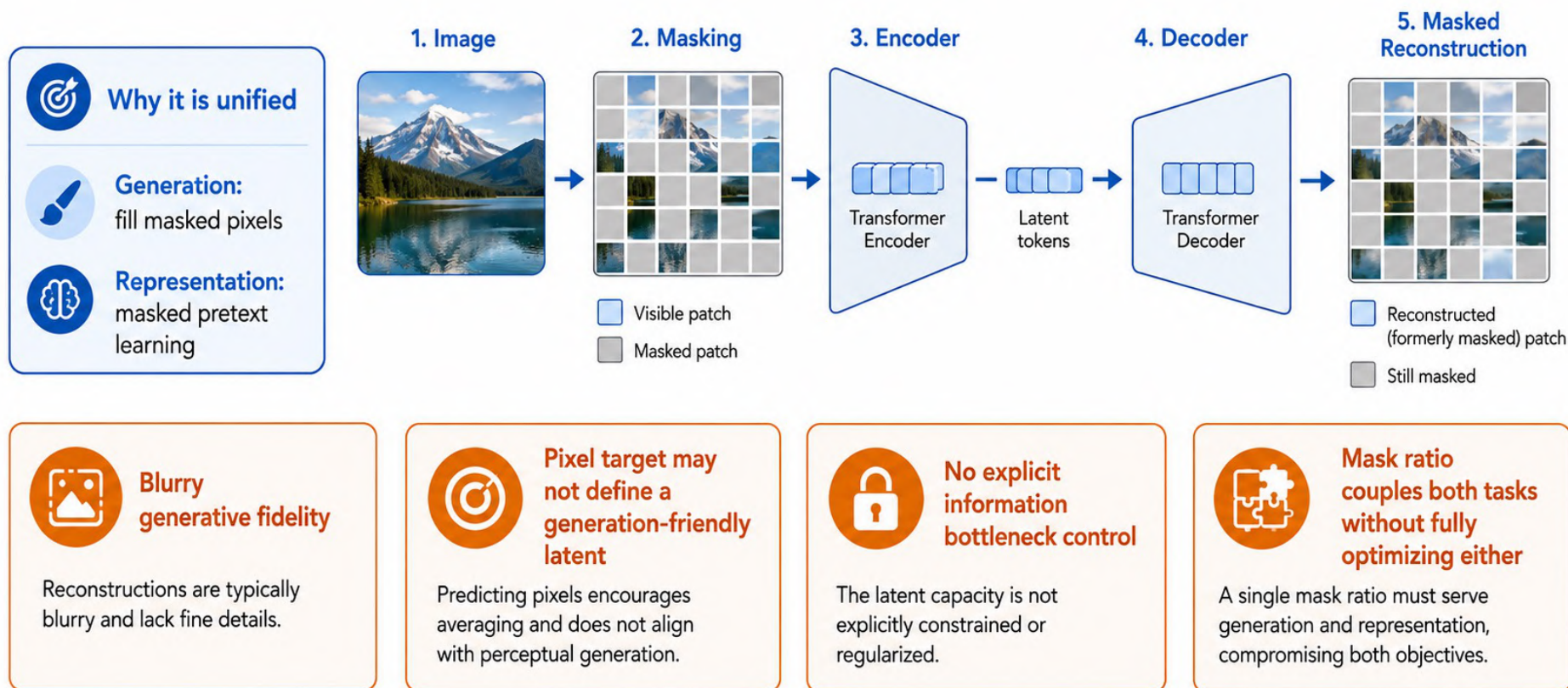
Existing attempts like MAE.

[1 Outline](#)[2 Motiv](#)[3 Intro](#)[4 Gen→Rep](#)[5 Rep→Gen](#)[6 Quick Tour](#)[7 Unified](#)[8 Summary](#)

## §7.0 MAE — the closest to a native unified model, and its limits



- MAE is *nearly* a native unified model: masked reconstruction is simultaneously a generation task (fill masked pixels) and a representation pretext.
- Limits: weak generative fidelity (blurry samples); a pixel-reconstruction target may not define a generation-friendly latent; no explicit information-bottleneck control; mask ratio couples both tasks but doesn't optimize either fully.

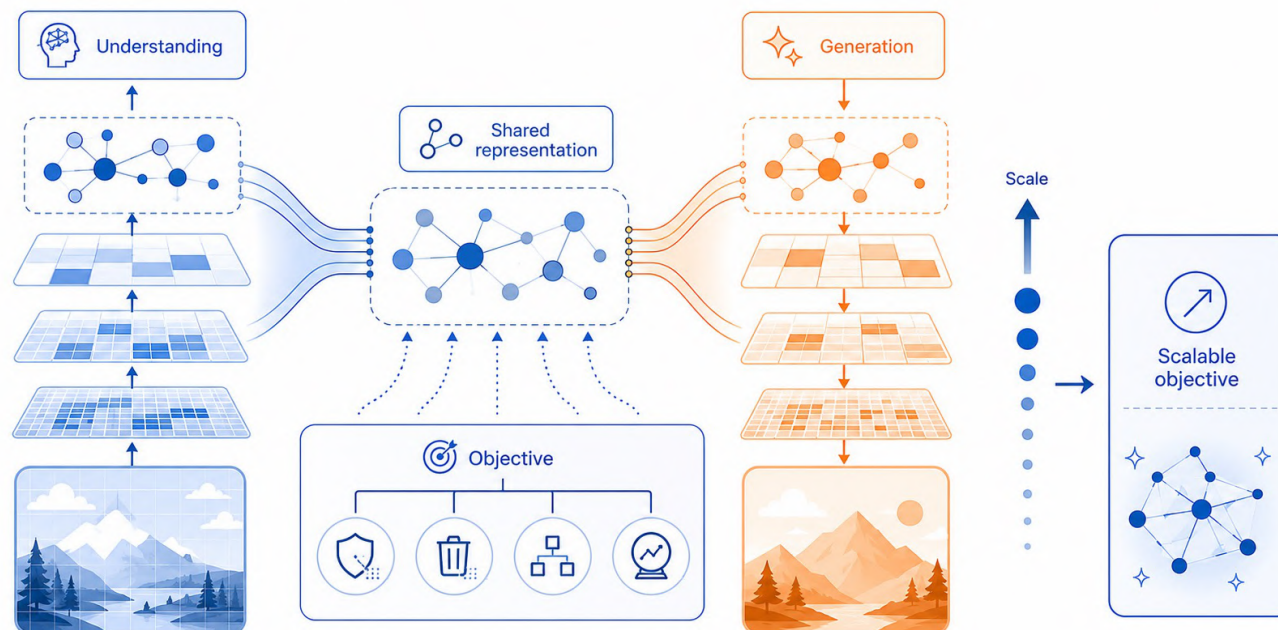


So, how?

## §7.1 The Missing Link: A Unified Learning Objective



- A unified model needs a shared representation that supports both understanding and generation—not merely a shared architecture.
- Representation is not given; it is shaped by the learning objective: the loss determines what information is preserved, discarded, organized, and made predictable.
- Therefore, a natural route to unification is to design an objective that learns representations that are simultaneously abstract enough for understanding and sufficiently informative for generation.
- This brings us to the key question: **what properties should a *scalable objective* enforce so that better representations emerge as model and data scale?**

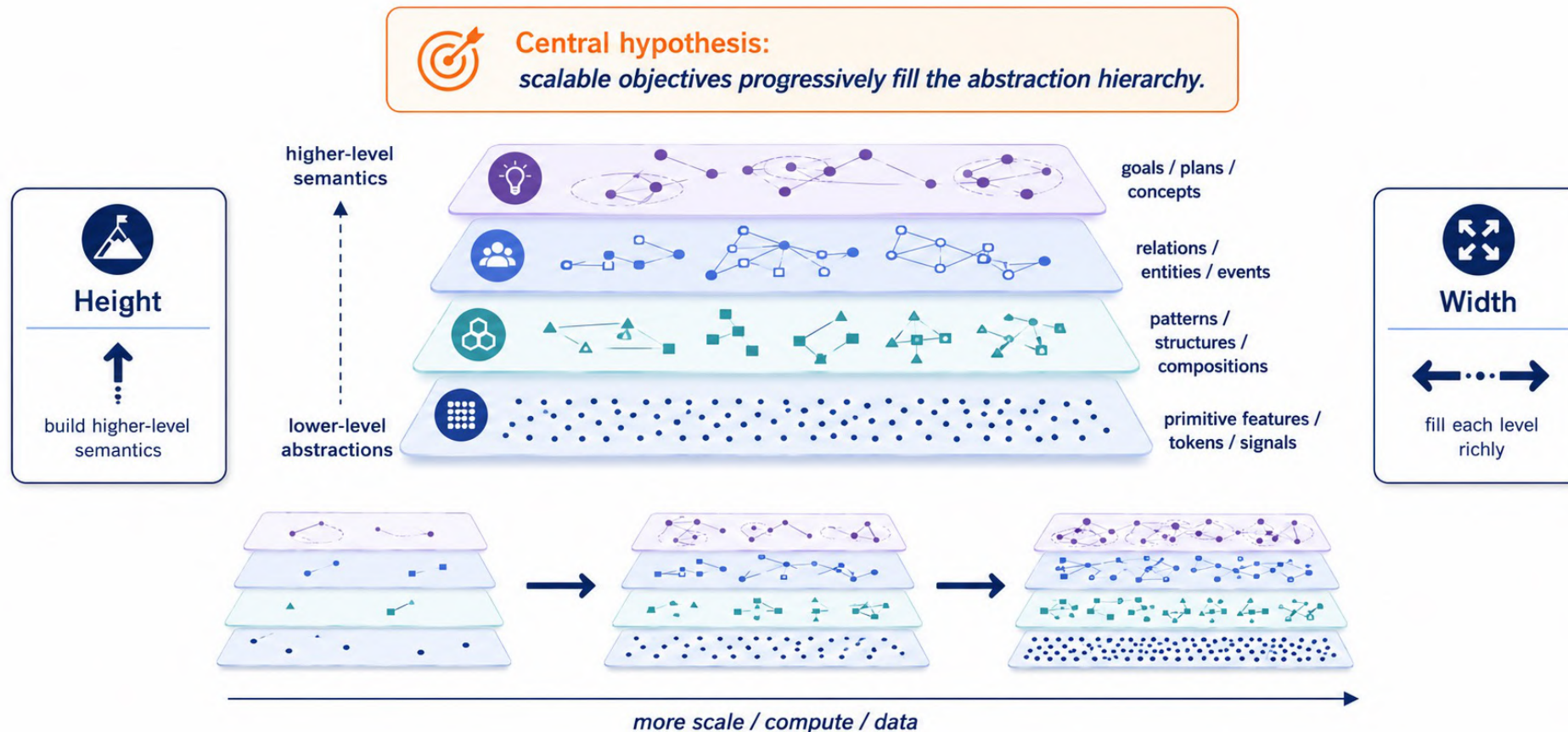


**Our thesis: unify the objective → shape the representation → unify understanding and generation.**

# §7.1 Scalable objective: Are All Scalable Objectives Alike?



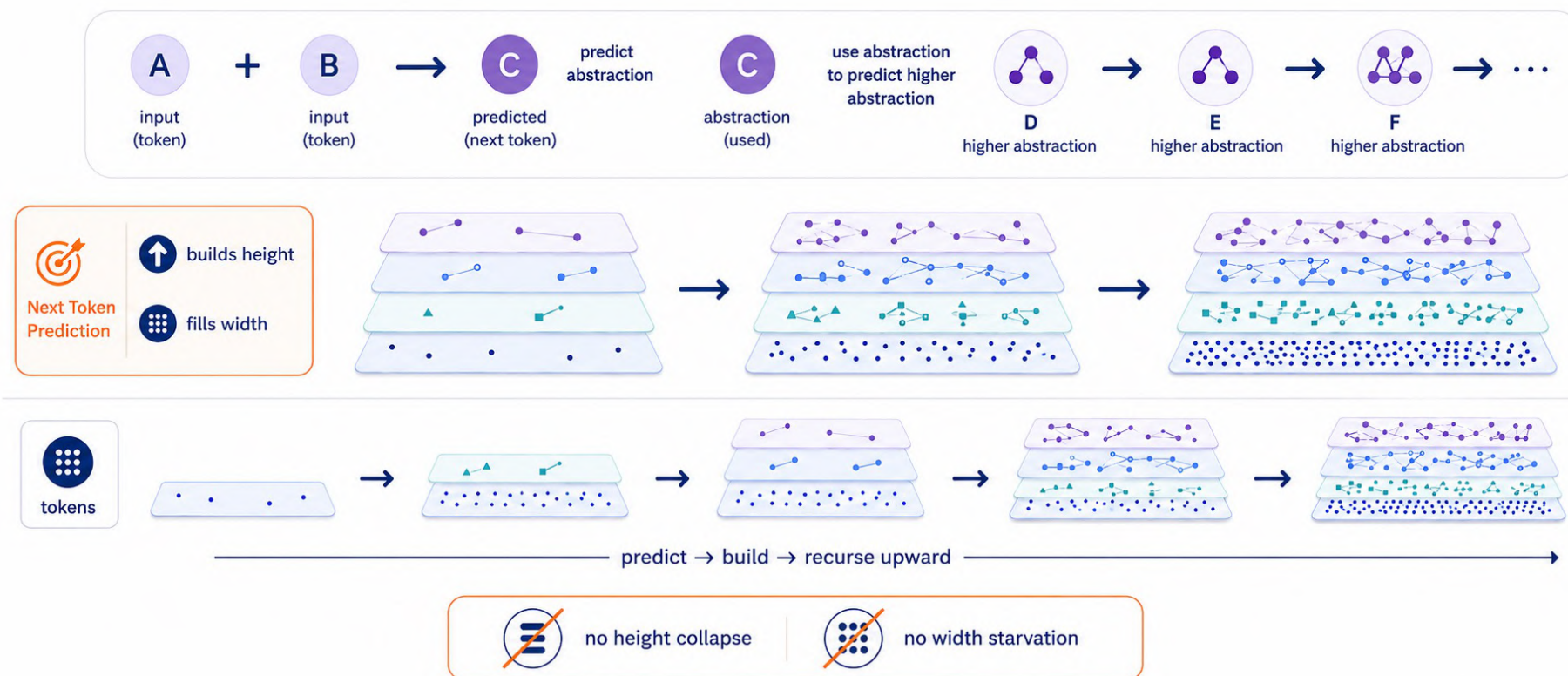
- **Hypothesis:** scalable objectives progressively fill the **abstraction hierarchy**.
- Scaling should both **build higher-level semantics** and **strengthen lower-level abstractions**.
- Good scaling therefore requires both **height** and **width**.



# §7.1 Scalable objective: Why NTP Fills the Hierarchy?



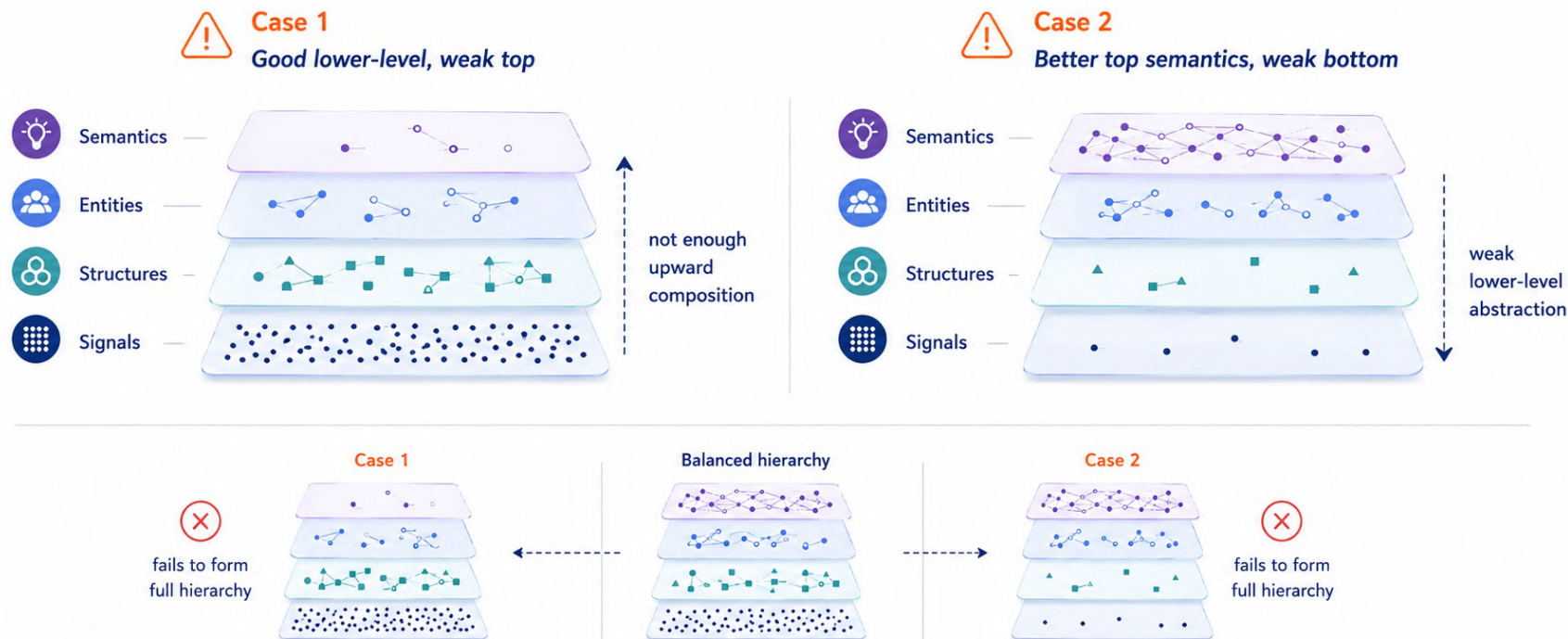
- NTP predicts **high-SNR, semantically meaningful targets**, pushing representations upward.
- **Predicting an abstraction** teaches how it is composed; **predicting with it** teaches how it composes further.
- This recursive supervision effectively learns **semantic construction across levels**.



# §7.1 Scalable objective : Why Today's SSL Falls Short



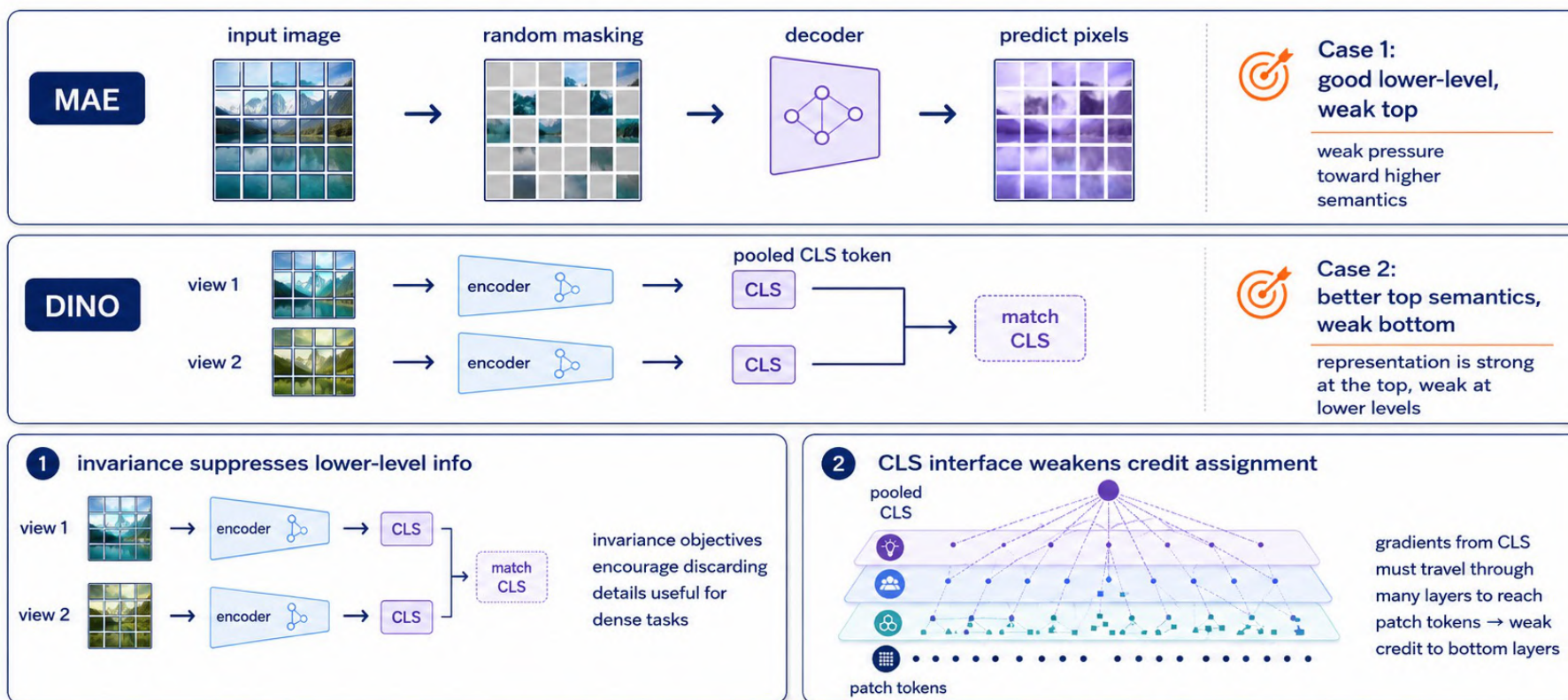
- **MAE**: pixel reconstruction preserves lower-level information, but provides weak pressure toward higher semantics.
- **DINO**: invariance promotes strong semantics, but can **underspecify or suppress lower-level abstractions**.
- Its CLS-based interface also makes **credit assignment to earlier representations** difficult.



# §7.1 Scalable objective : Two Ways Scaling Fails



- **Case 1:** strong lower levels, but insufficient **upward composition** → weak high-level semantics.
- **Case 2:** strong top semantics, but weak lower levels → fragile or incomplete composition.
- A scalable objective must avoid **both height collapse and width starvation**.

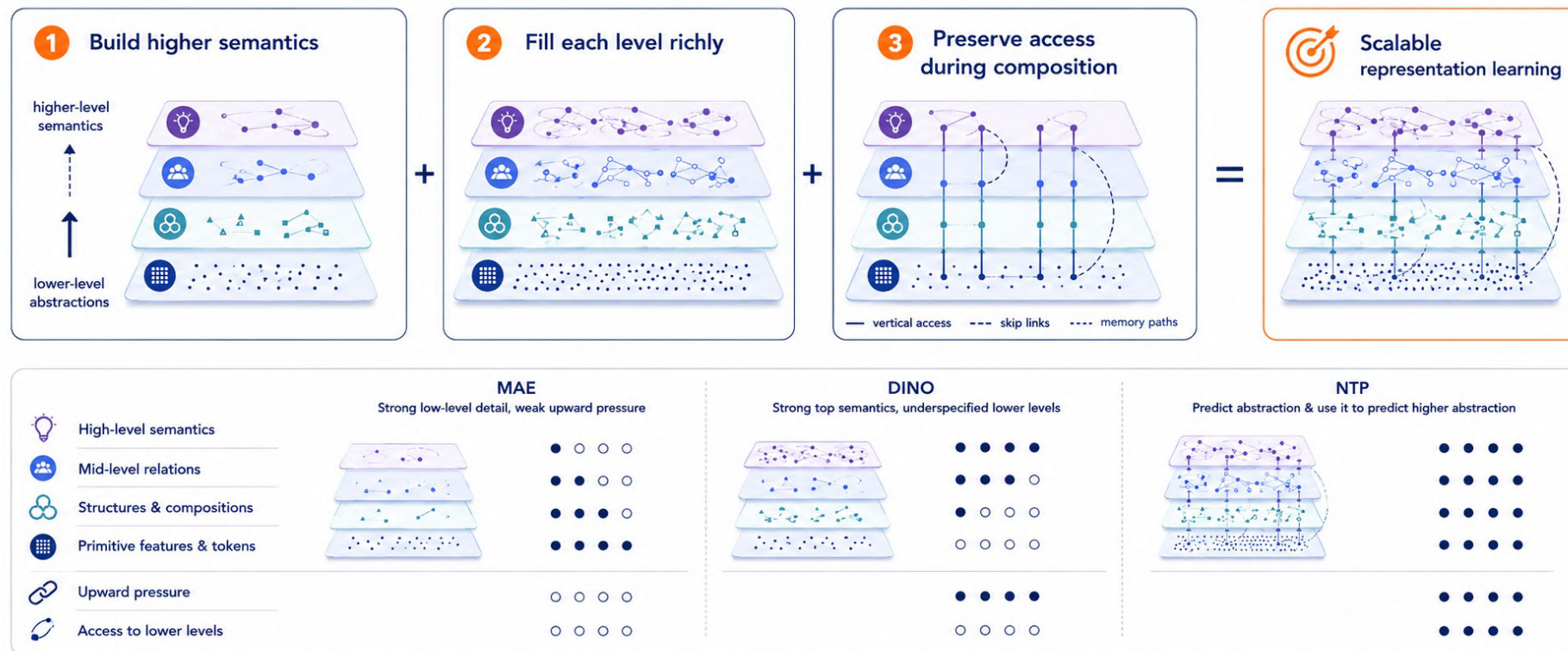


Prolonged training can hurt dense-task performance.

## §7.1 Scalable objective : What Should a Scalable Objective Do?



- **Build up:** progressively compose abstractions into higher-level semantics.
- **Fill out:** continuously improve the richness of every abstraction level.
- **Preserve access:** keep useful lower-level abstractions available throughout composition.
- **Core principle:** *build up, fill out, and preserve access.*



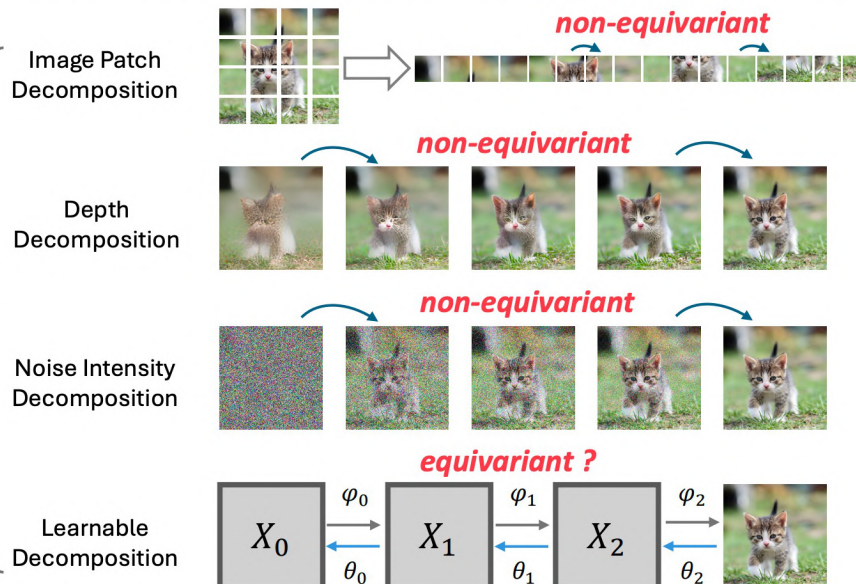
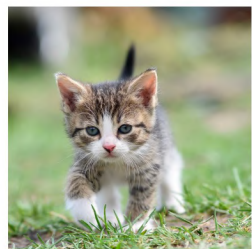
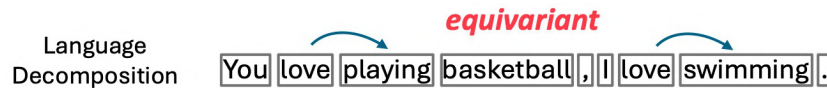
## §3.2 Generation —DDPM: some analysis (Recall)



### How Should Visual Signals Be Decomposed?

- Complex visual distributions should be split into **simpler conditional subproblems** rather than fitted monolithically.
- Language AR works well because position-wise subtasks are approximately **equivariant**: similar conditional mappings can reuse one shared model.
- Image **patch-, depth-, and noise-level decompositions are generally non-equivariant**, so shared parameters face conflicting distributions; fully separate models instead explode capacity.
- The real goal is therefore to discover **equivariant—or nearly independent—visual subproblems**, possibly through learnable or hybrid decomposition.

You love playing  
basketball,  
I love swimming.

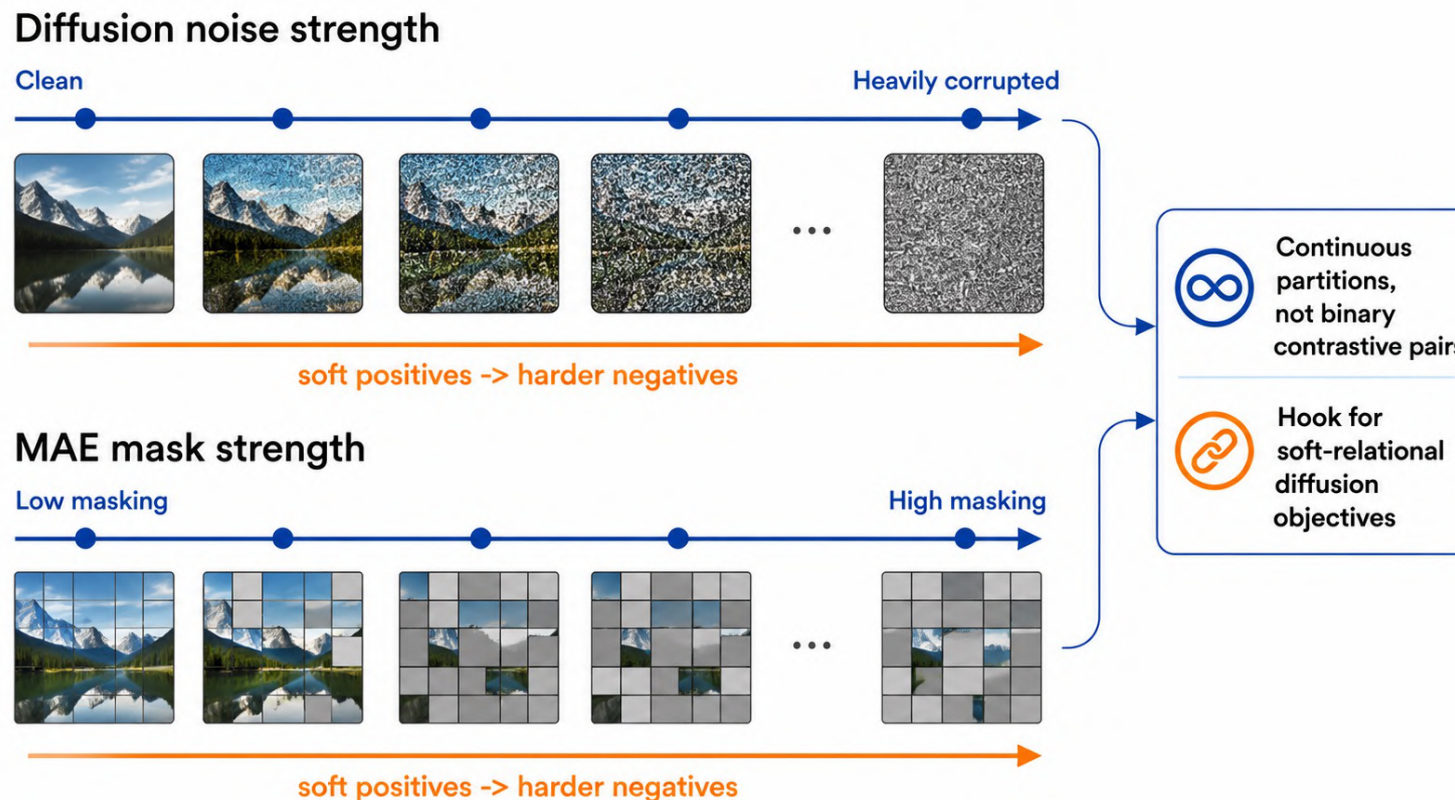


<https://cientgu.github.io/files/VisualSignalDecomposition.pdf>

## §7.2 The continuum thesis — noise builds a continuum of soft positive/negative



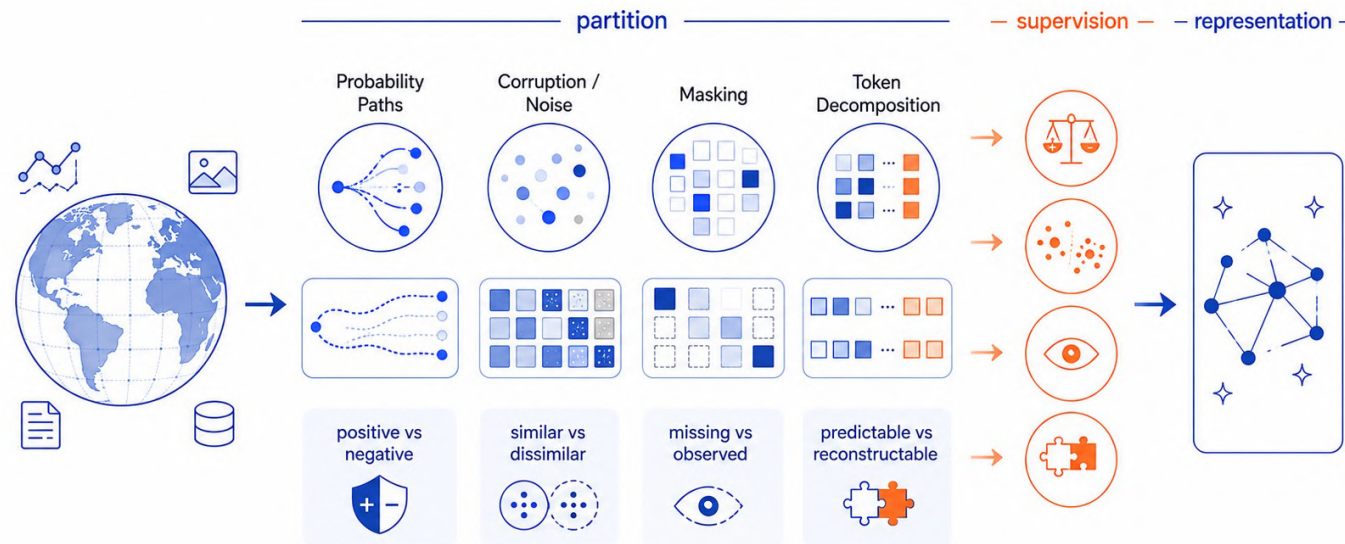
- Diffusion naturally builds a continuous spectrum of similarity via varying-strength corruption — a continuum of soft positive/negative pairs across noise levels.
- MAE's mask-strength does the same on the mask axis. → both define *continuous* partitions, unlike binary contrastive.
- This is exactly the hook for §4/§5's soft-relational diffusion objectives.



## §7.2 Back to the Partition: How Do We Split the World?



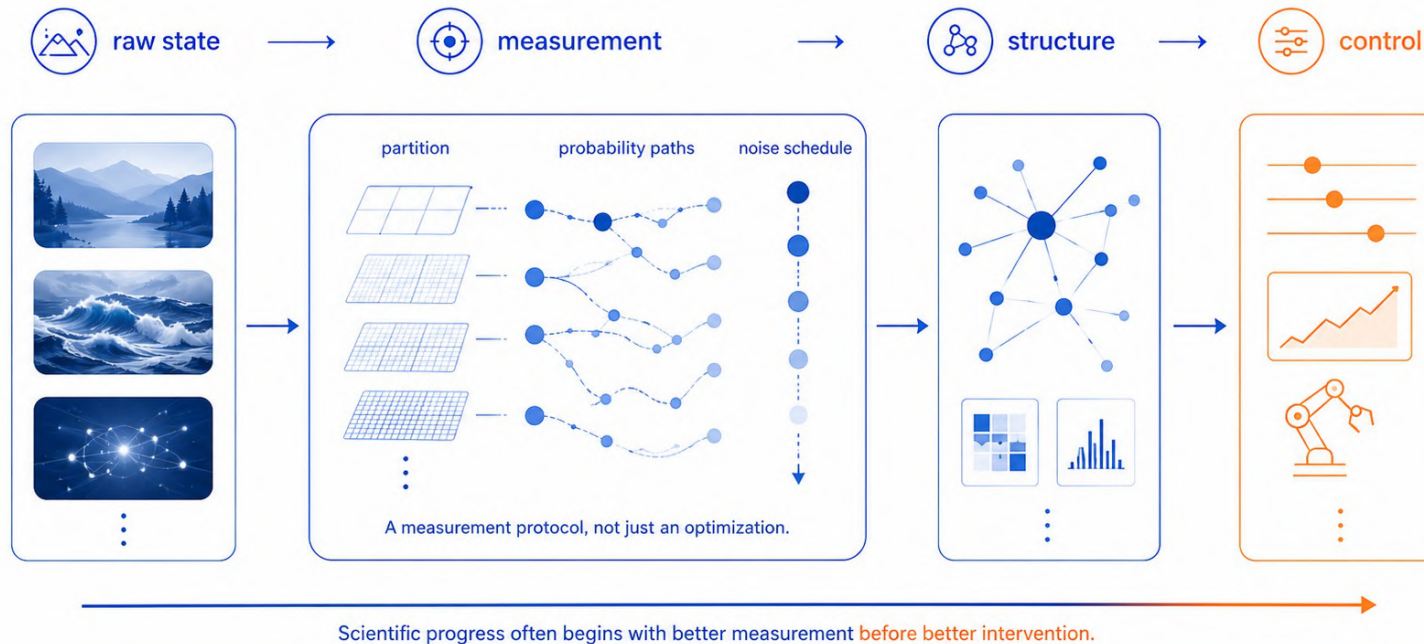
- The key design returns to the **partition strategy**: probability paths, corruption processes, masking schemes, or token decompositions.
- This partition decides what counts as positive, negative, similar, missing, predictable, or reconstructable.
- Therefore, it directly shapes the loss, the supervision signal, and ultimately the representation learned by the model.



## §7.2 Partition as Measurement



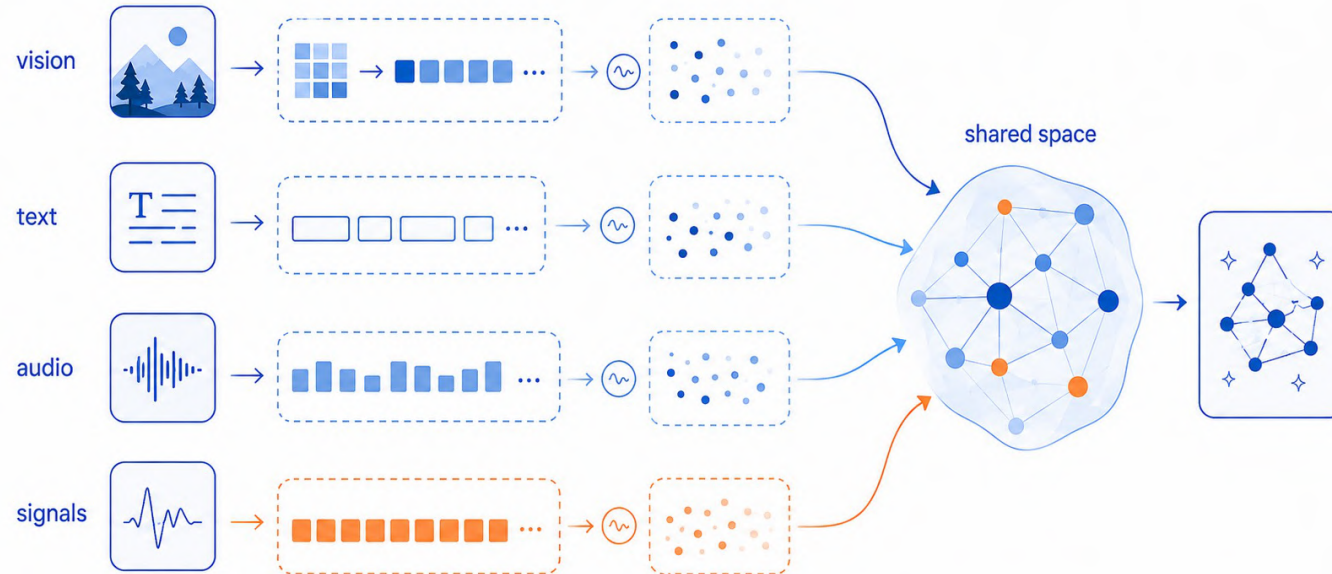
- A probability path is not only an optimization trick; it is also a **measurement protocol** for exposing structure in data.
- Scientific progress often begins with better measurement before better intervention.
- Images can be seen as raw physical states, but useful measurement requires representation learning to extract controllable structure.



## §7.2 From Single-Modality States to Unified Representation Space



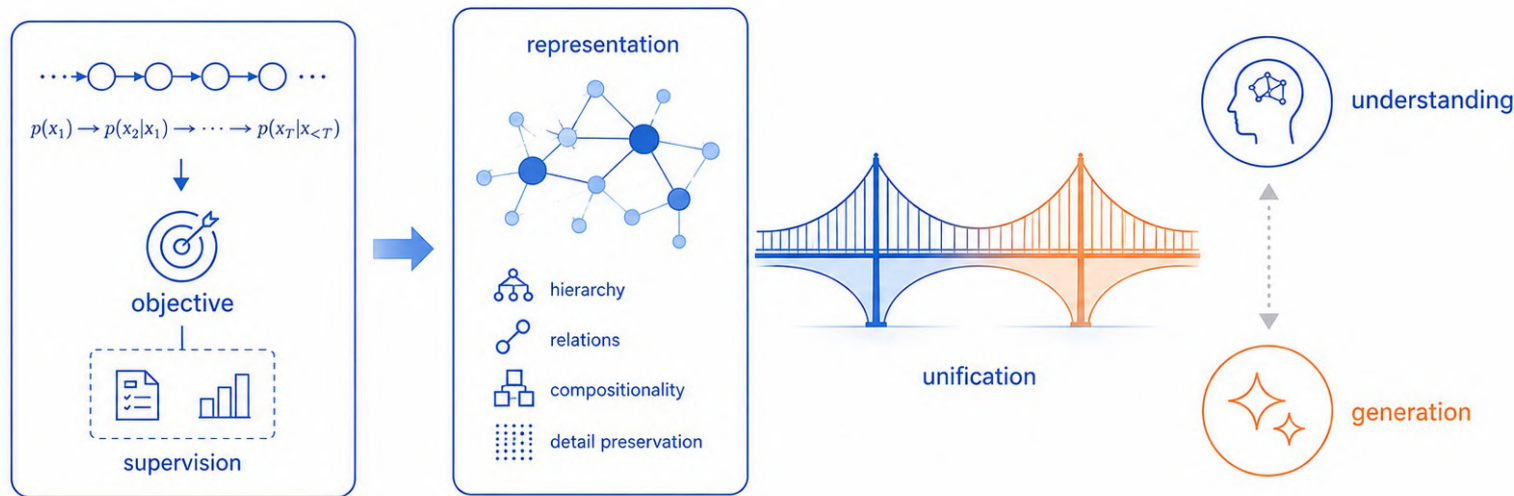
- In multimodal generation, “understanding” is often defined by generating text, but single-modality representations may already contain rich structure.
- The harder question is alignment: different modalities use different tokenizers, noise processes, and decompositions.
- A unified model therefore needs a shared representation space where visual, textual, and other modality-specific partitions become comparable.



## §7.2 The Core Logic: Objective → Representation → Unification



- To unify understanding and generation, we need representation as the bridge, not merely one shared architecture.
- To obtain scalable representations, the objective must induce hierarchy, relations, compositionality, and detail preservation beyond trivial MSE.
- The key is decomposition: the probability path defines supervision, supervision shapes representation, and representation enables unified understanding and generation.

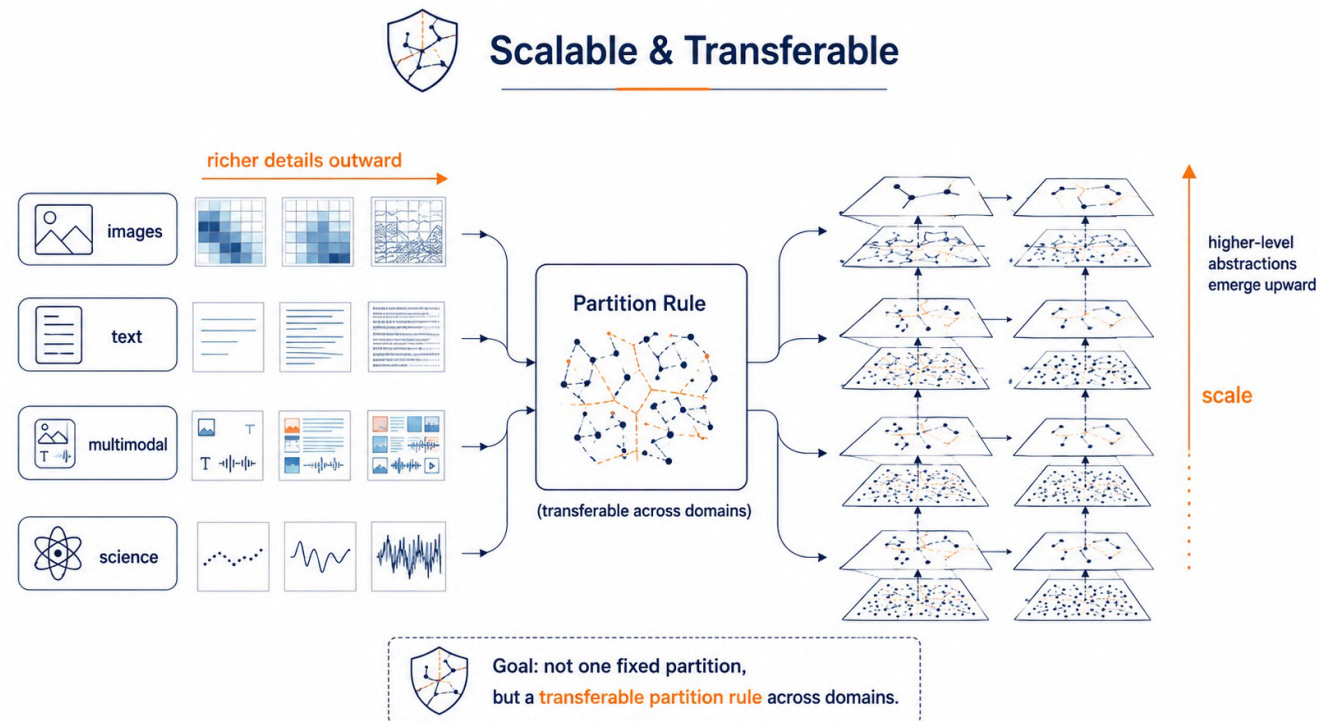


## §7.3 What requirements are needed for the Partition?



### Requirement I: Scalable & Transferable

- A good decomposition should scale with data, model size, and modality, rather than depend on a fragile hand-crafted tokenizer or external teacher.
- It should induce representations that become richer with scale: building higher-level abstractions while strengthening lower-level ones.
- The goal is not a fixed “best partition,” but a partition rule that transfers across images, text, multimodal data, and scientific domains.

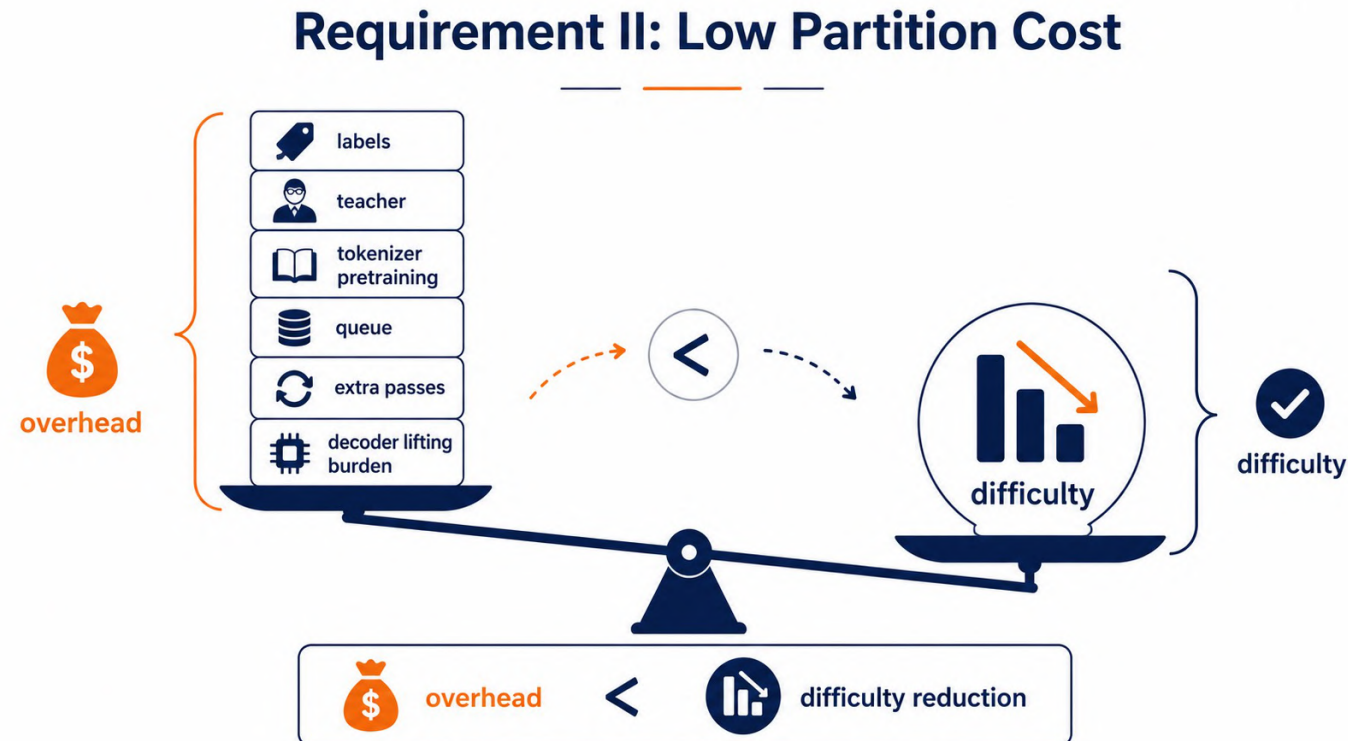


## §7.3 What requirements are needed for the Partition?



### Requirement II: Low Partition Cost

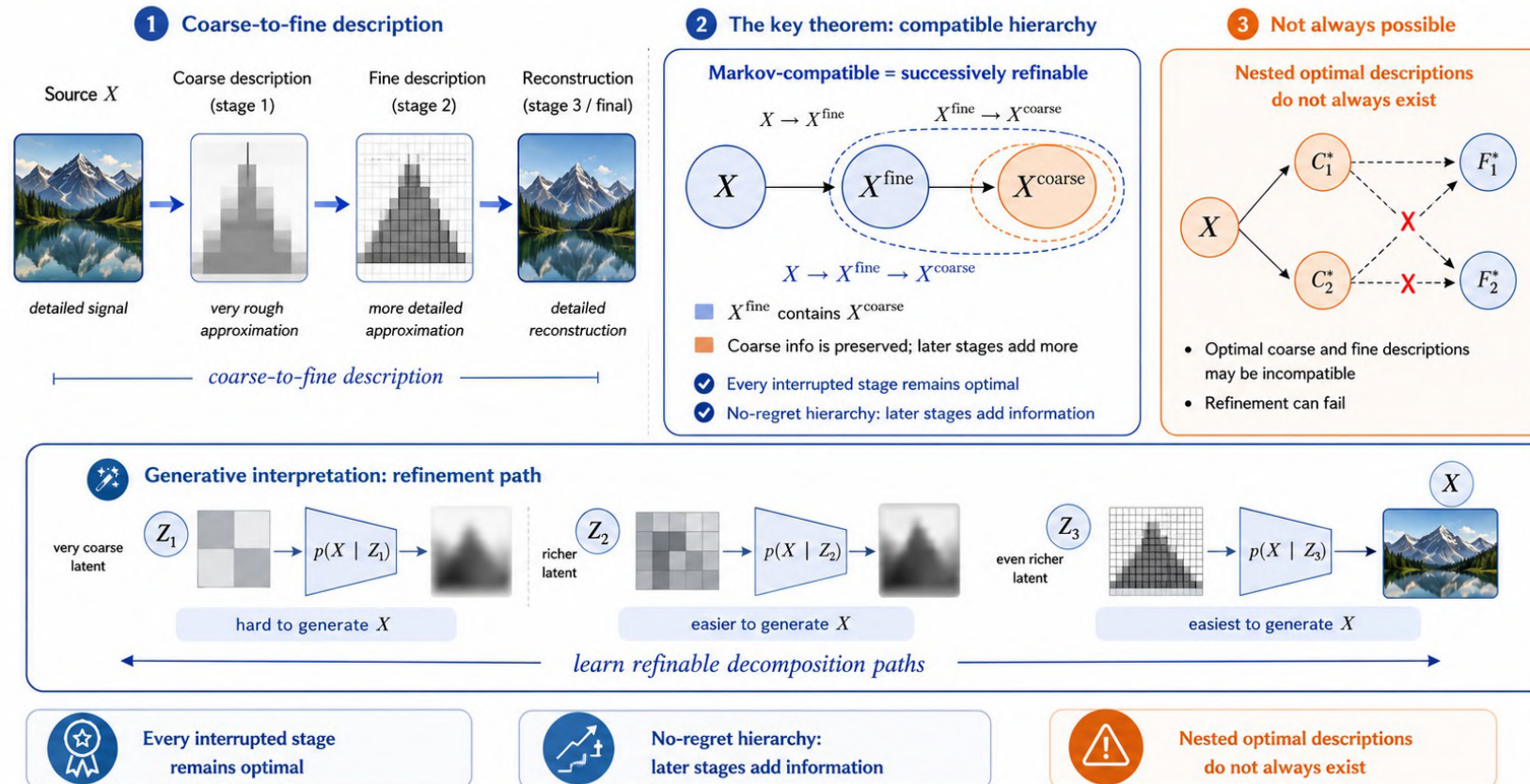
- The partition should be cheap to construct: no heavy labels, no expensive teacher for every domain, and no large extra inference-time module.
- Its cost must be counted honestly: tokenizer pretraining, teacher data, memory queue, extra forward passes, and decoder lifting burden.
- A useful decomposition should reduce total modeling difficulty more than it adds preprocessing or supervision overhead.



## §7.3 What requirements are needed for the Partition?

- **Question:** Can data be generated/compressed **coarse-to-fine**, while every intermediate representation remains optimal for its current information budget?
- **Key theorem:** Successive refinement is possible iff the optimal coarse and fine descriptions are **Markov-compatible**,  $X \rightarrow X^{fine} \rightarrow X^{coarse}$ .

### Equitz & Cover (1991) Successive Refinement of Information



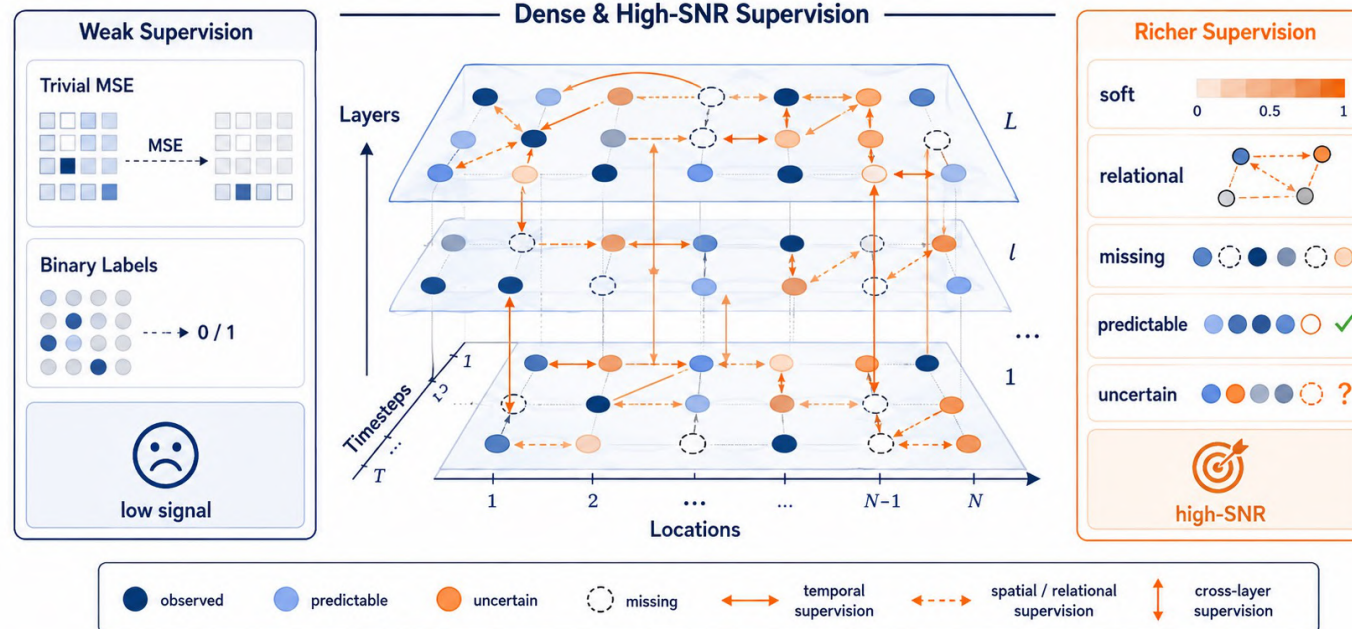
## §7.3 What requirements are needed for the Partition?



### Requirement III: Dense & High-SNR Supervision

- The partition should create meaningful supervision at many locations, timesteps, tokens, and abstraction levels.
- It should avoid trivial MSE targets and binary positive/negative labels; instead, it should provide soft, relational, and uncertainty-aware signals.
- Good supervision tells the model what is similar, what is missing, what is predictable, and what should remain uncertain.

### Requirement III: Dense & High-SNR Supervision

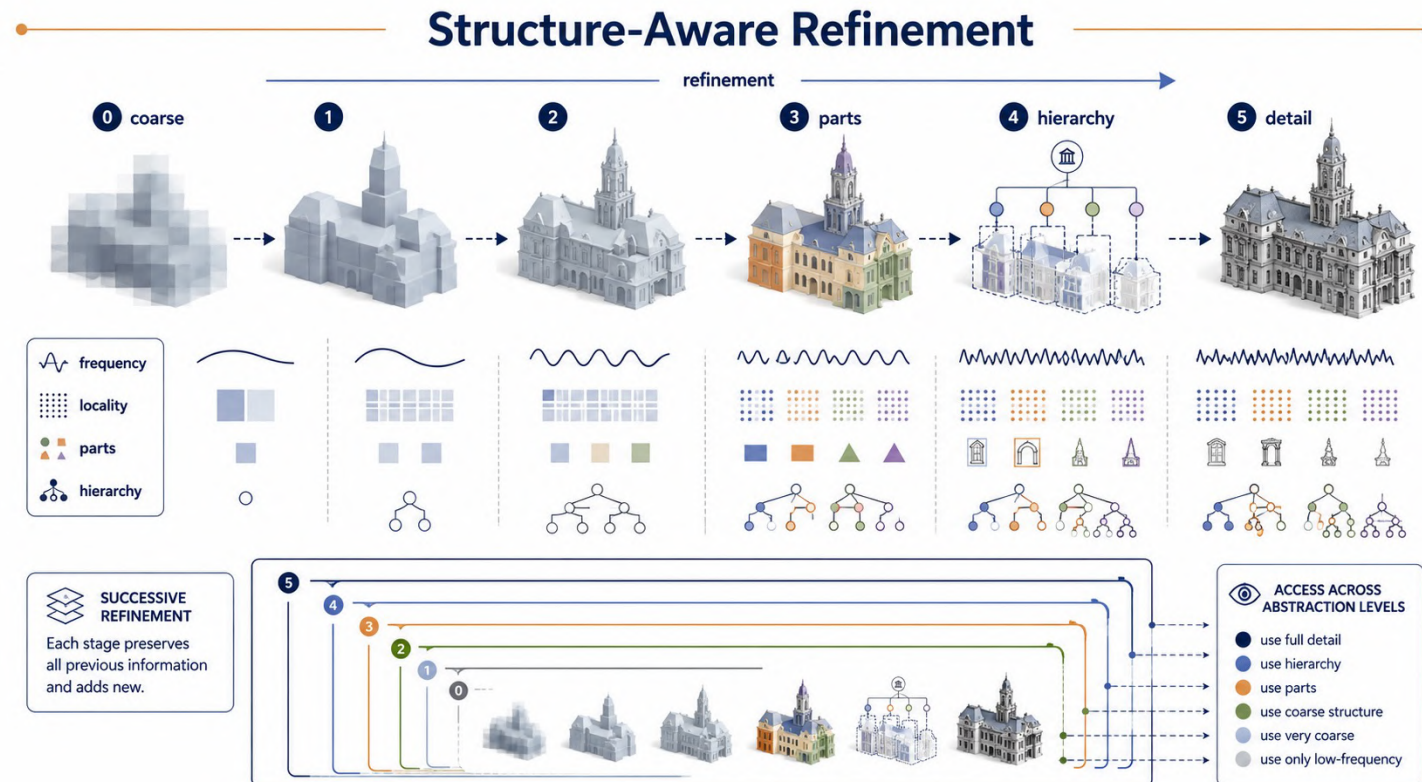


## §7.3 What requirements are needed for the Partition?



### Requirement IV: Structure-Aware Refinement

- The partition should respect the object's natural structure: frequency, locality, parts, hierarchy, and coarse-to-fine organization.
- Ideally, each intermediate state is useful under its own information budget, not merely a stepping stone to the final output.
- This connects to successive refinement: good decompositions should build up, fill out, and preserve access across abstraction levels.

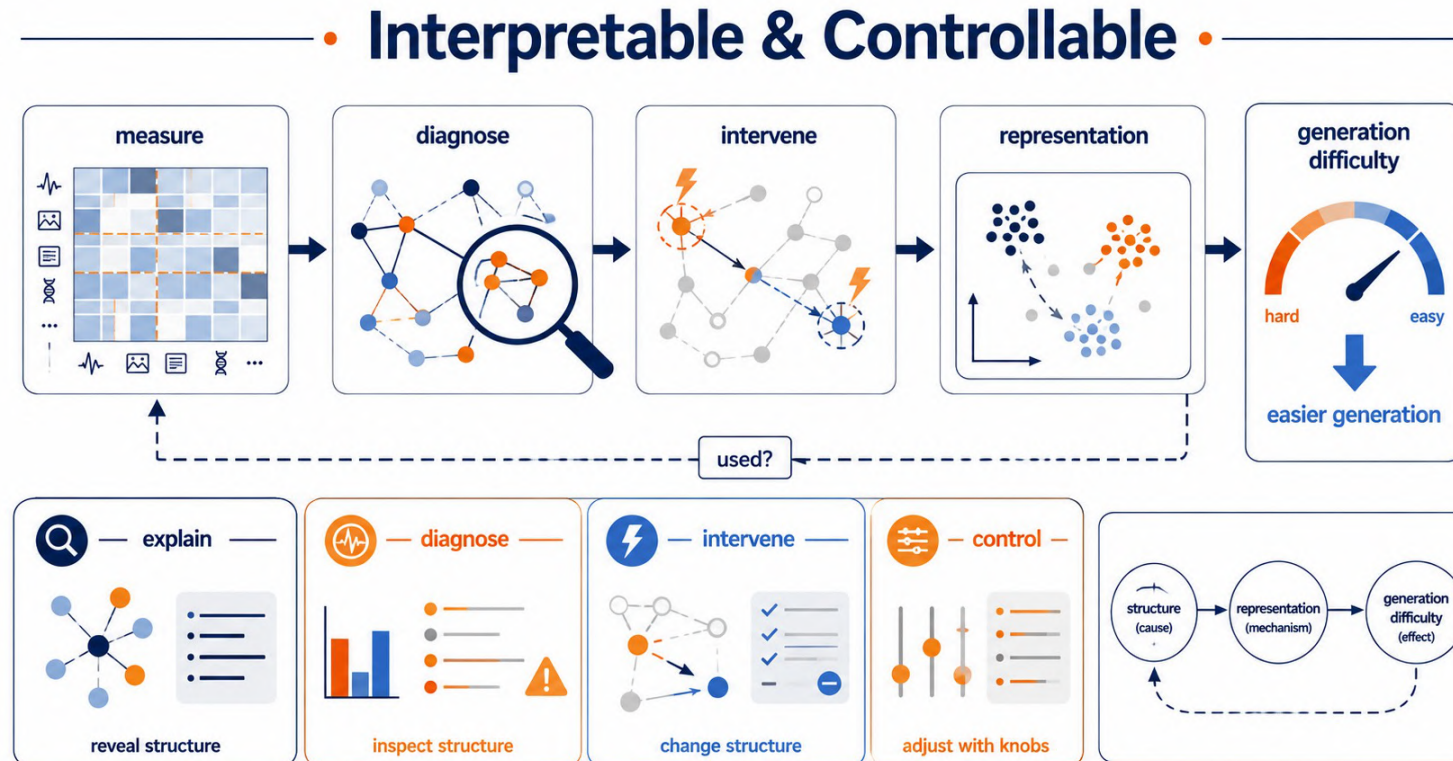


## §7.3 What requirements are needed for the Partition?



### Requirement V: Interpretable & Controllable

- A decomposition is also a measurement protocol: it decides which structure becomes visible to the model.
- We should be able to explain what each partition measures, diagnose whether it helps generation, and intervene when it fails.
- The final criterion is causal: does this partition shape representations that are actually used to reduce generative difficulty?



## §8 Summary & Wild idea & Open question

§8.1 Other wild ideas

§8.2 Answering the root question

1 Outline

2 Motiv

3 Intro

4 Gen→Rep

5 Rep→Gen

6 Quick Tour

7 Unified

8 Summary

## §8.1 Other wild ideas (brief)

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- Self-contained relation alignment: drop external encoder + pointwise + fixed mask + isotropic Gaussian at once.
- Enactive / actionable latents (behavioral similarity via RL) — my view (c).
- Computational Complexity theory ↔ representation-space (my view (b)).
  - Study few-step generation mechanism through **computational, information, and sequential complexity**
  - identify representation spaces with lower sequential complexity,
  - and optimize the **generation order** to reduce irreducible serial computation.
- To improve the NN architecture or design novel one with fully understanding of Gen  $\Leftrightarrow$  Rep, especially for the skip connection design.

## §8.2 Answering the root question

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- **Can one model do both, natively?** Not yet — but the path is visible: understanding and generation are one coin (§2), generation learns understanding for free but fragile (§4), understanding helps generation but crudely (§5).
- A unified model = borrowed/self semantic geometry + transport dynamics + light decoder, with a representation geometry jointly optimized for understanding and generation.

## §8 Main take-aways



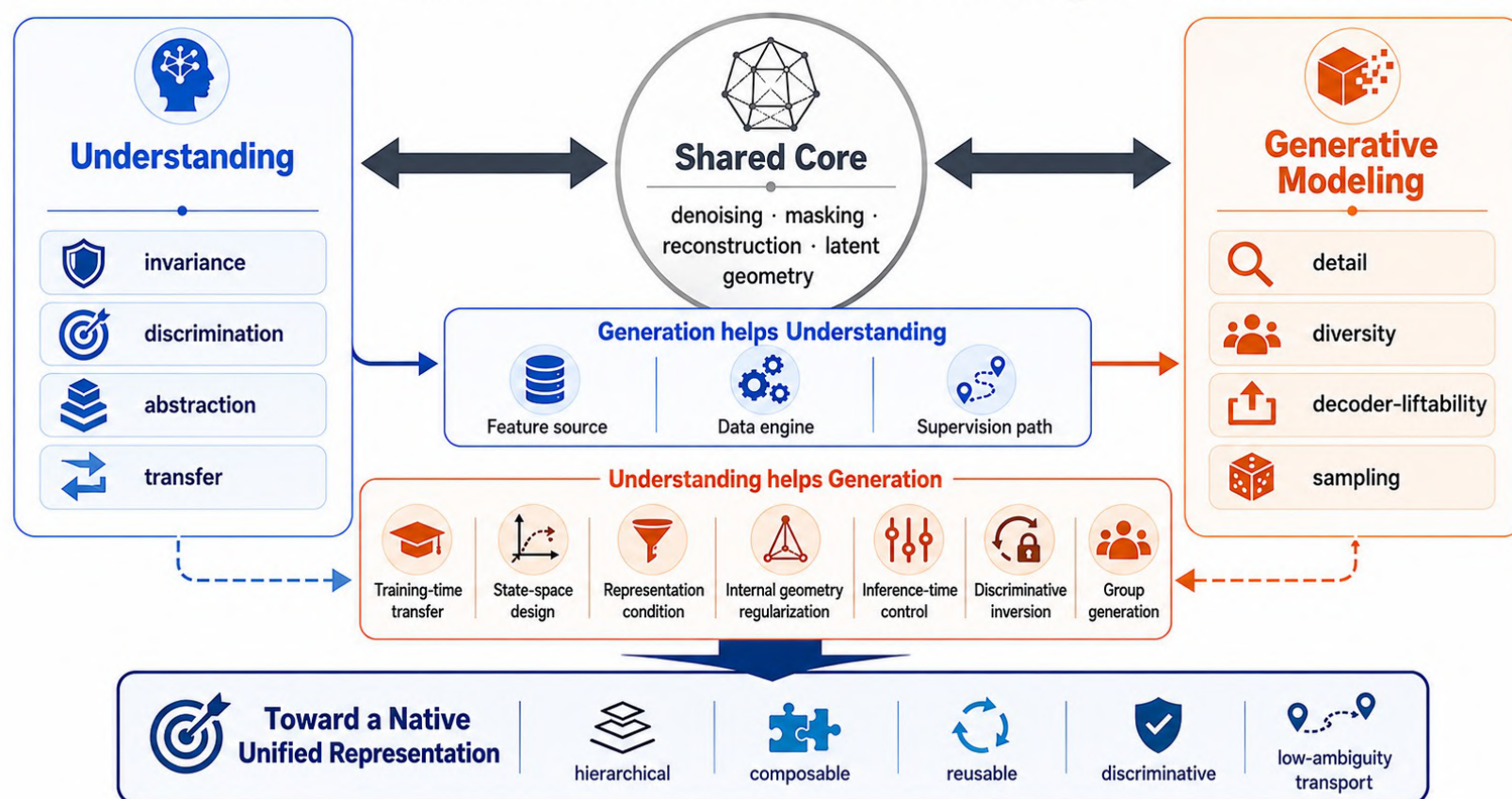
- 1 Understanding & generation are one problem seen from two ends — the root question is fusing them.
- 2 Generation learns understanding for free, but they lag (Gap 1) — decoupled from generation quality.
- 3 Understanding helps generation, but crudely (Gap 2) — the frontier is which property / what granularity.
- 4 MAE are near-native unified; none jointly optimizes the right rep geometry.
- 5 Our bet: representation is the bridge between understanding and generation.

# §8 Closing

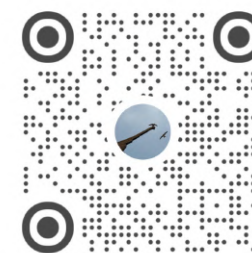


- "Representation is winning; generation paradigms stall; the two are one coin. The prize is a native unified model — and the key may be to stop matching points and start matching relations."
- Discussion / collaboration invite (per the notes' purpose: 交流协作, 跳出当局者迷).

## Toward a Unified Model for Understanding and Generation



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挑灯看剑

小红书号: 9757223776

扫描二维码, 在小红书找到我



Github repo

Related papers, blogs, textbooks, researchers can be found in

<https://github.com/Zhangtao167/Awesome-Representation-bridges-Generation-and-Understanding-Together>